Advanced Theory of Statistics

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Preface

This is the lecture note of the year course *Advanced Theory of Statistics* taught by Professor Wenlong Mou in the first half and Professor Xin Bing in the second half in academic year 2025-26 at University of Toronto. The note is later revised with the support of ChatGPT. Details missing from the presentations are complemented according to my understandings and thus may contain unavoidable errors and mistakes.

Contents

Chapter 1 Classical Decision Theory

1.1 Lecture 1, 19: Basic Concentration Inequalities

We start with the introduction to concentration inequalities.

There are several refinements to the Chebyshev inequality. For many random variables, the moment generating function will exist in a neighborhood around 0. In these cases, we can use the mgf to produce a tail bound.

Define $\mu = \mathbb{E}[X]$. For any $t > 0$, we have that,

$$\mathbb{P}((X-\mu) \geq u) = \mathbb{P}(\exp(t(X-\mu)) \geq \exp(tu)) \leq \frac{\mathbb{E}[\exp(t(X-\mu))]}{\exp(tu)}$$

by applying Markov's inequality. Now t is a parameter we can choose to get a tight upper bound, i.e., we can write this bound as:

$$\mathbb{P}((X-\mu) \geq u) \leq \inf_{0 \leq t \leq b} \exp(-t(u+\mu))\mathbb{E}[\exp(tX)]$$

This bound is known as Chernoff's bound. Suppose that, $X \sim N\left(\mu, \sigma^2\right)$, then a simple calculation gives that the mgf of X is:

$$M_X(t) = \mathbb{E}[\exp(tX)] = \exp\left(t\mu + t^2\sigma^2/2\right).$$

The mgf is defined for all t. To apply the Chernoff bound we then need to compute:

$$\inf_{t \geq 0} \exp(-t(u+\mu)) \exp\left(t\mu + t^2\sigma^2/2\right) = \inf_{t \geq 0} \exp\left(-tu + t^2\sigma^2/2\right),$$

which is minimized when $t = u/\sigma^2$ which in turn yields the tail bound,

$$\mathbb{P}(X - \mu \geq u) \leq \exp\left(-u^2/\left(2\sigma^2\right)\right)$$

This is often referred to as a one-sided or upper tail bound. We also have the two-sided Gaussian tail bound:

$$\mathbb{P}(|X-\mu| \geq u) \leq 2\exp\left(-u^2/\left(2\sigma^2\right)\right)$$

The main thing to observe is that this inequality is much sharper than Chebyshev's inequality. In particular, suppose we consider the average of i.i.d. Gaussian random variables, i.e. we have $X_1, \ldots, X_n \sim N\left(\mu, \sigma^2\right)$ and we construct the estimate:

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Using the fact that the average of Gaussian RVs is Gaussian, we obtain that $\widehat{\mu}$ has a $N\left(\mu, \sigma^2/n\right)$ distribution. In this case, using the Gaussian tail bound we derived, we obtain that,

$$\mathbb{P}(|\widehat{\mu} - \mu| \geq t\sigma/\sqrt{n}) \leq 2\exp\left(-t^2/2\right)$$

This is an example of an exponential tail inequality. It turns out that the Gaussian tail inequality from the previous section is much more broadly applicable to a class of random variables called sub-Gaussian random variables.

Definition 1.1.1. A random variable X with mean μ is sub-Gaussian if there exists a positive number σ such that,

$$\mathbb{E}[\exp(t(X-\mu))] \leq \exp\left(\sigma^2 t^2/2\right),$$

for all $t \in \mathbb{R}$. Such a random variable is called σ-sub-Gaussian.

Remark. Gaussian random variables with variance σ^2 satisfy the above condition with equality, so a σ-sub-Gaussian random variable basically just has an mgf that is dominated by a Gaussian with variance σ.

Example. (Rademacher random variables are 1-sub-Gaussian) Let X be a Rademacher random variable, which takes the values $\{+1, -1\}$ equiprobably. In this case, we can see that,

$$\begin{aligned}
\mathbb{E}[\exp(tX)] &= \frac{1}{2}[\exp(t) + \exp(-t)] \\
&= \frac{1}{2}\left[\sum_{k=0}^{\infty} \frac{(-t)^k}{k!} + \sum_{k=0}^{\infty} \frac{t^k}{k!}\right] \\
&= \sum_{k=0}^{\infty} \frac{t^{2k}}{(2k)!} \leq \sum_{k=0}^{\infty} \frac{t^{2k}}{2^k k!} \\
&= \exp\left(t^2/2\right)
\end{aligned}$$

This shows that Rademacher random variables are 1-sub-Gaussian.

Example. (Bounded random variables are sub-Gaussian) Let X be a random variable with zero mean and with support on some bounded interval $[a, b]$. WLOG, one may assume that X has a zero mean. Let X' denote an independent copy of X then we have that,

$$\mathbb{E}_X[\exp(tX)] = \mathbb{E}_X\left[\exp\left(t\left(X - \mathbb{E}\left[X'\right]\right)\right] \leq \mathbb{E}_{X,X'}\left[\exp\left(t\left(X - X'\right)\right]\right.\right.,$$

using Jensen's inequality and the convexity of the function $g(x) = \exp(x)$. Now, let ϵ be a Rademacher random variable. Then note that the distribution of $X - X'$ is identical to the distribution of $X' - X$ and, more importantly, of $\epsilon\left(X - X'\right)$. So we obtain that,

$$\begin{aligned}
\mathbb{E}_{X,X'}\left[\exp\left(t\left(X - X'\right)\right]\right. &= \mathbb{E}_{X,X'}\left[\mathbb{E}_{\epsilon}\left[\exp\left(t\epsilon\left(X - X'\right)\right)\right]\right] \\
&\leq \mathbb{E}_{X,X'}\left[\exp\left(t^2\left(X - X'\right)^2/2\right]\right.
\end{aligned}$$

where we just use the result that Rademacher random variables are 1-sub-Gaussian. Now using the boundedness of $X - X'$, we obtain that

$$\mathbb{E}_X[\exp(tX)] \leq \exp\left(t^2(b-a)^2/2\right),$$

which in turn shows that bounded random variables are $(b-a)$-sub-Gaussian. Using some refined analysis, one can show that X is indeed $(b-a)/2$-sub-Gaussian.

It is straightforward to go through the above Chernoff bound to conclude that for a sub-Gaussian random variable, we have the same two-sided exponential tail bound.

Theorem 1.1.2. Let X be a σ-sub-Gaussian random variable, then

$$\mathbb{P}(|X-\mu| \geq u) \leq 2\exp\left(-u^2/\left(2\sigma^2\right)\right)$$

The following elementary result will be useful.

Lemma 1.1.3. Suppose that $X = (X_1, \ldots, X_n)$ is a random vector such that X_i is σ_i-sub-Gaussian for $i = 1, \ldots, n$. If X_i are independent and $u \in \mathbb{R}^n$, then the random variable $u^\top X$ is sub-Gaussian with parameter $\sigma = \sqrt{\sigma_1^2 u_1^2 + \ldots + \sigma_n^2 u_n^2}$.

We list a couple of useful observations that follow directly from this lemma. They show that the σ parameter behaves similarly to the standard deviation.

Corollary 1.1.4. For a collection of independent σ-sub-Gaussian random variables $X_1, \ldots, X_n$, their average $\bar{X}_n$ is sub-Gaussian with parameter $\sigma/\sqrt{n}$.

Corollary 1.1.5. If X is σ-sub-Gaussian, then $aX + b (a \in \mathbb{R}, b \in \mathbb{R})$ is sub-Gaussian with parameter $|a|\sigma$. In particular, $-X$ is σ-sub-Gaussian.

Corollary 1.1.6. If $X = (X_1, \ldots, X_n)$ is a vector of independent random variables such that each X_i is σ-sub-Gaussian, and $\|\boldsymbol{u}\| = 1$, then $\boldsymbol{u}^T X$ is σ-sub-Gaussian.

Corollary 1.1.7. Suppose we have n i.i.d. σ-sub-Gaussian random variables $X_1, X_2, \ldots, X_n$ with a common mean μ. Then we also have the following tail estimate for their average $\widehat{\mu} = n^{-1} \sum_i X_i$:

$$\mathbb{P}(|\widehat{\mu} - \mu| \geq k\sigma/\sqrt{n}) \leq 2\exp\left(-k^2/2\right).$$

Remark. Actually, we have a tighter Hoeffding bound when the random variables are supported on $[a, b]$ and i.i.d.:

$$\mathbb{P}\left(|\hat{\mu} - \mu| \geq t/\sqrt{n}\right) \leq 2\exp\left(-\frac{2t^2}{(b-a)^2}\right)$$

The following result applies Chernoff bounds to a sequence of sub-Gaussian variables.

Proposition 1.1.8 (Hoeffding inequality)**.** If X_i are independent σ_i-subGaussian then for all $t \geq 0$ we have

$$\mathbb{P}\left(|\bar{X}_n - \mathbb{E}\left(\bar{X}_n\right)| \geq t\right) \leq 2e^{-t^2n^2/\left(2\sum_i \sigma_i^2\right)}$$

Proof. It sufficies to notice that $\bar{X}_n$ is sub-Gaussian with parameter

$$\sigma = \frac{1}{n}\sqrt{\sum_{i=1}^{n} \sigma_i^2}.$$

□

Remark. If $\sigma_i = \sigma$ for all i we also note that the Hoeffding inequality implies

$$\mathbb{P}\left(\left|\sqrt{n}\left(\bar{X}_n - \mathbb{E}\left(\bar{X}_n\right)\right)\right| \geq t\right) \leq 2e^{-t^2/\left(2\sigma^2\right)}$$

Another general type of bound on the moment generating function is the following.

Definition 1.1.9. A random variable with mean $\mu = \mathbb{E}\mathbb{X}$ is sub-exponential if there are non-negative parameters v, α such that

$$\mathbb{E}\left(e^{\lambda(X-\mu)}\right) \leq e^{v^2\lambda^2/2} \quad \text{for all } |\lambda| < \frac{1}{\alpha}.$$

Remark. This condition is rather mild, and it is essentially equivalent to the existence of the cumulant generating function in a neighbourhood of zero.

Example. Let $Z \sim N(0,1)$ and let $X = Z^2$. For $\lambda < 1/2$, we have

$$\mathbb{E}\left(e^{\lambda(X-1)}\right) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{\lambda\left(z^2-1\right)} e^{-z^2/2}\, \mathrm{d}z = \begin{cases} \frac{e^{-\lambda}}{\sqrt{1-2\lambda}} & \text{if } \lambda < \frac{1}{2}, \\ +\infty & \text{if } \lambda \geq \frac{1}{2}. \end{cases}$$

In particular, since the moment generating function is not defined everywhere, X is not sub-Gaussian. With a bit of calculus, we see that

$$\frac{e^{-\lambda}}{\sqrt{1-2\lambda}} \leq e^{4\lambda^2/2} \quad \text{for all } |\lambda| < \frac{1}{4}$$

and so X is sub-exponential with parameters $(v, \alpha) = (2, 4)$.

With essentially the same proof, the linearity of sub-Gaussian random variables generalizes to subexponential variables.

Lemma 1.1.10. Suppose that $X = (X_1, \ldots, X_n)$ is a random vector such that X_i is (v_i, α_i)-sub-exponential for $i = 1, \ldots, n$. If X_i are independent and $u \in \mathbb{R}^n$, then the random variable $u^\top X$ is sub-exponential with parameters $v = \sqrt{v_1^2 u_1^2 + \ldots + v_n^2 u_n^2}$ and $\alpha = \max_i |u_i|\, \alpha_i$.

We obtain simple concentration inequalities for sub-exponential variables.

Proposition 1.1.11. Suppose that X is sub-exponential with parameters (v, α) and $t \geq 0$. Then

$$\mathbb{P}(X - \mu \geq t) \leq \begin{cases} e^{-\frac{t^2}{2v^2}} & \text{if } 0 \leq t \leq \frac{v^2}{\alpha} \\ e^{-\frac{t}{2\alpha}} & \text{if } t > \frac{v^2}{\alpha} \end{cases}$$

and

$$\mathbb{P}(|X - \mu| \geq t) \leq \begin{cases} 2e^{-\frac{t^2}{2v^2}} & \text{if } 0 \leq t \leq \frac{v^2}{\alpha} \\ 2e^{-\frac{t}{2\alpha}} & \text{if } t > \frac{v^2}{\alpha} \end{cases}$$

Proof. Using the Chernoff bound, we know that

$$\mathbb{P}(X - \mu \geq t) \leq \inf_{\lambda \in \left[0, \frac{1}{\alpha}\right)} \left\{\mathbb{E}e^{\lambda(X - \mathbb{E}X)} e^{-\lambda t}\right\} \leq \inf_{\lambda \in \left[0, \frac{1}{\alpha}\right)} \left\{e^{v^2\lambda^2/2 - \lambda t}\right\}.$$

The global optimum of $e^{v^2\lambda^2/2-\lambda t}$ is $\lambda^* = t/v^2 \geq 0$. Thus, if $t/v^2 < 1/\alpha$ (equiv. $v^2 > t\alpha$), we get the sub-Gaussian bound $e^{-t^2/(2v^2)}$. Otherwise, if $v^2 \leq t\alpha$, the infimum is obtained on the boundary $\lambda^* = 1/\alpha$ and it is equal to $e^{v^2/(2\alpha^2)-t/\alpha}$. Note however that the fact that $v^2 \leq t\alpha$, allows to bound this by $e^{-t/(2\alpha)}$. $\square$

Remark. These propositions now give a handful of useful results. For example, if X_i are independent sub-exponential with parameters (v, α) then $\bar{X}_n$ is sub-exponential with parameters $\left(\frac{v}{\sqrt{n}}, \frac{\alpha}{n}\right)$ and so

$$\mathbb{P}\left(|\bar{X}_n - \mu| \geq t\right) \leq \begin{cases} 2e^{-\frac{t^2 n}{2v^2}} & \text{if } 0 \leq t \leq \frac{v^2}{\alpha} \\ 2e^{-\frac{tn}{2\alpha}} & \text{if } t > \frac{v^2}{\alpha} \end{cases}$$

To conclude the introduction to concentration inequalities, we present one of its applications in probabilistic combinatorics.

Definition 1.1.12. For any integer $n > 0$, define the Hamming distance on the hypercube $\{0,1\}^n$

$$d_H(x,y) := \sum_{i=1}^{n} \mathbf{1}_{x_i \neq y_i}, \qquad \text{for any } x, y \in \{0,1\}^n$$

Theorem 1.1.13. (GilbertVarshamov bound for Hamming packing) There exists a universal constant $c > 0$, such that for any $n > 0$, there exists a subset $A \subseteq \{0,1\}^n$ with $|A| \geq e^{cn}$, satisfying

$$d_H(x,y) \geq \frac{n}{4}, \qquad \text{for any pair } x, y \in A.$$

Proof. We use a brute-force search. Let M be some constant to be determined, and $X_1, \cdots, X_M$ be i.i.d. samples from the uniform distribution on $\{0,1\}^n$. For any $i \neq j$, since $d_H(X_i, X_j)$ is the sum of n independent Rademacher random variables, we have

$$\mathbb{P}\left(d_H(X_i, X_j) \leq n/4\right) \leq 2\exp(-n/8).$$

Therefore,

$$\mathbb{P}\left(d_H(x,y) \geq \frac{n}{4}, \text{ for any pair } x, y \in A\right) \leq M^2 \mathbb{P}\left(d_H(X_i, X_j) \leq n/4\right) \leq 2M^2 \exp(-n/8)$$

and it suffices to take $M = \left\lfloor \frac{1}{2} \exp(n/16) \right\rfloor$. Hence, $c = 1/16$ suffices. $\square$

1.2 Lecture 2, 15: Criteria for Decision Rules

Now we turn to the framework of decision theory. The goal of classical decision theory is to compare different statistical models and to find or prove the optimality of a specific model.

We start with formulating a statistical model and the method we use to evaluate it.

Definition 1.2.1. A statistical model is a collection of probability distributions $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$ of the observed data X, parametrized smoothly by the parameter Θ.
Suppose that we observe the data $X \sim \mathbb{P}_{\theta^*}$ for some $\theta^* \in \Theta$, and we have the following attributes of a statistical task:

(a). A decision rule δ that maps data X to an action $a \in \mathcal{A}$.

(b). A loss function $L : \Theta \times \mathcal{A} \mapsto \mathbb{R}_+$ that determines the loss of an action under this model.

(c). The risk of a decision rule δ under the model θ defined as

$$R(\theta; \delta) = \mathbb{E}_\theta[L(\theta; \delta(X))].$$

where $\mathbb{E}_\theta$ denotes the expectation under $\mathbb{P}_\Theta$.

We then illustrate some examples of statistical tasks.

Example. (Estimation) The goal is to estimate some quantity of a model, specified as $g(\theta^*)$. To this end, the decision rule δ is now an estimator $\hat{g}$ that operates on data, and usually one uses the mean square error to quantify the error of this estimation. Thus, in this task, $L(\theta; \hat{g}(X)) = (g(\theta) - \hat{g}(X))^2$.

Example. (Testing) The goal is to decide whether the true parameter θ^* lies in some subset Θ_0 of probability models. To formulate this, let the action set be $\mathcal{A} = \{0, 1\}$, where 0 denotes acceptance and 1 denotes rejection. The loss function is then

$$L(\theta; a) = \begin{cases} 1, & \theta \in \Theta_0, a = 1 \text{ or } \theta \notin \Theta_0, a = 0, \\ 0, & \text{otherwise.} \end{cases}$$

As a consequence, $R(\theta; \delta)$ is then the probability of the occurrence of type-I error if $\theta \in \Theta_0$ and the probability of the occurrence of type-II error otherwise.

In this lecture, we will go over three criteria for comparison of decision rules, including the admissibility of decision rules, Bayes procedures, and minimax procedures, and demonstrate some of their properties.

We start with the definition of admissibility, which is simply the decision rules that are not dominated by any other decision rules under any $\theta \in \Theta$.

Definition 1.2.2. Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ and that δ is a decision rule. Then δ is called inadmissible if there exists another decision rule $\tilde{\delta}$ such that

$$R(\tilde{\delta}; \theta) \leq R(\delta; \theta), \quad \forall \theta \in \Theta$$

and strict inequality holds for some $\theta \in \Theta$. If δ is not inadmissible, then δ is admissible.

The problem with this straightforward definition is that, sometimes, naive decision rules may be admissible and classical decision rules may be inadmissible. See the following examples.

Example. (Naive decision rules may be admissible) Suppose we have i.i.d. random variables $X_1, X_2, \ldots, X_n \sim \mathcal{N}(\theta, \sigma^2)$ where $\theta \in \mathbb{R}$ is unknown and σ^2 is known. Our goal is to estimate θ with an estimator $\hat{\theta}$ under the mean-square loss.
We now show that the naive estimator $\hat{\delta} \equiv 0$ is admissible. Indeed, if some $\tilde{\delta}$ strictly dominates $\hat{\delta}$, we know that $\tilde{\delta}$ will achieve zero loss in the case that $\theta = 0$. This indicates that $\tilde{\delta}(X) = 0$ for all possible $X = (X_1, X_2, \ldots, X_n)$. Since the Gaussian distribution has full support over the real line, we know that $\tilde{\delta}$ is identically zero and hence $\hat{\delta}$ is admissible.

Based on this drawback, it is reasonable to find some features that capture the risk $R(\delta; \theta)$ for all θ instead of considering them simultaneously. There are two main approaches: the first one is to consider a weighted average of the loss function among θ, which leads to a Bayes procedure; the other one is to consider the worst case among all θ, which leads to a minimax procedure.

Definition 1.2.3. Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ and that we impose a prior π on Θ. The Bayes risk is then defined as

$$r_\pi(\delta) = \int_\Theta \mathbb{E}_\theta[L(\theta; \delta(X))]\pi(\mathrm{d}\theta),$$

and the Bayes decision rule is then

$$\delta_{\text{Bayes},\pi} = \operatorname*{argmin}_\delta r_\pi(\theta)$$

Proposition 1.2.4. Suppose that the distribution of $X \sim P_\theta$ has density p_θ with respect to some base measure λ. Then the Bayes decision rule is given by

$$\delta_{\text{Bayes},\pi}(\mathbf{x}) = \operatorname*{argmin}_{a \in A} \mathbb{E}_{\pi(\cdot|\mathbf{x})}[L(\theta, a)],$$

where $\pi(\theta|\mathbf{x}) = \frac{p_\theta(\mathbf{x})}{\int_\Theta p_{\theta'}(\mathbf{x})\pi(d\theta')}\pi(d\theta)$ is the posterior distribution of θ.

Proof. Since the density of X is given by p_θ, we have

$$r_\pi(\delta) = \int_\Theta \int_X L(\theta; \delta(\mathbf{x}))p_\theta(x)\lambda(d\mathbf{x})\pi(d\theta) = \int_{\mathbf{X}} \left[\int_\Theta L(\theta; \delta(\mathbf{x}))p_\theta(\mathbf{x})\pi(d\theta)\right] \lambda(d\mathbf{x}),$$

and the result can be seen from that for each fixed $\mathbf{x}$, $\pi(\theta|\mathbf{x}) \propto p_\theta(\mathbf{x})\pi(d\theta)$. $\square$

Definition 1.2.5. Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. The minimax decision rule is defined as

$$\delta_{\text{minimax}} = \operatorname*{argmin}_\delta \sup_\theta R(\theta; \delta)$$

Remark. From a game-theoretic perspective, δ_{minimax} can be viewed as the optimal strategy of the zero-sum game between the statistician and the environment with the reward $R(\theta; \delta)$. As a result, we have the following propositions from game theory.

Proposition 1.2.6. (Weak duality) Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. Then it holds that

$$\sup_\theta \inf_\delta R(\theta; \delta) \le \inf_\delta \sup_\theta R(\theta; \delta)$$

Proposition 1.2.7. (Strong duality) Suppose in addition that the spaces Θ, $\mathcal{A}$, and X are finite, then strong duality holds:

$$\sup_\theta \inf_\delta R(\theta; \delta) = \inf_\delta \sup_\theta R(\theta; \delta)$$

Now we turn to relations and properties between these procedures.

Proposition 1.2.8. Bayes rules with constant risks are minimax.

Proof. Let θ be a Bayes rule under prior π with constant risk. Consider another decision rule δ', then

$$\sup_{\theta\in\Theta} R\left(\theta;\delta'\right) \geq \int_{\Theta} R\left(\theta,\delta'\right)\pi(d\theta) \geq \int_{\Theta} R\left(\theta,\delta\right)\pi(d\theta) = \sup_{\theta\in\Theta} R(\theta,\delta)$$

$\square$

The next result gives a connection between minimax and Bayesian rules as a generalization of the previous results.

Theorem 1.2.9. Let δ^* be a Bayes rule for some prior π. Suppose that

$$R\left(\theta,\delta^*\right) \leq r\left(\pi,\delta^*\right) \quad \text{for all} \quad \theta\in\Theta.$$

Then δ^* is minimax.

Proof. We prove the contrapositive statement. If δ^* is not minimax then there exists δ_0 such that $\bar{R}\left(\delta_0\right) < \bar{R}\left(\delta^*\right)$. However, the fact that $r\left(\pi,\delta_0\right) \leq \sup_\theta R\left(\theta,\delta_0\right)$ implies that

$$r\left(\pi,\delta_0\right) \leq \bar{R}\left(\delta_0\right) < \bar{R}\left(\delta^*\right) \leq r\left(\pi,\delta^*\right).$$

But this contradicts the fact that δ^* is a Bayesian rule. $\square$

Proposition 1.2.10. Suppose that $\exists\left(\pi_j,r_j\right)_{j=1}^{\infty}$ such that r_j is the Bayes risk under π_j using the Bayes decision rule. Suppose that there exists another decision rule δ such that $\liminf_{j\to+\infty} r_j \geq \sup_\theta R(\theta,\delta)$, then δ is minimax.

Proof. For another decision rule δ', we have

$$\begin{aligned}
\sup_{\theta} R\left(\theta,\delta'\right) &\geq \liminf_{j\to+\infty}\int_{\Theta} R\left(\theta,\delta'\right)\pi_j(d\theta)\\
&\geq \liminf_{j\to+\infty}\int_{\Theta} R\left(\theta,\delta_{\text{Bayes}}\right)\pi_j(d\theta)\\
&= \liminf_{j\to+\infty} r_j \geq \sup_{\theta} R(\theta,\delta).
\end{aligned}$$

$\square$

Proposition 1.2.11. Unique Bayes rules δ are admissible.

Proof. Suppose δ' dominates δ, then we have

$$r_\pi\left(\delta'\right) = \int R\left(\theta;\delta'\right)\pi(d\theta) \leq \int R(\theta;\delta)\pi(d\theta) = r_\pi(\delta)$$

So $\delta' = \delta$ by uniqueness. $\square$

Proposition 1.2.12. When $|\Theta| < +\infty$, admissible rules are Bayes (under some prior π).

Proof. Let δ_0 be admissible and $|\Theta| = k$. Consider the convex set

$$T := \left\{\left(R(\theta_i;\delta) - R(\theta_i;\delta_0)\right)_{i=1,\cdots,k} : \ \delta \text{ a rule}\right\} \subset \mathbb{R}^k.$$

Because $R(\theta;\cdot)$ is convex (mixtures of randomized rules produce convex combinations of risk vectors), T is convex and contains the origin 0 (corresponding to $\delta = \delta_0$). Admissibility of δ_0

means precisely that there is no $t \in T$ with $t \le 0$ (component-wise) and $t \neq 0$. Equivalently,

$$T \cap \left(-\mathbb{R}^k_{++}\right) = \varnothing,$$

where $\mathbb{R}^k_{++} = \{x \in \mathbb{R}^k :\ x_i > 0,\ \forall i\}$.
The sets T and $-\mathbb{R}^k_{++}$ are convex and disjoint, and $-\mathbb{R}^k_{++}$ is an open convex cone. By the separating hyperplane theorem, there exists a nonzero vector $\pi \in \mathbb{R}^k$ and a scalar c such that

$$\pi \cdot t \ge c \quad \text{for all } t \in T, \qquad \text{and} \qquad \pi \cdot y \le c \quad \text{for all } y \in -\mathbb{R}^k_{++}.$$

Because $0 \in T$, plugging $t = 0$ gives $0 = \pi \cdot 0 \ge c$, so $c \le 0$. Replace π by $-\pi$ if necessary so that $c \le 0$ still holds; we may then take $c = 0$ by scaling the separating hyperplane (the sign and scale of π are not important for separation). Thus we obtain a nonzero π with

$$\pi \cdot t \ge 0 \quad \text{for all } t \in T,$$

and

$$\pi \cdot y \le 0 \quad \text{for all } y \in -\mathbb{R}^k_{++}.$$

The second condition implies that $\pi_i \ge 0$ for every i: indeed, taking $y = -e_i \in -\mathbb{R}^k_{++}$ (where e_i is the ith standard basis vector) yields $-\pi_i \le 0$, hence $\pi_i \ge 0$. Not all π_i are zero because $\pi \neq 0$. Normalize π by dividing by $\sum_i \pi_i$ to obtain a probability vector (prior) which we again denote by $\pi = (\pi_1, \ldots, \pi_k)$ with $\pi_i \ge 0, \sum_i \pi_i = 1$.
Now for any decision rule δ let $t = R(\delta) - R(\delta_0) \in T$. The inequality $\pi \cdot t \ge 0$ becomes

$$\sum_{i=1}^{k} \pi_i \big(R(\theta_i, \delta) - R(\theta_i, \delta_0)\big) \ge 0,$$

or equivalently

$$r_\pi(\delta) \ge r_\pi(\delta_0).$$

Therefore δ_0 attains the minimal Bayes risk with respect to the prior π, i.e. δ_0 is a Bayes rule for π. This completes the proof. □

1.3 Lecture 3, 16: (Unbiased) Estimation

1.3.1 Construction of UMVUE

An estimator is called unbiased if $\mathbb{E}_\theta \delta(X) = \theta$ for all θ. Sometimes the following definition is used instead.

Definition 1.3.1. For a loss $L(\theta, a)$ a decision rule δ is unbiased with respect to L if

$$\mathbb{E}_\theta \left(L\left(\theta', \delta(X)\right)\right) \ge \mathbb{E}_\theta(L(\theta, \delta(X))) = R(\theta, \delta) \quad \text{for all } \theta, \theta' \in \Theta.$$

We start with the introduction of sufficient statistics.

Example. Suppose that we have observations $X_1, X_2 \ldots X_n \overset{\text{iid}}{\sim} \text{Ber}(p)$ from a Bernoulli distribution with an unknown parameter p. Then we know that the statistics $T(X_1, \cdots, X_n) = X_1 + \cdots + X_n \sim \text{Binom}(n, p)$ follows a Binomial distribution.
Moreover, based on this statistics, we can obtain the distribution of the data without knowing

the exact value of p: the conditional distribution of $X = (X_1, \cdots, X_n)$ conditioned on $\{T = t\}$ is the uniform distribution among all subsets with t ones in $\{0,1\}^n$.

This idea can be further generalized to other cases.

Definition 1.3.2. A statistics T is called sufficient if the distribution of the data X is independent of θ given $T(X)$. In other words, there exists a distribution μ_X such that

$$\mu_X(B) = \mathbb{P}_\theta\left(X \in B | T(X) = t\right), \quad \forall \theta \in \Theta, B \in \mathcal{B}, t \in \mathbb{R}.$$

We have the following nice characterization of sufficient statistics.

Definition 1.3.3. A family of distributions $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is dominated if there exists a measure μ with P_θ absolutely continuous with respect to μ, for all $\theta \in \Omega$.

Theorem 1.3.4. (Factorization Theorem). Let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a family of distributions dominated by μ. A necessary and sufficient condition for a statistic T to be sufficient is that there exist functions $g_\theta \geq 0$ and $h \geq 0$ such that the densities p_θ for the family satisfy

$$p_\theta(x) = g_\theta(T(x))h(x), \text{ for a.e. } x \text{ under } \mu.$$

Proof. Necessity: We construct g_θ and h as

$$g_\theta(t) = P_\theta(T(x) = t), \quad h(x) = P_{\theta_0}(X = x \mid T(x) = t)$$

Then by definition of sufficiency, $h(x)$ is independent of t. Therefore,

$$P_\theta(x) = P_\theta(X = x) = P_\theta(T(x) = t)P_\theta\left(X = x(T(x) = t\right) = g_\theta(T(x))h(x).$$

Sufficiency: Assume $P_\theta(x) = \rho_\theta(T(x))h(x)$. Then $\forall \theta_0 \in \mathbb{H}, t = T(x)$

$$\begin{aligned} P_{\theta_0}(X = x \mid T(X) = t) &= \frac{P_{\theta_0}(X = x, T(X) = t)}{P_{\theta_0}(T(X) = t)} = \frac{P_{\theta_0}(X = x)}{\int_{T(x)=t} P_{\theta_0}(x)dx} \\ &= \frac{P_{\theta_0}(x)}{\int_{T(x)=t} P_{\theta_0}(x)dx} = \frac{g_{\theta_0}(T(x))h(x)}{g_{\theta_0}(t) \int_{T(x)=t} h(x)dx} \\ &= \frac{h(x)}{\int_{T(x)=t} h(x)dx} \end{aligned}$$

which is independent of $\theta_0 \in \Theta$. □

Example. Let $P_\theta(x) = h(x)\exp\left(\eta(\theta)^\top T(x) - B(\theta)\right)$ be an exponential family. Then $T(X)$ is sufficient. Moreover, if $X_1, \cdots, X_n \sim P_\theta(x)$. then $\sum_{i=1}^n T\left(X_i\right)$ is sufficient.

In essence, the complete statistics capture all the information available in the distribution family $\mathcal{P}$. To gather this information with minimal effort, we need the notion of minimal statistics.

Definition 1.3.5. $T(X)$ is a minimal sufficient statistic if

- It is a sufficient statistic for θ.
- For any other sufficient statistic $S(X)$, there exists a function g such that $T(X) = g(S(X))$.

This gives rise to the so-called complete statistics.

Definition 1.3.6. A statistic T is complete for a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ if

$$E_\theta f(T) = 0, \text{ for all } \theta,$$

imply $f(T) = 0$ (a.e. $\mathcal{P}$).

Example. Suppose $X_1, \ldots, X_n$ are i.i.d. from a uniform distribution on $(0, \theta)$. Using indicator functions, the joint density is $I\{\min x_i > 0\} I\{\max x_i < \theta\}/\theta^n$, and so $T = \max\{X_1, \ldots, X_n\}$ is sufficient by the factorization theorem. Note that T has density $nt^{n-1}/\theta^n, t \in (0, \theta)$. Suppose $E_\theta f(T) = 0$ for all $\theta > 0$; then

$$E_\theta[f(T)] = \frac{n}{\theta^n} \int_0^\theta [f(t)] t^{n-1} dt = 0$$

From this, $[f(t) - c]t^{n-1} = 0$ for a.e. $t > 0$. So $f(T) = c$ (a.e. $\mathcal{P}$), and T is complete.

Theorem 1.3.7. Suppose $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is a dominated family with densities $p_\theta(x) = g_\theta(T(x))h(x)$. If $p_\theta(x) \propto_\theta p_\theta(y)$ implies $T(x) = T(y)$, then T is minimal sufficient.

Proof. A proper proof of this result unfortunately involves measure-theoretic niceties, but here is the basic idea. Suppose $\tilde{T}$ is sufficient. Then $p_\theta(x) = \tilde{g}_\theta(\tilde{T}(x))\tilde{h}(x)$ (a.e. μ). Assume this equation holds for all x. If T is not a function of $\tilde{T}$, then there must be two data sets x and y that give the same value for $\tilde{T}, \tilde{T}(x) = \tilde{T}(y)$, but different values for $T, T(x) \neq T(y)$. But then

$$p_\theta(x) = \tilde{g}_\theta(\tilde{T}(x))\tilde{h}(x) \propto_\theta \tilde{g}_\theta(\tilde{T}(y))\tilde{h}(y) = p_\theta(y),$$

and from the condition on T in the theorem, $T(x)$ must equal $T(y)$. Thus T is a function of $\tilde{T}$. Because $\tilde{T}$ was an arbitrary sufficient statistic, T is minimal. □

Theorem 1.3.8. If T is complete and sufficient, then T is minimal sufficient.

Proof. Let $\tilde{T}$ be a minimal sufficient statistic, and assume T and $\tilde{T}$ are both bounded random variables. Then $\tilde{T} = f(T)$. Define $g(\tilde{T}) = E_\theta[T \mid \tilde{T}]$, noting that this function is independent of θ because $\tilde{T}$ is sufficient. By smoothing, $E_\theta g(\tilde{T}) = E_\theta T$, and so $E_\theta[T - g(\tilde{T})] = 0$, for all θ. But $T - g(\tilde{T}) = T - g(f(T))$, a function of T, and so by completeness, $T = g(\tilde{T})$ (a.e. $\mathcal{P}$). This establishes a one-to-one relationship between T and $\tilde{T}$. From the definition of minimal sufficiency, T must also be minimal sufficient.
For the general case, first note that sufficiency and completeness are both preserved by one-to-one transformations, so two statistics can be considered equivalent if they are related by a one-to-one (bimeasurable) function. But there are one-to-one bi-measurable functions from $\mathbb{R}^n$ to $\mathbb{R}$, and so any random vector is equivalent to a single random variable. Using this, if T and $\tilde{T}$ are random vectors, the result follows easily from the one-dimensional case, transforming both of them to equivalent random variables. □

We now extend the previous example and show that the statistics T of a (sufficiently regular) exponential family are minimal.

Definition 1.3.9. An exponential family with densities $p_\theta(x) = \exp\{\eta(\theta) \cdot T(x) - B(\theta)\}h(x), \theta \in \Omega$, is said to be of full rank if the interior of $\eta(\Omega)$ is not empty and if $T_1, \ldots, T_s$ do not satisfy a linear constraint of the form $v \cdot T = c$ (a.e. μ).

Remark. If $\Omega \subset \mathbb{R}^s$ and η is continuous and one-to-one (injective), and the interior of Ω is nonempty, then the interior of $\eta(\Omega)$ cannot be empty. This follows from the "invariance of domain" theorem of Brouwer.

If the interior of $\eta(\Omega)$ is not empty, then the linear span of $\eta(\Omega) \ominus \eta(\Omega)$ will be all of $\mathbb{R}^s$, and, T will be minimal sufficient. The following result shows that in this case T is also complete.

Theorem 1.3.10. In an exponential family of full rank, T is complete.

Proof. Let $\eta(\theta_0)$ be an interior point of $\eta(\Theta)$. Suppose $\mathbb{E}_\theta[f(T(x))] = 0$ for some f, then $\forall \theta \in \Theta$,

$$\begin{aligned} 0 &= \int_{\mathbb{R}^d} f(t)h(t) \exp\left(\eta^\top(\theta)t - A(\theta)\right) dt \\ \Rightarrow 0 &= \int_{\mathbb{R}^d} f(t)h(t) \exp\left(\eta^\top(\theta)t\right) dt \end{aligned}$$

So the Laplace transform of $f \cdot v$ is 0, by the uniqueness of the Laplace transform $f(t)v(t) \equiv 0$ and hence $f \equiv 0$. $\square$

Definition 1.3.11. A statistic V is called ancillary if its distribution does not depend on θ.

An ancillary statistic V, by itself, provides no useful information about θ. But in some situations V can be a function of a minimal sufficient statistic T. The following result of Basu shows that when T is complete it will contain no ancillary information.

Theorem 1.3.12 (Basu)**.** If T is complete and sufficient for $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$, and if V is ancillary, then T and V are independent under P_θ for any $\theta \in \Omega$.

Proof. Define $q_A(t) = P_\theta(V \in A \mid T = t)$, so that $q_A(T) = P_\theta(V \in A \mid T)$, and define $p_A = P_\theta(V \in A)$. By sufficiency and ancillarity, neither p_A nor $q_A(t)$ depend on θ. Also, by smoothing,

$$p_A = P_\theta(V \in A) = E_\theta P_\theta(V \in A \mid T) = E_\theta q_A(T),$$

and so, by completeness, $q_A(T) = p_A$ (a.e. $\mathcal{P}$). By smoothing,

$$\begin{aligned} P_\theta(T \in B, V \in A) &= E_\theta 1_B(T) 1_A(V) \\ &= E_\theta E_\theta \left(1_B(T) 1_A(V) \mid T\right) \\ &= E_\theta 1_B(T) E_\theta \left(1_A(V) \mid T\right) \\ &= E_\theta 1_B(T) q_A(T) \\ &= E_\theta 1_B(T) p_A \\ &= P_\theta(T \in B) P_\theta(V \in A). \end{aligned}$$

Here A and B are arbitrary Borel sets, and so T and V are independent. $\square$

Example. Suppose $X_1, \ldots, X_n$ are i.i.d. from $N(\mu, \sigma^2)$, and take $\mathcal{P} = \mathcal{P}_\sigma = \left\{N(\mu, \sigma^2)^n : \mu \in \mathbb{R}\right\}$. (Thus $\mathcal{P}_\sigma$ is the family of all normal distributions with standard deviation the fixed value σ.) With

$\bar{x} = (x_1 + \cdots + x_n)/n$, the joint density can be written as

$$\frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left[\frac{n\mu}{\sigma^2}\bar{x} - \frac{n\mu^2}{2\sigma^2} - \frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2\right].$$

These densities for $\mathcal{P}_\sigma$ form a full rank exponential family, and so the average $\bar{X} = (X_1 + \cdots + X_n)/n$ is a complete sufficient statistic for $\mathcal{P}_\sigma$. Define

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

called the sample variance. For the family $\mathcal{P}_\sigma$, S^2 is ancillary. To see this, let $Y_i = X_i - \mu, i = 1, \ldots, n$. Because

$$\begin{aligned} P_\mu(Y_i \le y) = P_\mu(X_i \le y + \mu) &= \int_{-\infty}^{y+\mu} \exp\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right]\frac{dx}{\sqrt{2\pi\sigma^2}} \\ &= \int_{-\infty}^{y} \exp\left[-\frac{1}{2\sigma^2}u^2\right]\frac{du}{\sqrt{2\pi\sigma^2}} \end{aligned}$$

and the integrand is the density for $N(0,\sigma^2)$, $Y_i \sim N(0,\sigma^2)$. Then $Y_1, \ldots, Y_n$ are i.i.d. from $N(0,\sigma^2)$. Because $\bar{Y} = (Y_1 + \cdots + Y_n)/n = \bar{X} - \mu$, $X_i - \bar{X} = Y_i - \bar{Y}, i = 1, \ldots, n$, and

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

Because the joint distribution of $Y_1, \ldots, Y_n$ depends on σ but not μ, S^2 is ancillary for $\mathcal{P}_\sigma$. Hence, by Basu's theorem, $\bar{X}$ and S^2 are independent.

The existence of sufficient statistics allows us to operate on the statistics, which have a fixed distribution based on T, instead of the data itself, which may have different distributions as θ varies.

Theorem 1.3.13. Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ and that $T = T(X)$ is sufficient. Then, for any decision rule $\delta(X)$ of $g(\theta)$, there exists a randomized decision rule $\tilde{\delta}$ based on T that has the same distribution as $T(X)$. Consequently, it has the same risk function as $\delta(X)$.

Proof. It suffices to take $\tilde{\delta}(T(X)) = \delta(Y)$, where Y is distributed as $X|T(X)$. □

Theorem 1.3.14. (Rao-Blackwellization) Suppose X has distribution from a family $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ and that $T = T(X)$ is sufficient. If $L(\theta, \cdot)$ is convex in the action space $\mathcal{A}$, then for any decision rule δ, the new decision rule $\eta(T(X)) := \mathbb{E}[\delta(X) \mid T(x)]$ that operates on $T(X)$ satisfies $R(\theta;\eta) \le R(\theta;\delta)$.

Proof. Follows from an application of Jensen's inequality. □

Remark. The estimated risk $R(\theta;\eta)$ is computable since it does not rely on the model parameter θ.

Definition 1.3.15. An estimator δ is called unbiased for $g(\theta)$ if

$$E_\theta\delta(X) = g(\theta), \quad \forall\theta \in \Omega.$$

If an unbiased estimator exists, g is called U-estimable.

Definition 1.3.16. An unbiased estimator δ is uniformly minimum variance unbiased (UMVU) if

$$\operatorname{Var}_\theta(\delta) \leq \operatorname{Var}_\theta\left(\delta^*\right), \quad \forall \theta \in \Omega,$$

for any competing unbiased estimator δ^*.

Since the space of unbiased estimators is a linear space and UMVU acts as a projection under the L^2 norm, we have the following theorem:

Theorem 1.3.17. An unbiased estimator T is the UMVUE for its expected value if and only if its covariance with any unbiased estimator of zero is exactly zero. That is, $\operatorname{Cov}(T, U) = 0$ for all statistics U where $E[U] = 0$.

In a general setting, there is no reason to suspect that there will be a UMVU estimator. However, if the family has a complete sufficient statistic, a UMVU will exist, at least when g is U-estimable.

Theorem 1.3.18. Suppose g is U-estimable and T is complete sufficient. Then there is an essentially unique unbiased estimator based on T that is UMVU.

Proof. Let $\delta = \delta(X)$ be any unbiased estimator and define

$$\eta(T) = E[\delta \mid T],$$

as in the Rao-Blackwell theorem. By smoothing,

$$g(\theta) = E_\theta \delta = E_\theta E_\theta[\delta \mid T] = E_\theta \eta(T),$$

and thus $\eta(T)$ is unbiased. Suppose $\eta^*(T)$ is also unbiased. Then

$$E_\theta\left[\eta(T) - \eta^*(T)\right] = 0, \quad \forall \theta \in \Omega,$$

and by completeness, $\eta(T) - \eta^*(T) = 0$ (a.e. $\mathcal{P}$). This shows that the estimator $\eta(T)$ is essentially unique; any other unbiased estimator based on T will equal $\eta(T)$ except on a $\mathcal{P}$-null set. The estimator $\eta(T)$ has minimum variance by the Rao-Blackwell theorem with squared error loss. Specifically, if δ^* is any unbiased estimator, then $\eta^*(T) = E_\theta\left(\delta^* \mid T\right)$ is unbiased by the calculation above. With squared error loss, the risk of δ^* or $\eta^*(T)$ is the variance, and so

$$\operatorname{Var}_\theta\left(\delta^*\right) \geq \operatorname{Var}_\theta\left(\eta^*(T)\right) = \operatorname{Var}_\theta(\eta(T)), \quad \forall \theta \in \Omega$$

Thus, $\eta(T)$ is UMVU. $\square$

Remark. This also gives a way to compute the UMVU.

1.3.2 Lower Bounds of Risks of Unbiased Estimators

We now introduce the famous Cramer-Rao lower bound of unbiased estimators. We start with the definition of properties of score functions.

Lemma 1.3.19 (Fisher's identity). Let $l(\theta; X) = \log p_\theta(X)$. Then

$$\mathbb{E}_\theta \left[\nabla_\theta l(\theta; X)\right] = \int \nabla_\theta l(\theta; x) p_\theta(x) d\mu(x) = \nabla_\theta \left(\int p_\theta(x) d\mu(x) \right) = 0.$$

Lemma 1.3.20. From Fisher's identity, we have

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta_j} l(\theta; X) \right] = 0.$$

Therefore,

$$\begin{aligned} 0 = \frac{\partial}{\partial \theta_i} \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta_j} l(\theta; x) \right] &= \frac{\partial}{\partial \theta_i} \int \frac{\partial}{\partial \theta_j} l(\theta; x) \cdot e^{l(\theta; x)} d\mu(x) \\ &= \int \left(\frac{\partial^2 l(\theta; x)}{\partial \theta_j \partial \theta_i} + \frac{\partial l(\theta; x)}{\partial \theta_i} \cdot \frac{\partial l(\theta, x)}{\partial \theta_j} \right) e^{l(\theta; x)} d\mu(x). \end{aligned}$$

and we have

$$\mathbb{E}_\theta \left[\nabla \ell(\theta; X) \nabla \ell(\theta; X)^\top \right] = -\mathbb{E}_\theta \left[\nabla^2 \ell(\theta; X) \right].$$

Theorem 1.3.21 (Cramer-Rao lower bound). Let δ be an unbiased estimator for $g(\theta) \in \mathbb{R}$. Assume that $(p_\theta : \theta \in \Theta)$ has a shared support for all θ, and

$$\log p_\theta(x) \in C^2, \quad \mathbb{E} \left| \nabla \log p_\theta(x) \right|^2 < +\infty.$$

Then we have the Cramer-Rao lower bound

$$\mathrm{Var}_\theta(\delta(X)) \geq \nabla g(\theta)^\top I(\theta)^{-1} \nabla g(\theta)$$

where the Fisher information matrix is given by

$$I(\theta) := \mathbb{E}_\theta \left[\nabla \log p_\theta(x) \nabla \log p_\theta(x)^\top \right].$$

Proof. From Fisher's identity, we have

$$\int \nabla_\theta l(\theta; x) e^{l(\theta; x)} d\mu(x) = 0.$$

and therefore,

$$\begin{aligned} \nabla g(\theta) &= \int (\delta(x) - g(\theta)) \cdot \nabla_\theta l(\theta; x) \cdot e^{l(\theta; x)} d\mu(x) \\ &= \mathbb{E}_\theta \left[(\delta(x) - g(\theta)) \cdot \nabla_\theta l(\theta; x) \right]. \end{aligned}$$

Consider the quadratic form

$$u \in \mathbb{R}^d \mapsto \mathbb{E}_0 \left[\left(u^\top \nabla_\theta l(\theta; x) - (\delta(x) - g(\theta)) \right)^2 \right]$$

which is nonnegative, a direct expansion with respect to u suggests that

$$0 \leq u^\top I(\theta) u - 2u^\top \mathbb{E}_\theta \left[(\delta(x) - g(\theta)) \cdot \nabla_\theta l(\theta; x) \right] + \mathrm{Var}_\theta[\delta(X)]$$

The conclusion then follows from noting that $\mathbb{E}_\theta \left[(\delta(x) - g(\theta)) \cdot \nabla_\theta l(\theta; x) \right] = \nabla g(\theta)$. □

Corollary 1.3.22. Let $g : \mathbb{R}^d \to \mathbb{R}^k$ and δ be a multi-dimensional unbiased estimator of $g(\theta)$. Then

$$\mathbb{E}\left[(\delta(X) - g(\theta))^{\otimes 2}\right] \succeq \nabla g(\theta)^\top I(\theta)^{-1} \nabla g(\theta).$$

In particular, when $g(\theta) = \theta$,

$$\mathbb{E}\left[\|\delta(X) - \theta\|_2^2\right] \geq \mathrm{Tr}\left(I(\theta)^{-1}\right).$$

Remark. The CR lower bound is usually not achieved by unbiased estimation. However, it is achieved asymptotically by the maximum likelihood estimator.

Remark. To attain the CRLB, an estimator must satisfy the condition for equality in the Cauchy-Schwarz inequality used to derive the bound. This occurs if and only if the estimator is a linear function of the score function.
For regular exponential families, the score function is linear in the sufficient statistic S. Therefore, an unbiased estimator can only achieve the CRLB if it is a linear function of the sufficient statistic S.

Remark. When $p_\theta(x)$ is singular or has discontinuities, the CR lower bound may not hold. For example, consider $X_1, X_2, \cdots, X_n \sim \mathrm{Unif}([0, \theta])$. Then the UPVU is given by $f(X) = \frac{n+1}{n} \max_i X_i$, and $\mathrm{Var}_\theta[f(X)] = \frac{\theta^2}{n(n+2)}$, which is better than the CR lower bound.

We now introduce the Van-Trees inequality, also known as the Bayesian Cramér-Rao bound, which extends the classical Cramér–Rao inequality to the Bayesian setting. Consider the Bayes risk

$$r_\pi(\delta) := \int_\Theta \mathbb{E}_\theta\left[(\delta(x) - g(\theta))^2\right] \pi(d\theta).$$

Theorem 1.3.23 (van-Trees inequality)**.** For any estimator δ and prior π compactly supported on Θ, we have

$$r_\pi(\delta) \geq \left(\int_\Theta \nabla g(\theta)\pi(d\theta)\right)^\top \left(\int_\Theta I(\theta)\pi(d\theta) + J(\pi)\right)^{-1} \left(\int_\Theta \nabla g(\theta)\pi(d\theta)\right),$$

where $J(\pi) := \int_\Theta \nabla_\theta \log \pi(\theta) \nabla_\theta \log \pi(\theta)^\top \pi(d\theta)$ is the Information theorist's Fisher information.

Similar to the proof of the Cramer-Rao lower bound, we start with two lemmas.

Lemma 1.3.24. Suppose that the distribution of X under θ has density p_θ with respect to some base measure μ, then

(a). $\iint \nabla_\theta(p_\theta(x)\pi(\theta))d\mu(x)d\theta = 0$

(b). $\iint g(\theta) \cdot \nabla_\theta(p_\theta(x)\pi(\theta))d\mu(x)d\theta = -\iint \nabla_\theta g(\theta) \cdot p_\theta(x)\pi(\theta)d\mu(x)d\theta$

Proof. (a) Using $\int p_\theta(x)\, d\mu(x) = 1$ and differentiation under the integral,

$$\iint \nabla_\theta(p_\theta(x)\pi(\theta))\, d\mu(x)\, d\theta = \int \nabla_\theta\Big(\pi(\theta) \int p_\theta(x)\, d\mu(x)\Big)\, d\theta = \int \nabla_\theta \pi(\theta)\, d\theta.$$

By compact support and vanishing boundary terms, $\int \nabla_\theta \pi(\theta)\, d\theta = 0$.
(b) Observe that

$$\nabla_\theta\big(g(\theta)p_\theta(x)\pi(\theta)\big) = \nabla_\theta g(\theta)\, p_\theta(x)\pi(\theta) + g(\theta)\, \nabla_\theta(p_\theta(x)\pi(\theta)).$$

Integrating over (x, θ) yields

$$\iint g(\theta)\,\nabla_\theta(p_\theta(x)\pi(\theta))\,d\mu\,d\theta + \iint \nabla_\theta g(\theta)\,p_\theta(x)\pi(\theta)\,d\mu\,d\theta = \iint \nabla_\theta\big(g(\theta)p_\theta(x)\pi(\theta)\big)\,d\mu\,d\theta.$$

As in (a), differentiation under the integral gives

$$\iint \nabla_\theta\big(g(\theta)p_\theta(x)\pi(\theta)\big)\,d\mu\,d\theta = \int \nabla_\theta\left(\pi(\theta)g(\theta)\int p_\theta(x)\,d\mu(x)\right)\,d\theta = \int \nabla_\theta(\pi(\theta)g(\theta))\,d\theta = 0,$$

again by compact support/vanishing boundary terms. Rearranging yields (b). □

Proof of Theorem. Define the joint density of (X, θ) (w.r.t. $\mu \otimes d\theta$) by

$$m(x, \theta) := p_\theta(x)\pi(\theta),$$

and write $\mathbb{E}_\pi[\cdot]$ for expectation under this joint law:

$$\mathbb{E}_\pi[h(X, \theta)] := \iint h(x, \theta)\,p_\theta(x)\pi(\theta)\,d\mu(x)\,d\theta.$$

Let

$$U := \delta(X) - g(\theta), \qquad s_\theta(x) := \nabla_\theta \log p_\theta(x), \qquad \rho(\theta) := \nabla_\theta \log \pi(\theta),$$

and define the *Bayesian score*

$$V := s_\theta(X) + \rho(\theta) = \nabla_\theta \log\big(p_\theta(X)\pi(\theta)\big).$$

Note that $V(x, \theta)m(x, \theta) = \nabla_\theta m(x, \theta)$.

Step 1: Identify $\mathbb{E}_\pi[UV]$. Using $Vm = \nabla_\theta m$,

$$\mathbb{E}_\pi[UV] = \iint (\delta(x) - g(\theta))\,\nabla_\theta\big(p_\theta(x)\pi(\theta)\big)\,d\mu(x)\,d\theta.$$

Split into two terms:

$$\mathbb{E}_\pi[UV] = \iint \delta(x)\,\nabla_\theta(p_\theta(x)\pi(\theta))\,d\mu\,d\theta - \iint g(\theta)\,\nabla_\theta(p_\theta(x)\pi(\theta))\,d\mu\,d\theta.$$

Since $\delta(x)$ does not depend on θ, part (a) of lemma gives the first term equals 0 and part (b) gives

$$\iint g(\theta)\,\nabla_\theta(p_\theta(x)\pi(\theta))\,d\mu\,d\theta = -\iint \nabla_\theta g(\theta)\,p_\theta(x)\pi(\theta)\,d\mu\,d\theta = -\int_\Theta \nabla g(\theta)\,\pi(\theta)\,d\theta.$$

Hence

$$\mathbb{E}_\pi[UV] = \int_\Theta \nabla g(\theta)\,\pi(\theta)\,d\theta. \tag{1.1}$$

Step 2: Identify $\mathbb{E}_\pi[VV^\top]$. We expand

$$\mathbb{E}_\pi[VV^\top] = \mathbb{E}_\pi[s_\theta(X)s_\theta(X)^\top] + \mathbb{E}_\pi[\rho(\theta)\rho(\theta)^\top] + \mathbb{E}_\pi[s_\theta(X)\rho(\theta)^\top] + \mathbb{E}_\pi[\rho(\theta)s_\theta(X)^\top].$$

First,

$$\mathbb{E}_\pi[s_\theta(X)s_\theta(X)^\top] = \int_\Theta \mathbb{E}_\theta[s_\theta(X)s_\theta(X)^\top]\,\pi(\theta)\,d\theta = \int_\Theta I(\theta)\,\pi(\theta)\,d\theta.$$

Second, by definition,

$$\mathbb{E}_\pi[\rho(\theta)\rho(\theta)^\top] = J(\pi).$$

For the cross term, condition on θ:

$$\mathbb{E}_\pi[s_\theta(X)\rho(\theta)^\top] = \int_\Theta \Big(\mathbb{E}_\theta[s_\theta(X)]\Big)\rho(\theta)^\top \, \pi(\theta)\, d\theta.$$

But $\mathbb{E}_\theta[s_\theta(X)] = 0$ since

$$\mathbb{E}_\theta[s_\theta(X)] = \int \nabla_\theta \log p_\theta(x)\, p_\theta(x)\, d\mu(x) = \int \nabla_\theta p_\theta(x)\, d\mu(x) = \nabla_\theta \int p_\theta(x)\, d\mu(x) = \nabla_\theta 1 = 0,$$

so $\mathbb{E}_\pi[s_\theta(X)\rho(\theta)^\top] = 0$ and similarly $\mathbb{E}_\pi[\rho(\theta)s_\theta(X)^\top] = 0$. Therefore

$$\mathbb{E}_\pi[VV^\top] = \int_\Theta I(\theta)\, \pi(\theta)\, d\theta + J(\pi). \tag{1.2}$$

Step 3: A quadratic lower bound via Cauchy–Schwarz. For any $a \in \mathbb{R}^d$,

$$0 \le \mathbb{E}_\pi\big[(U - a^\top V)^2\big] = \mathbb{E}_\pi[U^2] - 2a^\top \mathbb{E}_\pi[UV] + a^\top \mathbb{E}_\pi[VV^\top]a.$$

Minimizing the right-hand side over a gives the optimizer

$$a^\star = \big(\mathbb{E}_\pi[VV^\top]\big)^{-1}\mathbb{E}_\pi[UV],$$

and the minimal value equals

$$\mathbb{E}_\pi[U^2] - \mathbb{E}_\pi[UV]^\top \big(\mathbb{E}_\pi[VV^\top]\big)^{-1}\mathbb{E}_\pi[UV] \;\ge\; 0.$$

Hence

$$\mathbb{E}_\pi[U^2] \ge \mathbb{E}_\pi[UV]^\top \big(\mathbb{E}_\pi[VV^\top]\big)^{-1}\mathbb{E}_\pi[UV]. \tag{1.3}$$

Finally, note that $\mathbb{E}_\pi[U^2] = r_\pi(\delta)$ by definition, and substitute (1.1)–(1.2) into (1.3) to obtain

$$r_\pi(\delta) \;\ge\; \Big(\int_\Theta \nabla g(\theta)\, \pi(\theta)\, d\theta\Big)^\top \Big(\int_\Theta I(\theta)\, \pi(\theta)\, d\theta + J(\pi)\Big)^{-1} \Big(\int_\Theta \nabla g(\theta)\, \pi(\theta)\, d\theta\Big),$$

which is the claimed van–Trees inequality. □

Remark. Compared to the Cramer-Rao lower bound, we remove the assumption of unbiasedness by adding an extra $J(\pi)$ term.

The Bayesian CRLB is a powerful tool for establishing lower bounds on the minimax risk over a local neighborhood of a parameter point θ_0.

Example. The **local minimax risk** is defined as:

$$R_{\text{local-minimax}}(\theta_0, \varepsilon) = \inf_{\hat\theta_n} \sup_{|\theta-\theta_0|\le\varepsilon} E_\theta[(\hat\theta_n - \theta)^2]$$

This risk is bounded below by the Bayes risk for any prior π supported on $[\theta_0 - \varepsilon, \theta_0 + \varepsilon]$. Consider a prior $\nu(x)$ on $[\theta_0 - \varepsilon, \theta_0 + \varepsilon]$ by scaling a base prior ν_0 on $[-1, 1]$. If $\nu_0(x) = \cos^2(\frac{\pi x}{2})$ for $|x| \le 1$, then $J(\nu_0) = \pi^2$. Scaling this to $\nu(x) = \frac{1}{\varepsilon}\nu_0\left(\frac{x-\theta_0}{\varepsilon}\right)$ gives $J(\nu) = \frac{\pi^2}{\varepsilon^2}$. Applying the Bayesian CRLB with this prior yields:

$$R_{\text{local-minimax}}(\theta_0, \varepsilon) \ge \left(n \int I(\theta)\nu(\theta)d\theta + \frac{\pi^2}{\varepsilon^2}\right)^{-1}$$

Corollary 1.3.25 (The Local Asymptotic Minimax (LAM) Theorem)**.** By letting the local neighborhood shrink as the sample size grows (e.g., $\varepsilon = c/\sqrt{n}$), we derive the famous Hájek-Le Cam asymptotic lower bound:

$$\lim_{c\to\infty}\lim_{n\to\infty}\inf_{\hat{\theta}_n}\sup_{|\theta-\theta_0|\le c/\sqrt{n}} E[n(\hat{\theta}_n-\theta)^2] \ge \frac{1}{I(\theta_0)}$$

1.3.3 Beyond Unbiased Estimators

Let us first introduce a motivating example.

Consider a sample $X^{(1)},\dots,X^{(n)} \sim N_d(\mu,\Sigma)$ with $\Sigma = I_d$. Using the square loss $L(\mu,a) = \|\mu - a\|^2$ we easily show that the risk of any estimator $\widehat{\mu}_n$ admits the bias-variance decomposition

$$R(\mu,\widehat{\mu}_n) = \mathbb{E}\,\|\widehat{\mu}_n - \mathbb{E}[\widehat{\mu}_n]\|^2 + \|\mathbb{E}[\widehat{\mu}_n] - \mu\|^2.$$

The MLE estimator of μ is the sample average $\bar{X}_n$. Its bias is zero, and its variance is

$$\mathbb{E}\left\|\frac{1}{n}\sum_i\left(X^{(i)}-\mu\right)\right\|^2 = \frac{1}{n}\mathbb{E}\|\epsilon\|^2 = \frac{d}{n}.$$

In the special case when $n = 1$, the MLE is X which is an unbiased estimator with large variance d. Consider first as an alternative a simple linear estimator $\widehat{\mu}_C = CX$ with $C = \mathrm{diag}(c)$ diagonal and $c = (c_1,\dots,c_d)$. In this case

$$R(\mu,\widehat{\mu}_C) = \sum_{i=1}^d (1-c_i)^2\mu_i^2 + \sum_{i=1}^d c_i^2.$$

Suppose we restrict ourselves to the hyperrectangular model class $|\mu_i| \le \tau_i$. In this case we can easily find that the minimax risk is

$$\inf_{c\in\mathbb{R}^d}\sup_{-\tau\le\mu\le\tau} R(\mu,\widehat{\mu}_C) = \inf_c\sum_{i=1}^d(1-c_i)^2\tau_i^2 + \sum_{i=1}^d c_i^2 = \sum_{i=1}^d\frac{\tau_i^2}{1+\tau_i^2} < d.$$

This computation shows that for sparse model classes diagonal estimators $\widehat{\mu}_C$ may strictly dominate the MLE if c is chosen carefully.

In this section, we show a famous surprising result that the observation $X = (X_1,\dots,X_d)$ is not an admissible estimator for the parameter $\mu = (\mu_1,\dots,\mu_d)$ unless $d \le 2$. We start introducing the Stein's Unbiased Risk Estimates (SURE).

For a function $h : \mathbb{R}^d \to \mathbb{R}^d$ denote by $\mathrm{J}h(x)$ the Jacobian of h at $x \in \mathbb{R}^d$, that is, the matrix of partial derivatives with

$$(Jh(x))_{ij} = \frac{\partial h_i}{\partial x_j}.$$

Proposition 1.3.26 (Stein's Unbiased Risk Estimates (SURE))**.** Let $X_1,\dots,X_d$ be independent, $X_i \sim N(\mu_i,1)$. Consider an estimator $\widehat{\mu}(X)$ of $\mu = (\mu_1,\dots,\mu_d)$ and let

$$h(x) = x - \widehat{\mu}(x).$$

Suppose that $h(x)$ satisfies

(a). h is (weakly) differentiable,

(b). $\mathbb{E}_\mu \|Jh(X)\| < \infty$.

Define

$$\hat{R} = d + \|h(X)\|^2 - 2\,\mathrm{tr}(\mathrm{Jh}(X)).$$

Then

$$R(\mu, \widehat{\mu}) = \mathbb{E}_\mu \|\widehat{\mu}(X) - \mu\|^2 = \mathbb{E}_\mu R.$$

We start with Stein's lemma.

Lemma 1.3.27 (Stein's lemma)**.** Let $X \sim N\left(\mu, \sigma^2\right)$ and let $h : \mathbb{R} \to \mathbb{R}$ be differentiable with $\mathbb{E}\left|h'(X)\right| < \infty$. Then

$$E[(X-\mu)h(X)] = \sigma^2 \mathbb{E}h'(X).$$

Proof. First assume $\mu = 0, \sigma^2 = 1$. In this case we equivalently show that $\mathbb{E}(Xh(X)) = \mathbb{E}h'(X)$. Without loss of generality, assume $h(0) = 0$. The proof is an application of integration by parts. We have

$$\left(h(x)e^{-x^2/2}\right)' = h'(x)e^{-x^2/2} - xh(x)e^{-x^2/2}$$

and so

$$0 = \mathbb{E}h'(X) - \mathbb{E}(Xh(X)),$$

which establishes the result. The fact that we get zero on the left follows from the fact that $\lim_{x\to\pm\infty} h(x)e^{-x^2/2} = 0$, which can be argued since $\mathbb{E}\left|h'(X)\right| < +\infty$. Indeed, first note that

$$\left|h'(t)\right| e^{-x^2/2}\mathbb{1}_{[0,x]}(t) \le \left|h'(t)\right| e^{-t^2/2}$$

and the dominating function on the right is integrable by assumption. Thus, we can use the dominating convergence theorem

$$\begin{aligned}
\lim_{x\to\pm\infty} h(x)e^{-x^2/2} &= \lim_{x\to\pm\infty} \int_0^x h'(t)e^{-x^2/2}\,\mathrm{d}t = \lim_{x\to\pm\infty} \int_{-\infty}^{\infty} h'(t)e^{-x^2/2}\mathbb{1}_{[0,x]}(t)\mathrm{d}t \\
&= \int_{-\infty}^{\infty} \lim_{x\to\pm\infty} h'(t)e^{-x^2/2}\mathbb{1}_{[0,x]}(t)\mathrm{d}t = 0.
\end{aligned}$$

This establishes the result in the standard normal case. For general μ and σ define $Z = (X-\mu)/\sigma \sim N(0,1)$. Define $\tilde{h}(z) = h(\mu + \sigma z)$. We have

$$\begin{aligned}
\mathbb{E}[(X-\mu)h(X)] &= \sigma\mathbb{E}(Zh(\mu+\sigma Z)) = \sigma\mathbb{E}(Z\tilde{h}(Z)) \\
&= \sigma\mathbb{E}\tilde{h}'(Z) = \sigma^2\mathbb{E}h'(\mu+\sigma Z) = \sigma^2\mathbb{E}h'(X),
\end{aligned}$$

where moving from the first to the second line we used the case $\mu = 0, \sigma = 1$ proved earlier. □

We need another technical lemma.

Lemma 1.3.28. Let $X = (X_1, \dots, X_d)$ be a random vector with independent entries, $X_i \sim N(\mu_i, 1)$. If $h : \mathbb{R}^d \to \mathbb{R}^d$ satisfies $\mathbb{E}\|Jh(X)\| < \infty$. Then

$$\mathbb{E}\left((X-\mu)^\top h(X)\right) = \mathbb{E}\,\mathrm{tr}(\mathrm{J}h(X))$$

Proof. Denote by $X_{\setminus i} \in \mathbb{R}^{d-1}$ the vector X with X_i removed. Using the first lemma, for every $i = 1, \dots, d$

$$\begin{aligned}\mathbb{E}\left((X_i - \mu_i)\, h_i(X)\right) &= \mathbb{E}\left(\mathbb{E}\left[(X_i - \mu_i)\, h_i(X) \mid X_{\setminus i}\right]\right) \\ &= \mathbb{E}\left(\mathbb{E}\left[\frac{\partial h_i(X)}{\partial x_i}\,\middle|\, X_{\setminus i}\right]\right) = \mathbb{E}\left(\frac{\partial h_i(X)}{\partial x_i}\right).\end{aligned}$$

Summing over i we get the result. □

Now the formula for the risk of Stein's Unbiased Risk Estimates follows easily.

Proof. We have

$$\begin{aligned}R(\mu, \widehat{\mu}) &= \mathbb{E}\left((\widehat{\mu}(X) - \mu)^\top (\widehat{\mu}(X) - \mu)\right) \\ &= \mathbb{E}\left((\widehat{\mu}(X) - X + X - \mu)^\top (\widehat{\mu}(X) - X + X - \mu)\right) \\ &= \mathbb{E}\|h(X)\|^2 - 2\mathbb{E}\left((X-\mu)^\top h(X)\right) + \mathbb{E}\|X - \mu\|^2 \\ &= \mathbb{E}\|h(X)\|^2 - 2\mathbb{E}(\mathrm{tr}(Jh(X))) + \mathbb{E}\|X - \mu\|^2 \\ &= \mathbb{E}\hat{r},\end{aligned}$$

where the last equality follows because $\mathbb{E}\|X - \mu\|^2 = d$, which follows because $X - \mu$ is standard normal. □

A natural estimator of μ is X and it has constant risk $R(\mu, X) = \mathbb{E}\|X - \mu\|^2 = d$. Although it is unbiased, the variance is large if d is large. The James-Stein estimator of $\mu = (\mu_1, \dots, \mu_d)$ is defined as

$$\delta_{JS}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right) X.$$

which dominates the natural estimator when $d \geq 3$.

Theorem 1.3.29. The risk of the James-Stein estimator is

$$R\left(\mu, \delta_{\mathrm{JS}}\right) = d - \mathbb{E}_\mu \left(\frac{d-2}{\|X\|}\right)^2.$$

In particular, the natural estimator is not admissible if $d \geq 3$.

Proof. For the James-Stein estimator, the function h in SURE is

$$h(x) = \frac{d-2}{\|x\|^2} x$$

and so

$$\hat{R} = d + \frac{(d-2)^2}{\|X\|^2} - 2\,\mathrm{tr}(Jh(X)).$$

The diagonal entries of the Jacobian have a simple form

$$\frac{\partial h_i}{\partial x_i} = \frac{d-2}{\|x\|^2} - 2\frac{d-2}{\|x\|^4} x_i^2,$$

giving

$$\operatorname{tr}(Jh(X)) = \left(\frac{d-2}{\|X\|}\right)^2 \quad \text{and} \quad \hat{R} = d - \left(\frac{d-2}{\|X\|}\right)^2.$$

Therfore, if $d \geq 3$

$$R\left(\mu, \delta_{\mathrm{JS}}\right) = \mathbb{E}_\mu R = d - \mathbb{E}_\theta \left(\frac{d-2}{\|X\|}\right)^2 < d = R(\mu, X)$$

proving inadmissibility. □

It is interesting to study the risk of a general linear estimator $\widehat{\mu}_C = CX$ for an arbitrary square matrix C. To use SURE, note that $h(X) = (I_d - C)\,X$ and so $\mathrm{J}h(x) = (I_d - C)$ giving $\operatorname{tr}(\mathrm{J}h(x)) = d - \operatorname{tr}(C)$. Therefore

$$R\left(\mu, \widehat{\mu}_C\right) = \mathbb{E}_\mu \left[d + \|(I_d - C)\,X\|^2 - 2d + 2\operatorname{tr}(C)\right] = \mathbb{E}\,\|(C - I_d)\,X\|^2 - d + 2\operatorname{tr}(C).$$

We formulate the following result without a proof.

Proposition 1.3.30. The linear estimator $\widehat{\mu}_C = CX$ is admissible if and only if

(a). C is symmetric,

(b). the eigenvalues satisfy $0 \leq \rho_i(C) \leq 1$,

(c). $\rho_i(C) = 1$ for at most two i.

In many situations in the expression, $C = C(\lambda)$ depends on a regularization parameter λ so one could find optimal λ^* by minimizing the MSE over λ. One example is given by ridge regression.

Example. In this case for some $Z \in \mathbb{R}^{d \times p}$

$$C = Z\left(Z^\top Z + \lambda I_d\right)^{-1} Z^\top$$

and the corresponding estimator is obtained as $\widehat{\mu}_Z = Z\widehat{\beta}$ where $\widehat{\beta} \in \mathbb{R}^p$ is obtained by solving

$$\min_\beta \frac{1}{2}\|X - Z\beta\|^2 + \frac{\lambda}{2}\|\beta\|^2, \quad \lambda > 0.$$

1.4 Lecture 4: Testing

We start with the basic framework of hypothesis testing.

Definition 1.4.1. Hypothesis testing is used to determine whether the underlying parameter θ is in some specific test. Given a parameter space Θ, we consider two disjoint sets Θ_0 and Θ_1 with the following hypothesis:

- **Null Hypothesis:** $H_0 : \theta \in \Theta_0$,
- **Alternative Hypothesis:** $H_1 : \theta \in \Theta_1$.

A test function is defined by a function $\phi(x)$ that maps the data to the interval $[0, 1]$ that determines the outcome the test:

$$\phi(x) = \begin{cases} 1 & \text{reject } H_0 \\ \pi \in (0,1) & \text{reject } H_0 \text{ with probability } \pi \\ 0 & \text{do not reject } H_0 \end{cases}$$

Definition 1.4.2. Let $\phi(X)$ be a test function, where $\phi(X) = 1$ means we reject the null hypothesis (H_0) and $\phi(X) = 0$ means we do not reject it. We define

- **Significance Level (α_ϕ):** The worst case of probability of a Type I error (rejecting H_0 when it is true).

$$\alpha_\phi = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\phi(X)]$$

  For a level-α test, we require $\alpha_\phi \leq \alpha$ for a pre-specified α (e.g., 5%).

- **Power ($\beta_\phi(\theta)$):** The probability of correctly rejecting the null hypothesis.

$$\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)] \quad \text{for } \theta \in \Theta_1$$

Remark. The goal is to keep $\alpha_\phi \leq \alpha$ while maximizing β_ϕ.

Definition 1.4.3. A test $\phi^\star$ is **Uniformly Most Powerful** (**UMP**) (level-α) if for any other level-α test ϕ,

$$\beta_{\phi^\star}(\theta) \geq \beta_\phi(\theta) \quad (\forall \theta \in \Theta_1).$$

Remark. A UMP test is not always achievable.

1.4.1 Simple Testing

Definition 1.4.4. For a **Simple Testing Problem**, the parameter set of the null and alternative hypotheses are singleton:

$$\begin{aligned} H_0 &: \theta \in \Theta_0 = \{\theta_0\} \quad (\text{under distribution } P_0) \\ H_1 &: \theta \in \Theta_1 = \{\theta_1\} \quad (\text{under distribution } P_1) \end{aligned}$$

Definition 1.4.5. The **Likelihood Ratio** is $L(X) = \frac{p_1(X)}{p_0(X)}$, where $p_i(x)$ is the density of the data under P_i. The **Likelihood Ratio Test** is defined as:

$$\phi(X) = \begin{cases} 1 & L(X) > c, \\ \pi & L(X) = c, \\ 0 & L(X) < c. \end{cases}$$

We now state and prove the famous Neyman-Pearson lemma, which claims that to find a UMP test, it suffices to consider likelihood ratio tests.

Lemma 1.4.6 (Neyman-Pearson lemma)**.** Assuming P_0, P_1 have common support. Given $\alpha \in (0,1)$, there exists an LRT $\phi^\star$ such that $\alpha_{\phi^*} = \alpha$, and any LRT ϕ^* such that $\alpha_{\phi^*} = \alpha$ is an UMP.

Proof. By picking c and π appropriately, we can make the likelihood ratio test ϕ^* a level-α test. Since ϕ^* maximizes

$$\max_{\phi} \{\mathbb{E}_1[\phi(X)] - c\mathbb{E}_0[\phi(X)]\}$$

because $L(X) > c$ is equivalent to $p_1(X) > cp_0(X)$. For any other level-α test ϕ, since $\phi^\star$ maximizes the objective, we have

$$\mathbb{E}_1[\phi^\star] - c\mathbb{E}_0[\phi^\star] \geq \mathbb{E}_1[\phi] - c\mathbb{E}_0[\phi].$$

Therefore,

$$\mathbb{E}_1[\phi] \leq \mathbb{E}_1[\phi^\star] + c(\underbrace{\mathbb{E}_0[\phi]}_{\leq \alpha} - \underbrace{\mathbb{E}_0[\phi^\star]}_{=\alpha}) \leq \mathbb{E}_1[\phi^\star]$$

Since $\mathbb{E}_i[\phi] = \beta_\phi(\theta_i)$, this shows $\beta_{\phi^\star} \geq \beta_\phi$. $\square$

Remark (Testing Lower Bound). The function that we constructed in the proof

$$\mathbb{E}_1[\phi(X)] - c\mathbb{E}_0[\phi(X)]$$

is related to the lower bound of the testing error. For instance, when $c = 1$, the objective function relates the minimum probability of error to the total variation distance. For any test ϕ,

$$\begin{aligned}
\text{Probability of Type I + Type II error} &= \mathbb{E}_0[\phi] + \mathbb{E}_1[1 - \phi] \\
&= \int (p_0(x)\phi(x) + p_1(x)(1 - \phi(x)))d\mu(x) \\
&\geq \int \min(p_0(x), p_1(x))d\mu(x) = 1 - d_{\text{TV}}(P_0, P_1)
\end{aligned}$$

This minimum is achieved by the LRT with $c = 1$.

Remark. Choosing $c = 1$ in the Neyman-Pearson objective yields d_{TV}. Choosing different values of c leads to an asymmetric notion of distances/indistinguishability.

Example. Consider $X_1, \ldots, X_n \sim N(\theta, 1)$ for testing $H_0 : \theta = 0$ vs. $H_1 : \theta = \mu$. The TV distance between the n-sample distributions is:

$$d_{\text{TV}}(P_1^{\otimes n}, P_0^{\otimes n}) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P_1^{\otimes n} \| P_0^{\otimes n})} = \sqrt{\frac{n}{2} D_{\text{KL}}(P_1 \| P_0)}$$

If $P_1 = N(\mu, 1)$ and $P_0 = N(0, 1)$, $D_{\text{KL}}(P_1 \| P_0) = \frac{\mu^2}{2}$.

$$d_{\text{TV}}(P_1^{\otimes n}, P_0^{\otimes n}) \leq \sqrt{\frac{n}{2} \cdot \frac{\mu^2}{2}} = \frac{\mu\sqrt{n}}{2}$$

When $\mu = \frac{1}{\sqrt{n}}$, the testing lower bound is $1 - \frac{1}{2} = \frac{1}{2}$.

Remark. To prove testing lower bounds, we often compute an upper bound on d_{TV}. The $\mathbf{d}_{\text{TV}}$ between two distributions P_0 and P_1 is defined as:

$$d_{\text{TV}}(P_0, P_1) = \frac{1}{2} \int |p_1(x) - p_0(x)| d\mu(x) d_{\text{TV}}(P_0, P_1) = \sup_{\|f\|_\infty \leq 1} |\mathbb{E}_1[f(X)] - \mathbb{E}_0[f(X)]|$$

The fundamental Pinsker's inequality relates d_{TV} and the **Kullback-Leibler Divergence** (D_{KL}):

$$d_{\text{TV}}(P, Q) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P \| Q)}.$$

where the Kullback-Leibler Divergence from Q to P is:

$$D_{\text{KL}}(P \| Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right]$$

The benefit of using KL divergence is that it admits the following tensorization property: for i.i.d. samples $X_1, \ldots, X_n$:

$$D_{\text{KL}}(P^{\otimes n} \| Q^{\otimes n}) = \sum_{i=1}^{n} D_{\text{KL}}(P \| Q) = n D_{\text{KL}}(P \| Q)$$

Remark. We can study estimation lower bounds using a testing framework. The core idea is that if two probability distributions, P_0 and P_1, are difficult to distinguish, then any estimator for a parameter $g(\theta)$ that differs under these two distributions must have a high risk.

Lemma 1.4.7 (Le Cam's Lemma)**.** Let $\delta(X)$ be an estimator for $g(\theta)$. For any two points $\theta_0, \theta_1 \in \Theta$, the minimax risk is bounded below as follows:

$$\inf_{\delta} \sup_{\theta \in \Theta} \mathbb{E}_\theta [(\delta(X) - g(\theta))^2] \geq \frac{(g(\theta_1) - g(\theta_0))^2}{8} (1 - d_{TV}(P_0, P_1))$$

where P_0 and P_1 are the distributions corresponding to θ_0 and θ_1, and d_{TV} is the total variation distance.

Proof. Let δ be an arbitrary test. Then, we have

$$\sup_{\theta \in \Theta} \mathbb{E}_\theta \left[(\delta(X) - g(\theta))^2 \right] \geq \frac{1}{2} \left(\mathbb{E}_{\theta_0} \left[(\delta(X) - g(\theta_0))^2 \right] + \mathbb{E}_{\theta_1} \left[(\delta(X) - g(\theta_1))^2 \right] \right)$$

Let's convert the estimation problem into a simple testing problem. Consider a test

$$\phi(X) := 1_{\{|\delta(X) - g(\theta_1)| \leq |\delta(X) - g(\theta_0)|\}}$$

Then the test δ make an error under θ_i implies that $|\delta(X) - g(\theta_i)| \geq \frac{1}{2} |g(\theta_0) - g(\theta_1)|$. Therefore, from the testing lower bound, we have

$$\begin{aligned} \frac{1}{2} \left(\mathbb{E}_{\theta_0} \left[(\delta - g_0)^2 \right] + \mathbb{E}_{\theta_1} \left[(\delta - g_1)^2 \right] \right) &\geq \frac{|g(\theta_0) - g(\theta_1)|^2}{8} (\text{Probability of Type I + Type II error}) \\ &\geq \frac{|g(\theta_0) - g(\theta_1)|^2}{8} \left(1 - d_{\text{TV}}(P_0, P_1) \right). \end{aligned}$$

□

1.4.2 One-sided and Two-sided Testing

We would like to generalize the conclusions in simple testing to one-sided and two-sided testing. A key property that facilitates the construction of optimal tests in these scenarios is the Monotone Likelihood Ratio (MLR).

Definition 1.4.8 (Monotone Likelihood Ratio). A family of distributions $\{P_\theta : \theta \in \mathbb{R}\}$ with densities $p_\theta(x)$ has a **Monotone Likelihood Ratio (MLR)** in a statistic $T(X)$ if for any $\theta_1 < \theta_2$, the likelihood ratio

$$\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)}$$

is a non-decreasing function of $T(x)$.

Example. Consider a one-parameter exponential family with density:

$$p_\theta(x) = \exp(\eta(\theta)T(x) - B(\theta))h(x)$$

The likelihood ratio is:

$$\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} = \exp\left((\eta(\theta_2) - \eta(\theta_1))T(x) - (B(\theta_2) - B(\theta_1))\right)$$

If $\eta(\theta)$ is a monotone non-decreasing function of θ, then for $\theta_1 < \theta_2$, we have $\eta(\theta_1) \le \eta(\theta_2)$. The exponent is a non-decreasing function of $T(x)$, and since $\exp(\cdot)$ is increasing, the entire ratio is non-decreasing in $T(x)$. Thus, the family has MLR in $T(X)$.

The MLR property is the crucial condition in the Karlin-Rubin Theorem, which guarantees the existence of a Uniformly Most Powerful (UMP) test for one-sided hypotheses.

Theorem 1.4.9 (Karlin-Rubin). Consider the one-sided hypothesis testing problem:

$$H_0 : \theta \le \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0$$

If the family of distributions $\{P_\theta\}$ has MLR in the statistic $T(X)$, then there exists a UMP test of size α. This test, $\phi^*(x)$, is of the form:

$$\phi^*(x) = \begin{cases} 1 & \text{if } T(x) > c \\ \gamma & \text{if } T(x) = c \\ 0 & \text{if } T(x) < c \end{cases}$$

where the constants c and $\gamma \in [0, 1]$ are chosen to satisfy the size constraint:

$$\sup_{\theta \le \theta_0} \mathbb{E}_\theta[\phi^*(X)] = \mathbb{E}_{\theta_0}[\phi^*(X)] = \alpha$$

Proof. The proof relies on the Neyman-Pearson Lemma. For any specific alternative $\theta_1 > \theta_0$, the test ϕ^* is the Likelihood Ratio Test (LRT) for the simple hypothesis problem $H_0' : \theta = \theta_0$ vs. $H_1' : \theta = \theta_1$. By the Neyman-Pearson Lemma, ϕ^* is the Most Powerful (MP) test for this simple problem. Since the form of the test (thresholding $T(X)$) does not depend on the specific choice of θ_1 (as long as $\theta_1 > \theta_0$), the test is uniformly most powerful for all $\theta > \theta_0$. □

For two-sided hypothesis testing problems, such as $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$, a Uniformly Most Powerful (UMP) test generally does not exist. A test that is optimal for alternatives $\theta > \theta_0$ is typically suboptimal for $\theta < \theta_0$. To find a meaningful "best" test in these situations, we can restrict our search to the class of **unbiased tests**.

Definition 1.4.10 (Unbiased Test). A test ϕ with significance level α is **unbiased** if its power function, $P_\phi(\theta) = \mathbb{E}_\theta[\phi(X)]$, is never smaller than its size for any parameter in the alternative hypothesis. That is:

$$P_\phi(\theta) \geq \alpha \quad \text{for all } \theta \in \Theta_1$$

This ensures that the probability of correctly rejecting the null is always at least as large as the probability of a Type I error. For a test of $H_0 : \theta = \theta_0$, this implies the power function has a minimum at θ_0.

This concept is analogous to unbiased estimation, where we seek estimators $\hat{\theta}$ that are correct on average. The search for a Uniformly Most Powerful Unbiased (UMPU) test is the search for a test that is uniformly most powerful among all unbiased tests. Our focus for constructing such tests is typically on **exponential families**.

For a multi-parameter exponential family, the probability density function is given by:

$$p_\eta(x) = \exp(\eta \cdot T(x) - A(\eta)) \cdot h(x)$$

where η is the natural parameter vector and $T(X)$ is the vector of sufficient statistics.

Example. Consider the test $H_0 : \eta = \eta_0$ vs. $H_1 : \eta \neq \eta_0$. For a test to be unbiased, its power function $P_\phi(\eta) = \mathbb{E}_\eta[\phi(X)]$ must have a local minimum at $\eta = \eta_0$. Assuming we can differentiate under the integral sign, this implies:

$$\frac{d}{d\eta}\mathbb{E}_\eta[\phi(X)]\bigg|_{\eta=\eta_0} = 0$$

For an exponential family, this derivative condition is equivalent to:

$$\mathrm{Cov}_{\eta_0}(T(X), \phi(X)) = \mathbb{E}_{\eta_0}\left[(T(X) - \mathbb{E}_{\eta_0}[T(X)])\phi(X)\right] = 0$$

To construct a two-sided test, we intuitively need two thresholds, c_1 and c_2. The proposed form of the test is:

$$\phi^*(X) = \begin{cases} 1 & \text{if } T(X) < c_1 \text{ or } T(X) > c_2 \\ \gamma_i & \text{if } T(X) = c_i \text{ for } i = 1, 2 \\ 0 & \text{if } T(X) \in (c_1, c_2) \end{cases}$$

The constants $c_1, c_2, \gamma_1, \gamma_2$ are found by solving the following two equations simultaneously:

(a). **Size constraint:** $\mathbb{E}_{\eta_0}[\phi^*(X)] = \alpha$

(b). **Unbiasedness constraint:** $\mathbb{E}_{\eta_0}[T(X)(\phi^*(X) - \alpha)] = 0$.

This motivates the following result, whose proof is omitted here.

Theorem 1.4.11. Suppose η_0 is in the interior of the natural parameter space $\mathcal{H}$. For any $\alpha \in (0, 1)$, there exists a test ϕ^* that satisfies the two equations above. Furthermore, this test ϕ^* is the UMPU test for $H_0 : \eta = \eta_0$ vs. $H_1 : \eta \neq \eta_0$.

1.4.3 Minimax Testing

When the parameter θ is a vector of dimension $d \geq 2$, UMP or UMPU tests often do not exist. Consider testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$ for $X \sim N(\theta, \sigma^2 I_d)$.

For $d = 2$, we could define different rejection regions. For example:

- ϕ_1: Reject H_0 when $\|X - \theta_0\|_2 > c_1$ (a circular rejection region).
- ϕ_2: Reject H_0 when $\|X - \theta_0\|_\infty > c_2$ (a square rejection region).

Neither test is uniformly more powerful than the other. Test ϕ_1 will be more powerful for alternatives along the diagonals, while ϕ_2 will be more powerful for alternatives along the axes. We cannot compare them uniformly.

Since a uniformly optimal test may not exist, we can change the criterion for optimality. The minimax framework seeks a test that minimizes the worst-case risk. First, we redefine the hypotheses to ensure the null and alternative are well-separated.

Definition 1.4.12 (Minimax Hypotheses). We test $H_0 : \theta \in \Theta_0$ against a composite alternative that is separated by a distance ϵ:

$$H_1 : \theta \in \Theta_1(\epsilon) = \{\theta \in \Theta_1 : \text{dist}(\theta, \Theta_0) \geq \epsilon\}$$

The distance $\epsilon > 0$ is a user's choice, representing the minimum signal strength of interest.

The risk of a test ϕ is defined as the sum of the maximum Type I and Type II error probabilities:

$$R_\epsilon(\phi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\phi(X)] + \sup_{\theta \in \Theta_1(\epsilon)} \mathbb{E}_\theta[1 - \phi(X)]$$

The objective in minimax testing is often to find the smallest possible separation ϵ that allows for a test ϕ^* to achieve a specified low risk level δ. The intuition is to find the minimum signal strength required for reliable detection.

Example (Univariate Mean). Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\theta, 1)$. We test $H_0 : \theta = 0$ vs. $H_1 : \theta \neq 0$. The minimum separation distance required to achieve a fixed risk level δ is:

$$\epsilon_n^* = \frac{c}{\sqrt{n}}$$

where c is a constant that depends on the desired risk δ.

We consider testing a hypothesis about the mean of a multivariate normal distribution. Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\theta, I_d)$, where $\theta \in \mathbb{R}^d$. The hypothesis testing problem is:

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1(\epsilon) : \|\theta\|_2 \geq \epsilon$$

This is a problem of detecting a signal of magnitude at least ϵ against the null hypothesis of no signal.

A natural first guess for the detection boundary, ϵ_n, might be based on the standard estimation error for θ. The sample mean $\hat{\theta}_n = \frac{1}{n}\sum X_i$ has an expected squared error of $\mathbb{E}[\|\hat{\theta}_n - \theta\|_2^2] = d/n$. This suggests that we can only distinguish θ from 0 when $\|\theta\|_2$ is larger than the "noise" level, perhaps leading to a conjecture that the minimax separation is $\epsilon_n \approx \sqrt{d/n}$. However, the actual rate is surprisingly faster.

Theorem 1.4.13. The minimax testing radius for the multivariate Gaussian mean problem is

$$\epsilon_n^* \asymp \frac{d^{1/4}}{\sqrt{n}}$$

This means that for a constant c_1, if $\epsilon \leq c_1 \frac{d^{1/4}}{\sqrt{n}}$, no test can reliably distinguish H_0 from $H_1(\epsilon)$ (the sum of errors is bounded below). Conversely, for a constant C_1, if $\epsilon \geq C_1 \frac{d^{1/4}}{\sqrt{n}}$, there exists a test that can reliably distinguish them (the sum of errors is bounded above).

Proof. The proof has two parts.

Part I: Upper bound / achievability.
Let $Y = \frac{1}{n}\sum_{i=1}^{n} X_i$, then $Y \sim N\left(\theta, \frac{1}{n} I_d\right)$. We use the statistic $T = \|Y\|_2^2$. Since the coordinates of Y are independent, we have

$$\mathbb{E}_\theta[T] = \sum_{j=1}^{d} \mathbb{E}_\theta[Y_j^2] = \sum_{j=1}^{d} \left(\frac{1}{n} + \theta_j^2\right) = \frac{d}{n} + \|\theta\|_2^2.$$

$$\mathrm{Var}_\theta(T) = \sum_{j=1}^{d} \mathrm{Var}_\theta(Y_j^2) = \sum_{j=1}^{d} \left(\frac{2}{n^2} + \frac{4\theta_j^2}{n}\right) = \frac{2d}{n^2} + \frac{4}{n}\|\theta\|_2^2.$$

Fix $a \geq 1$ and define the test

$$\phi_a(X) = \mathbf{1}\left\{\|Y\|_2^2 \geq \frac{d}{n} + a\frac{\sqrt{2d}}{n}\right\}.$$

The threshold is the null mean plus a null standard deviations.
Under H_0, we have

$$\mathbb{E}_0[T] = \frac{d}{n}, \qquad \mathrm{Var}_0(T) = \frac{2d}{n^2}.$$

Hence, by Chebyshev's inequality,

$$\mathbb{P}_0(\phi_a = 1) = \mathbb{P}_0\left(T - \mathbb{E}_0[T] \geq a\sqrt{\mathrm{Var}_0(T)}\right) \leq \frac{1}{a^2}.$$

We now bound the Type II error. Write $s = \|\theta\|_2^2, \qquad r = \frac{\sqrt{d}}{n}$. Then the threshold is

$$\frac{d}{n} + a\frac{\sqrt{2d}}{n} = \frac{d}{n} + a\sqrt{2}\,r.$$

Therefore,

$$\mathbb{P}_\theta(\phi_a = 0) = \mathbb{P}_\theta\left(T < \frac{d}{n} + a\sqrt{2}\,r\right) = \mathbb{P}_\theta\left(T - \mathbb{E}_\theta[T] < a\sqrt{2}\,r - s\right).$$

If $s > a\sqrt{2}\,r$, then Chebyshev's inequality gives

$$\mathbb{P}_\theta(\phi_a = 0) \leq \frac{\mathrm{Var}_\theta(T)}{(s - a\sqrt{2}\,r)^2} = \frac{\frac{2d}{n^2} + \frac{4}{n}s}{(s - a\sqrt{2}\,r)^2}.$$

Since $r = \sqrt{d}/n$, this may be written as

$$\mathbb{P}_\theta(\phi_a = 0) \leq \frac{2r^2 + \frac{4}{n}s}{(s - a\sqrt{2}\,r)^2}.$$

Now assume

$$s = \|\theta\|_2^2 \geq K_a r, \qquad K_a := 8a^2.$$

For $s > a\sqrt{2}\,r$, the function

$$s \mapsto \frac{2r^2 + \frac{4}{n}s}{(s - a\sqrt{2}\,r)^2}$$

is decreasing. Therefore,

$$\mathbb{P}_\theta(\phi_a = 0) \le \frac{2r^2 + \frac{4}{n}K_a r}{(K_a r - a\sqrt{2}\,r)^2}.$$

Using $d \ge 1$ and $K_a = 8a^2$, we obtain

$$\frac{2r^2 + \frac{4}{n}K_a r}{(K_a r - a\sqrt{2}\,r)^2} = \frac{2 + \frac{4K_a}{\sqrt{d}}}{(K_a - a\sqrt{2})^2} \le \frac{2 + 4K_a}{(K_a - a\sqrt{2})^2} \le \frac{1}{a^2}.$$

Thus, we have shown that

$$\|\theta\|_2^2 \ge 8a^2 \frac{\sqrt{d}}{n} \implies \mathbb{P}_\theta(\phi_a = 0) \le \frac{1}{a^2}.$$

Combining the two bounds, we have

$$\epsilon \ge 2\sqrt{2}\, a\, \frac{d^{1/4}}{\sqrt{n}} \implies R_\epsilon(\phi_a) \le \frac{1}{a^2} + \frac{1}{a^2} = \frac{2}{a^2},$$

This proves the upper bound.

Part II: Lower bound / impossibility.

We now show that no test can succeed when ϵ is smaller than a constant multiple of $d^{1/4}/\sqrt{n}$.
Let $\delta = \frac{\epsilon}{\sqrt{d}}$. Consider the prior π that is uniform on the hypercube

$$\Theta_\epsilon = \left\{\theta^\omega = \delta\omega : \omega = (\omega_1, \dots, \omega_d) \in \{-1, +1\}^d\right\}.$$

Every point in this prior satisfies

$$\|\theta^\omega\|_2^2 = \sum_{j=1}^{d} \delta^2 = d\delta^2 = \epsilon^2,$$

so the prior is supported on the sphere $\{\|\theta\|_2 = \epsilon\}$.
Let P_θ denote the joint law of $X_1, \dots, X_n$ under parameter θ, and let

$$P_\pi = \int P_\theta \, d\pi(\theta)$$

be the corresponding mixture distribution.
For any test ϕ, by the definition of total variation distance,

$$\begin{aligned}\mathbb{P}_0(\phi = 1) + \sup_{\|\theta\|_2 = \epsilon} \mathbb{P}_\theta(\phi = 0) &\ge \mathbb{P}_0(\phi = 1) + \mathbb{P}_\pi(\phi = 0) \\ &= 1 - (\mathbb{P}_\pi(\phi = 1) - \mathbb{P}_0(\phi = 1)) \ge 1 - \|P_\pi - P_0\|_{\mathrm{TV}}.\end{aligned}$$

Taking the infimum over all tests,

$$\inf_\phi \left\{ \mathbb{P}_0(\phi = 1) + \sup_{\|\theta\|_2 = \epsilon} \mathbb{P}_\theta(\phi = 0) \right\} \ge 1 - \|P_\pi - P_0\|_{\mathrm{TV}}.$$

Thus, it remains to show that P_π and P_0 are close in total variation.
We bound total variation by the chi-square divergence:

$$\|P_\pi - P_0\|_{\mathrm{TV}} = \frac{1}{2}\mathbb{E}_0 \left| \frac{dP_\pi}{dP_0} - 1 \right| \le \frac{1}{2} \left(\mathbb{E}_0 \left[\left(\frac{dP_\pi}{dP_0} - 1 \right)^2 \right] \right)^{1/2}.$$

Hence

$$\|P_\pi - P_0\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\chi^2(P_\pi\|P_0)}.$$

We now compute $\chi^2(P_\pi\|P_0)$. For a fixed θ, the likelihood ratio between P_θ and P_0 is

$$L_\theta(X) = \frac{dP_\theta}{dP_0}(X) = \exp\left(\theta^\top \sum_{i=1}^n X_i - \frac{n}{2}\|\theta\|_2^2\right).$$

Equivalently, since $Y = n^{-1}\sum_{i=1}^n X_i$ and $\theta^\omega = \delta\omega$,

$$L_\theta(X) = \exp\left(n\theta^\top Y - \frac{n}{2}\|\theta\|_2^2\right) = \exp\left(n\delta\,\omega^\top Y - \frac{n}{2}\epsilon^2\right).$$

The likelihood ratio of the mixture is

$$L_\pi(X) = \frac{dP_\pi}{dP_0}(X) = 2^{-d}\sum_{\omega\in\{-1,+1\}^d} L_\omega(X).$$

Therefore,

$$\mathbb{E}_0[L_\pi^2] = 2^{-2d}\sum_{\omega,\omega'}\mathbb{E}_0[L_\omega L_{\omega'}].$$

Under P_0, $Y \sim N(0, n^{-1}I_d)$. Hence, for fixed ω, ω',

$$\mathbb{E}_0[L_\omega L_{\omega'}] = \exp\left(n\delta^2\omega^\top\omega'\right).$$

Indeed, this follows from the Gaussian moment generating function.
Thus,

$$\mathbb{E}_0[L_\pi^2] = 2^{-2d}\sum_{\omega,\omega'}\exp\left(n\delta^2\sum_{j=1}^d \omega_j\omega_j'\right).$$

For independent uniformly distributed signs ω, ω', each product $\omega_j\omega_j'$ is again a uniform sign. Therefore,

$$\mathbb{E}_0[L_\pi^2] = \left(\frac{e^{n\delta^2} + e^{-n\delta^2}}{2}\right)^d$$

Since $\delta^2 = \epsilon^2/d$, using the elemenetary inequality $(e^x + e^{-x})/2 \le e^{x^2/2}$, we obtain

$$\chi^2(P_\pi\|P_0) = \mathbb{E}_0[L_\pi^2] - 1 \le \exp\left(\frac{d}{2}\left(\frac{n\epsilon^2}{d}\right)^2\right) - 1 = \exp\left(\frac{n^2\epsilon^4}{2d}\right) - 1..$$

Choose $c_1 > 0$ small enough, we have

$$\epsilon \le c_1\frac{d^{1/4}}{\sqrt{n}} \implies \frac{n^2\epsilon^4}{2d} \le \frac{c_1^4}{2} \implies \chi^2(P_\pi\|P_0) \le \frac{4}{9}$$

Therefore,

$$\|P_\pi - P_0\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\frac{4}{9}} = \frac{1}{3}.$$

and hence,

$$\inf_\phi\left\{\mathbb{P}_0(\phi = 1) + \sup_{\|\theta\|_2=\epsilon}\mathbb{P}_\theta(\phi = 0)\right\} \ge 1 - \frac{1}{3} = \frac{2}{3}.$$

Since the sphere $\{\|\theta\|_2 = \epsilon\}$ is contained in the alternative $\{\|\theta\|_2 \ge \epsilon\}$, the same lower bound also holds for the composite alternative $H_1(\epsilon) : \|\theta\|_2 \ge \epsilon$.
Combining the upper and lower bounds, we obtain

$$\epsilon_n^* \asymp \frac{d^{1/4}}{\sqrt{n}}.$$

□

Chapter 2 Asymptotic Theory

2.1 Lecture 5: M-estimation

2.1.1 Introduction to M-estimation Problems

We consider a general estimation problem where the goal is to find a parameter θ that minimizes a population-level objective (or risk) function, $F(\theta)$.

Definition 2.1.1 (Population-Level Problem)**.** Let $f(\theta; X)$ be a loss function for a parameter $\theta \in \Theta$ and a random variable $X \sim P$. The population-level risk is the expected loss:

$$F(\theta) = \mathbb{E}[f(\theta; X)]$$

The true or optimal parameter θ^* is the minimizer of this risk:

$$\theta^* = \operatorname*{argmin}_{\theta \in \Theta}\{F(\theta)\}$$

In practice, we do not know the true distribution P and cannot compute $F(\theta)$ directly. Instead, we use a sample of data $X_1, \dots, X_n \overset{i.i.d.}{\sim} P$ to construct an empirical version of the risk. This leads to the method of M-estimation.

Definition 2.1.2 (M-estimation / ERM / SAA)**.** The empirical risk function (or Sample Average Approximation) is defined as:

$$F_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} f(\theta; X_i)$$

The M-estimator, denoted $\hat{\theta}_n$, is the minimizer of the empirical risk:

$$\hat{\theta}_n = \operatorname*{argmin}_{\theta \in \Theta}\{F_n(\theta)\} = \operatorname*{argmin}_{\theta \in \Theta}\left\{\frac{1}{n}\sum_{i=1}^{n} f(\theta; X_i)\right\}$$

This is also known as Empirical Risk Minimization (ERM). Sometimes, a regularization term is added to the objective function.

Remark. Examples include Maximum Likelihood Estimation (MLE), regression, and classification.

We evaluate the quality of the estimator $\hat{\theta}_n$ by how close it is to the optimal θ^*. This can be measured in two main ways:

- **Function Value (Excess Risk):** The difference in the population risk, $F(\hat{\theta}_n) - F(\theta^*)$.
- **Parameter Estimation Error:** The distance between the parameters, $\|\hat{\theta}_n - \theta^*\|$.

Our goal is to show that as the sample size n increases, $\hat{\theta}_n$ converges to θ^*. We start by analyzing the excess risk, $F(\hat{\theta}_n) - F(\theta^*)$. We can decompose the excess risk into three terms:

$$F(\hat{\theta}_n) - F(\theta^*) = \underbrace{(F(\hat{\theta}_n) - F_n(\hat{\theta}_n))}_{\text{Term 1: Hard}} + \underbrace{(F_n(\hat{\theta}_n) - F_n(\theta^*))}_{\text{Term 2: } \leq 0} + \underbrace{(F_n(\theta^*) - F(\theta^*))}_{\text{Term 3: Easy}}$$

Let's analyze each term:

- **Term 2:** By the definition of $\hat{\theta}_n$ as the minimizer of $F_n(\theta)$, we have $F_n(\hat{\theta}_n) \leq F_n(\theta)$ for all $\theta \in \Theta$. In particular, $F_n(\hat{\theta}_n) \leq F_n(\theta^*)$, so this term is always less than or equal to zero.
- **Term 3:** This term compares the empirical risk and population risk at a *fixed* point θ^*. By the Law of Large Numbers (LLN), $F_n(\theta^*) \xrightarrow{P} F(\theta^*)$. Thus, this term converges to zero. Concentration inequalities (e.g., Hoeffding's inequality) can provide non-asymptotic bounds on this convergence.
- **Term 1:** This term is difficult because it involves the difference between the population and empirical risk evaluated at the *random* estimator $\hat{\theta}_n$, which itself depends on the data (and thus on F_n).

To handle the "hard" term, we bound it by the worst-case deviation over the entire parameter space:

$$|F(\hat{\theta}_n) - F_n(\hat{\theta}_n)| \leq \sup_{\theta \in \Theta} |F(\theta) - F_n(\theta)|$$

The key to proving consistency is showing that this supremum converges to zero. This is the essence of the Uniform Law of Large Numbers (ULLN).

Definition 2.1.3 (Uniform Law of Large Numbers)**.** The ULLN holds if the empirical process, $\sup_{\theta \in \Theta} |F_n(\theta) - F(\theta)|$, converges to zero in probability:

$$\sup_{\theta \in \Theta} |F_n(\theta) - F(\theta)| \xrightarrow{P} 0$$

Remark. Empirical process theory is the field that studies the properties of such processes, including uniform convergence, non-asymptotic bounds, and their limiting distributions.

If the ULLN holds, then Term 1 converges to 0. Since Term 3 also converges to 0 and Term 2 is non-positive, we can conclude that the excess risk converges to 0, i.e., $F(\hat{\theta}_n) \xrightarrow{P} F(\theta^*)$. This establishes consistency in terms of the function value.

2.1.2 Conditions for Consistency

Consistency in value $(F(\hat{\theta}_n) \to F(\theta^*))$ does not automatically guarantee consistency in the parameter ($\hat{\theta}_n \to \theta^*$). We need an additional identifiability condition.

Proposition 2.1.4. Suppose that for every $\epsilon > 0$, the true minimum is well-separated, i.e.,

$$\inf_{\theta: \|\theta - \theta^*\| \geq \epsilon} F(\theta) > F(\theta^*)$$

Then, if the ULLN holds (which implies $F(\hat{\theta}_n) \xrightarrow{P} F(\theta^*)$), it follows that $\hat{\theta}_n \xrightarrow{P} \theta^*$.

Proof Sketch. The condition implies that there exists some $\delta > 0$ such that $\inf_{\|\theta - \theta^*\| \geq \epsilon} F(\theta) > F(\theta^*) + \delta$. Therefore,

$$\mathbb{P}(\|\hat{\theta}_n - \theta^*\| \geq \epsilon) \leq \mathbb{P}(F(\hat{\theta}_n) \geq F(\theta^*) + \delta)$$

Since $F(\hat{\theta}_n) \xrightarrow{P} F(\theta^*)$, the probability on the right-hand side goes to 0 as $n \to \infty$. Thus, $\hat{\theta}_n \xrightarrow{P} \theta^*$. □

Now we consider the case when the ULLN holds.

(a). **Case 1: Finite Parameter Space.** If the parameter space Θ is finite (i.e., $|\Theta| < \infty$), the ULLN is straightforward to prove using a union bound.

Proof. For any $\epsilon > 0$,

$$\begin{aligned}\mathbb{P}\left(\sup_{\theta\in\Theta}|F_n(\theta) - F(\theta)| > \epsilon\right) &= \mathbb{P}\left(\bigcup_{\theta\in\Theta}\{|F_n(\theta) - F(\theta)| > \epsilon\}\right) \\ &\leq \sum_{\theta\in\Theta}\mathbb{P}(|F_n(\theta) - F(\theta)| > \epsilon)\end{aligned}$$

For each fixed θ, the standard LLN implies $\mathbb{P}(|F_n(\theta) - F(\theta)| > \epsilon) \to 0$ as $n \to \infty$. Since the sum is finite, the entire expression converges to 0. □

(b). **Case 2: Infinite/Uncountable Parameter Space.** When Θ is infinite (e.g., a compact subset of $\mathbb{R}^d$), the union bound argument fails. The idea is to use **discretization**. We argue that if $F_n(\theta)$ is close to $F(\theta)$ on a finite "net" of points, and the function f is smooth enough, then $F_n(\theta)$ will be close to $F(\theta)$ everywhere. The "size" or "complexity" of the space Θ can be measured using covering and packing numbers.

Let (K, ρ) be a metric space.

Definition 2.1.5 (Covering Number). The **covering number**, $N(K, \rho, \epsilon)$, is the minimum number of closed balls of radius ϵ needed to cover the set K.

$$N(K, \rho, \epsilon) := \min\left\{N : \exists\{\theta_i\}_{i=1}^N \text{ s.t. } K \subseteq \bigcup_{i=1}^N B_\rho(\theta_i, \epsilon)\right\}$$

where $B_\rho(\theta_i, \epsilon)$ is the closed ball of radius ϵ centered at θ_i.

Definition 2.1.6 (Packing Number). The **packing number**, $M(K, \rho, \epsilon)$, is the maximum number of points that can be placed in K such that the distance between any two distinct points is at least ϵ.

$$M(K, \rho, \epsilon) := \max\left\{M : \exists\{\theta_i\}_{i=1}^M \subseteq K \text{ s.t. } \rho(\theta_i, \theta_j) \geq \epsilon \text{ for all } i \neq j\right\}$$

These two concepts are closely related, as described by the following duality theorem.

Theorem 2.1.7 (Duality). For any set K and $\epsilon > 0$, the packing and covering numbers are related by:

$$M(K, \rho, 2\epsilon) \leq N(K, \rho, \epsilon) \leq M(K, \rho, \epsilon)$$

Proof. Let $\{\theta_j\}_{j=1}^M$ be a *maximal* ϵ-packing of K. By maximality, for any $\theta \in K$, there must exist some $j \in \{1, \ldots, M\}$ such that $\rho(\theta, \theta_j) < \epsilon$. If this were not the case, we could add θ to the packing, contradicting its maximality. Therefore, the set of balls $\{B(\theta_j, \epsilon)\}_{j=1}^M$ forms an ϵ-covering of K. Since the minimal covering $N(K, \rho, \epsilon)$ can only be smaller, we have $N(K, \rho, \epsilon) \leq M$.
Let $\{\theta_i\}_{i=1}^M$ be a 2ϵ-packing of K (so $\rho(\theta_i, \theta_k) \geq 2\epsilon$ for $i \neq k$) and let $\{\psi_j\}_{j=1}^N$ be an ϵ-covering of K. We want to show $M \leq N$. For each θ_i in the packing, it must be contained in at least one ball

from the covering, say $\theta_i \in B(\psi_j, \epsilon)$. Suppose two distinct points from the packing, θ_i and θ_k, fall into the same ball $B(\psi_j, \epsilon)$. Then by the triangle inequality:

$$\rho(\theta_i, \theta_k) \leq \rho(\theta_i, \psi_j) + \rho(\psi_j, \theta_k) \leq \epsilon + \epsilon = 2\epsilon$$

But this contradicts the fact that $\{\theta_i\}_{i=1}^M$ is a 2ϵ-packing, which requires $\rho(\theta_i, \theta_k) \geq 2\epsilon$. Therefore, each θ_i must map to a different ball in the covering, which implies $M \leq N$.

□

Example. Let $K = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq 1\}$ be the unit ball in $\mathbb{R}^d$. We can bound its covering number using volume arguments.

- **Upper bound (via packing):** Consider an $\epsilon/2$-packing $\{\theta_i\}_{i=1}^M$. The balls $B(\theta_i, \epsilon/4)$ are disjoint. All these balls are contained within a larger ball of radius $1 + \epsilon/4$. By comparing volumes:

$$\sum_{i=1}^M \mathrm{Vol}(B(\theta_i, \epsilon/4)) \leq \mathrm{Vol}(B(0, 1 + \epsilon/4))$$

$$M \cdot V_d(\epsilon/4)^d \leq V_d(1 + \epsilon/4)^d \implies M \leq \left(\frac{1 + \epsilon/4}{\epsilon/4}\right)^d = \left(1 + \frac{4}{\epsilon}\right)^d$$

  By the duality theorem, $N(K, \epsilon) \leq M(K, \epsilon) \leq (1 + 4/\epsilon)^d \approx (4/\epsilon)^d$.

- **Lower bound (via covering):** If $\{\psi_j\}_{j=1}^N$ is an ϵ-covering of K, then $K \subseteq \bigcup_{j=1}^N B(\psi_j, \epsilon)$. Comparing volumes:

$$\mathrm{Vol}(K) \leq \sum_{j=1}^N \mathrm{Vol}(B(\psi_j, \epsilon)) \implies V_d(1)^d \leq N \cdot V_d \epsilon^d \implies N \geq \left(\frac{1}{\epsilon}\right)^d$$

Combining these, we see that for the Euclidean unit ball, $N(K, \epsilon)$ scales like $(C/\epsilon)^d$ for some constant C.

2.1.3 Consistency of M-Estimators

We return to the M-estimation problem where $\hat{\theta}_n = \mathrm{argmin}_{\theta \in \Theta} F_n(\theta)$. We established previously that the consistency of the estimator, $\hat{\theta}_n \xrightarrow{P} \theta^*$, is implied by two main conditions:

(a). θ^* is a unique minimizer of the population risk $F(\theta) = \mathbb{E}[f(\theta; X)]$.

(b). The Uniform Law of Large Numbers (ULLN) holds: $\sup_{\theta \in \Theta} |F_n(\theta) - F(\theta)| \xrightarrow{P} 0$.

The following theorem provides a set of sufficient conditions for the ULLN to hold, which is often easier to verify than working with covering numbers directly.

Theorem 2.1.8 (Sufficient Conditions for ULLN)**.** Suppose the following conditions hold:

(a). The parameter space Θ is a compact set.

(b). The function $f(\cdot; x)$ is continuous on Θ for (almost) every x.

(c). There is a dominating function such that $\mathbb{E}[\sup_{\theta \in \Theta} |f(\theta; X)|] < \infty$.

Then the ULLN holds: $\sup_{\theta \in \Theta} |F_n(\theta) - F(\theta)| \xrightarrow{P} 0$.

Proof Sketch. The proof relies on a discretization argument enabled by the compactness of Θ.
1. Discretization: Since Θ is compact, for any $\delta > 0$, we can find a finite δ-covering, $\{\theta_j\}_{j=1}^N$. Any $\theta \in \Theta$ is within δ of some θ_j.

2. Decomposition: Using the triangle inequality, for any $\theta \in B(\theta_j, \delta)$, we can bound the deviation:

$$|F_n(\theta) - F(\theta)| \leq \underbrace{|F_n(\theta) - F_n(\theta_j)|}_{\text{Term A}} + \underbrace{|F_n(\theta_j) - F(\theta_j)|}_{\text{Term B}} + \underbrace{|F(\theta_j) - F(\theta)|}_{\text{Term C}}$$

Taking the supremum over all $\theta \in \Theta$:

$$\sup_{\theta\in\Theta} |F_n(\theta) - F(\theta)| \leq \max_{j\leq N} \sup_{\theta\in B(\theta_j,\delta)} \left(|F_n(\theta) - F_n(\theta_j)| + |F(\theta_j) - F(\theta)|\right) + \max_{j\leq N} |F_n(\theta_j) - F(\theta_j)|$$

3. Analyzing the Terms:

- **Terms C (Fluctuations in F):** From continuity of f, we know that F is also continuous and hence uniformly continuous on Θ. We can then choose δ such that this term is less than $\epsilon/3$.
- **Term B (Center Points):** For any fixed δ, the covering $\{\theta_j\}$ is finite. By the standard LLN, $|F_n(\theta_j) - F(\theta_j)| \xrightarrow{P} 0$ for each j. By a union bound, $\max_{j\leq N} |F_n(\theta_j) - F(\theta_j)| \xrightarrow{P} 0$ as $n \to \infty$.
- **Terms A (Fluctuations):** These terms measure the modulus of continuity of F_n and F. We need to show they can be made arbitrarily small by choosing a small δ. Let's analyze the expectation of their supremum:

$$\mathbb{E}\left[\sup_{\rho(\theta,\theta')\leq\delta} |f(\theta;X) - f(\theta';X)|\right]$$

  Let $M_\delta(X) = \sup_{\rho(\theta,\theta')\leq\delta} |f(\theta;X) - f(\theta';X)|$.
  - By the continuity of $f(\cdot;X)$, we have $M_\delta(X) \to 0$ pointwise as $\delta \to 0$.
  - By the triangle inequality, $|M_\delta(X)| \leq 2\sup_{\theta\in\Theta} |f(\theta;X)|$. Condition (3) of the theorem, $\mathbb{E}[\sup_{\theta\in\Theta} |f(\theta;X)|] < \infty$, ensures this upper bound is integrable.
  - By the Dominated Convergence Theorem (DCT), we can exchange the limit and expectation:
  $$\lim_{\delta\to 0} \mathbb{E}[M_\delta(X)] = \mathbb{E}\left[\lim_{\delta\to 0} M_\delta(X)\right] = \mathbb{E}[0] = 0$$

  This shows that for any $\epsilon > 0$, we can first choose δ small enough to make the expected fluctuation term less than $\epsilon/3$. Therefore, for each $j \leq N$, we have

$$\mathbb{E}\left[|F_n(\theta) - F_n(\theta_j)|\right] \leq \mathbb{E}\left[\sup_{\rho(\theta,\theta')\leq\delta} |f(\theta;X) - f(\theta';X)|\right] < \epsilon/3$$

  An application of Markov's inequality suggests that

$$\max_{j\leq N} \sup_{\theta\in B(\theta_j,\delta)} |F_n(\theta) - F_n(\theta_j)| \xrightarrow{P} 0.$$

This completes the proof that the overall supremum converges to zero in probability. □

2.2 Lecture 6: Techniques in Empirical Processes

2.2.1 The Symmetrization Technique

In the proof of the Uniform Law of Large Numbers (ULLN), a key step is to bound the deviation of the empirical process from its mean on the center points of a covering, i.e., bounding $\mathbb{P}(\max_{j\le N}|F_n(\theta_j)-F(\theta_j)|>\epsilon)$. A simple approach uses a union bound:

$$\mathbb{P}\left(\max_{j\le N}|F_n(\theta_j)-F(\theta_j)|>\epsilon\right)\le\sum_{j=1}^{N}\mathbb{P}(|F_n(\theta_j)-F(\theta_j)|>\epsilon)\le N\cdot\sup_{\theta\in\Theta}\mathbb{P}(|F_n(\theta)-F(\theta)|>\epsilon)$$

- If the loss function f is bounded, we can use Hoeffding's inequality, $\mathbb{P}(|F_n(\theta)-F(\theta)|>\epsilon)\le 2e^{-2n\epsilon^2}$. To make the total probability small, we need $N\cdot e^{-2n\epsilon^2}\to 0$, which leads to a rate of $\epsilon\approx\sqrt{\frac{\log N}{n}}$.
- If f is only assumed to have finite variance (heavy-tailed), a naive application of Chebyshev's inequality gives $\mathbb{P}(|F_n(\theta)-F(\theta)|>\epsilon)\le\frac{\mathrm{Var}(f(\theta;X))}{n\epsilon^2}$. This leads to a much slower rate of $\epsilon\approx\sqrt{\frac{N}{n}}$, which is often too slow for consistency.

To overcome this issue with heavy-tailed functions and to develop more powerful, data-dependent bounds, we introduce the **symmetrization** technique.

Let $\mathcal{F}$ be a class of functions. We adopt the notation $P_nf=\frac{1}{n}\sum_{i=1}^{n}f(X_i)$ for the empirical measure and $Pf=\mathbb{E}[f(X)]$ for the true expectation. Our goal is to bound the expected supremum deviation, $\mathbb{E}[\sup_{f\in\mathcal{F}}|P_nf-Pf|]$.

Definition 2.2.1 (Rademacher Complexity). Let $\mathcal{F}$ be a class of functions. Let $\{\epsilon_i\}_{i=1}^n$ be a sequence of i.i.d. Rademacher random variables, where $\mathbb{P}(\epsilon_i=1)=\mathbb{P}(\epsilon_i=-1)=1/2$. The Rademacher complexity of $\mathcal{F}$ is defined as:

$$\mathcal{R}_n(\mathcal{F}):=\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_if(X_i)\right|\right]$$

The expectation is taken over both the data $\{X_i\}$ and the Rademacher variables $\{\epsilon_i\}$.

The key insight of symmetrization is that the original problem can be bounded by this seemingly more complex quantity.

Theorem 2.2.2 (Symmetrization Lemma). For any function class $\mathcal{F}$, we have:

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}|P_nf-Pf|\right]\le 2\mathcal{R}_n(\mathcal{F})$$

Proof. Let $\{X_i'\}_{i=1}^n$ be a "ghost sample", an independent copy of the original data $\{X_i\}_{i=1}^n$.

$$\begin{aligned}\mathbb{E}\left[\sup_{f\in\mathcal{F}}|P_nf-Pf|\right]&=\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}f(X_i)-\mathbb{E}[f(X_i')]\right|\right]\\&=\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\mathbb{E}_{X'}\left[\frac{1}{n}\sum_{i=1}^{n}(f(X_i)-f(X_i'))\right]\right|\right]\end{aligned}$$

$$\leq \mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}(f(X_i)-f(X_i'))\right|\right] \quad \text{(by Jensen's inequality)}$$

Now, introduce Rademacher variables ϵ_i. Since $f(X_i)-f(X_i')$ is symmetric about 0, multiplying by ϵ_i does not change its distribution.

$$\begin{aligned}
&= \mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i(f(X_i)-f(X_i'))\right|\right] \\
&\leq \mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(X_i)\right|\right] + \mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(X_i')\right|\right] \quad \text{(triangle ineq.)} \\
&= 2\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(X_i)\right|\right] = 2\mathcal{R}_n(\mathcal{F})
\end{aligned}$$

The last step holds because $\{X_i\}$ and $\{X_i'\}$ are identically distributed. □

Remark. The Rademacher complexity $\mathcal{R}_n(\mathcal{F})$ can be analyzed by first conditioning on the data $\{X_i\}_{i=1}^n$. The inner expectation over $\{\epsilon_i\}$ is often much easier to handle using concentration inequalities for sums of independent (but not identically distributed) random variables.

2.2.2 The Chaining Technique

Let $A = \{(f(X_1),\ldots,f(X_n)) : f\in\mathcal{F}\}\subseteq\mathbb{R}^n$. The (conditional) Rademacher complexity is a property of this set A.

$$R_n(A) := \mathbb{E}_\epsilon\left[\sup_{a\in A}\left|\frac{1}{n}\sum_{i=1}^{n}\epsilon_i a_i\right|\right]$$

This quantity measures the complexity of the set A, interpreted as its maximum correlation with a random direction vector $(\epsilon_1,\ldots,\epsilon_n)$.

If the set A is finite, we can bound its Rademacher complexity using a moment-generating function argument.

Lemma 2.2.3. For a finite set $A\subseteq\mathbb{R}^n$,

$$R_n(A) \leq \frac{\max_{a\in A}\|a\|_2}{\sqrt{n}}\sqrt{2\log|A|}$$

Proof Sketch. The proof uses Jensen's inequality for the convex function $x\mapsto\exp(\lambda x)$, a union bound, and Hoeffding's lemma for Rademacher variables ($\mathbb{E}[e^{\lambda\sum\epsilon_i a_i}] = \prod\mathbb{E}[e^{\lambda\epsilon_i a_i}] \leq \prod e^{\lambda^2 a_i^2/2} = e^{\lambda^2\|a\|_2^2/2}$).

$$R_n(A) \leq \frac{1}{\lambda n}\log\mathbb{E}\left[\exp\left(\lambda\sup_{a\in A}\sum\epsilon_i a_i\right)\right] \leq \frac{1}{\lambda n}\left(\log|A| + \frac{\lambda^2}{2}\max_{a\in A}\|a\|_2^2\right)$$

Choosing the optimal λ yields the result. □

For an infinite set A, we can use a one-step discretization (or "chaining") argument. Let $\{a^{(j)}\}_{j=1}^M$ be a δ-covering of A under the empirical norm $\|a\|_n = \sqrt{\frac{1}{n}\sum a_i^2}$. Let $\pi(a) = \operatorname{argmin}_j\|a-a^{(j)}\|_n$.

$$R_n(A) = \mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\left|\sum\epsilon_i(a_i-\pi(a)_i+\pi(a)_i)\right|\right]$$

$$\leq \mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\left|\sum \epsilon_i(a_i - \pi(a)_i)\right|\right] + \mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\left|\sum \epsilon_i \pi(a)_i\right|\right]$$

$$\leq \mathbb{E}\left[\sup_{a\in A}\|\epsilon/n\|_n\|a-\pi(a)\|_n\right] + \mathbb{E}\left[\max_{j\leq M}\frac{1}{n}\left|\sum \epsilon_i a_i^{(j)}\right|\right] \quad \text{(by Cauchy-Schwarz)}$$

The first term is bounded by δ since $\|\epsilon/n\|_n = 1$ and $\|a-\pi(a)\|_n \leq \delta$. The second term is the Rademacher complexity of the finite covering, which we can bound using the previous lemma. This leads to Dudley's entropy integral bound, which for one-step discretization is:

$$R_n(A) \lesssim \inf_{\delta>0}\left(\delta + \frac{\text{diam}(A)}{\sqrt{n}}\sqrt{\log N(A, \|\cdot\|_n, \delta)}\right)$$

For a d-dimensional parametric model, the covering number $N(A,\delta) \approx (C/\delta)^d$. Optimizing for δ (roughly $\delta \propto 1/\sqrt{n}$) leads to a complexity bound:

$$R_n(A) \lesssim \sqrt{\frac{d\log n}{n}}$$

This demonstrates how the complexity of the function class (measured by dimension d) controls the rate of uniform convergence. For more complex non-parametric models, this gap can be larger.

Chaining is a powerful refinement of the single-step discretization method used to bound the Rademacher complexity of an infinite set $A \subseteq \mathbb{R}^n$. Instead of using a single approximation, chaining uses a sequence of increasingly accurate finite approximations.

Let $A_0, A_1, \ldots, A_m, \ldots$ be a sequence of finite sets that provide progressively better approximations to the set A. Let $\pi_m(a)$ be the best approximation to a vector $a \in A$ from the set A_m. We assume that the approximation error goes to zero as $m \to \infty$.

We can express any vector $a \in A$ as a telescoping sum. Let $\pi_0(a) = 0$. Then,

$$a = a - \pi_0(a) = \sum_{m=0}^{\infty}(\pi_{m+1}(a) - \pi_m(a))$$

Using this identity, we can decompose the Rademacher complexity:

$$\mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\epsilon^T a\right] = \mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\sum_{m=0}^{\infty}\epsilon^T(\pi_{m+1}(a)-\pi_m(a))\right] \leq \sum_{m=0}^{\infty}\mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\left|\epsilon^T(\pi_{m+1}(a)-\pi_m(a))\right|\right]$$

Let's assume that for each m, the set A_m is a δ_m-covering of A under the empirical norm $\|\cdot\|_n$. This means for any $a \in A$, $\|\pi_m(a) - a\|_n \leq \delta_m$. The term for each increment can be bounded. The pair $(\pi_m(a), \pi_{m+1}(a))$ takes values in a finite set of cardinality at most $|A_m|\cdot|A_{m+1}|$. The difference vector $\pi_{m+1}(a) - \pi_m(a)$ is bounded in norm:

$$\|\pi_{m+1}(a) - \pi_m(a)\|_n \leq \|\pi_{m+1}(a) - a\|_n + \|a - \pi_m(a)\|_n \leq \delta_{m+1} + \delta_m$$

Using the bound for the Rademacher complexity of a finite set, we have:

$$\mathbb{E}\left[\sup_{a\in A}\frac{1}{n}\left|\epsilon^T(\pi_{m+1}(a)-\pi_m(a))\right|\right] \leq \frac{\max\|\pi_{m+1}(a)-\pi_m(a)\|_n}{\sqrt{n}}\sqrt{2\log(|A_m|\cdot|A_{m+1}|)}$$

$$\leq \frac{\delta_m + \delta_{m+1}}{\sqrt{n}}\sqrt{2\log(|A_m|\cdot|A_{m+1}|)}$$

Summing these bounds over all m gives a total bound on the Rademacher complexity. This multi-scale argument is formalized by Dudley's theorem, which relates the Rademacher complexity to an integral involving the covering numbers of the set.

Theorem 2.2.4 (Dudley's Entropy Integral)**.** There exists a universal constant $C > 0$ such that for any set $A \subseteq \mathbb{R}^n$, its Rademacher complexity is bounded by:

$$R_n(A) \leq \frac{C}{\sqrt{n}} \int_0^{\mathrm{diam}(A)} \sqrt{\log N(A, \|\cdot\|_n, \delta)}\, d\delta$$

where $\mathrm{diam}(A) = \sup_{a \in A} \|a\|_n$ and $N(A, \|\cdot\|_n, \delta)$ is the δ-covering number of A with respect to the empirical norm.

Proof Sketch. Let $D = \mathrm{diam}(A)$. We choose a dyadic (geometric) sequence of scales $\delta_m = D/2^m$ for $m = 0, 1, 2, \ldots$. Let A_m be a minimal δ_m-covering of A, with $A_0 = \{0\}$. The bound on the m-th increment involves terms like $(\delta_m + \delta_{m+1})$ and $\log(|A_m||A_{m+1}|)$. Since $|A_m| = N(A, \delta_m)$ and $\delta_m + \delta_{m+1} = 3D/2^{m+1}$, substituting these into the sum gives:

$$R_n(A) \leq \sum_{m=0}^{\infty} \frac{3D/2^{m+1}}{\sqrt{n}} \sqrt{2\log(N(A, D/2^m)N(A, D/2^{m+1}))}$$

Since $N(A, \cdot)$ is a decreasing function, $N(A, D/2^{m+1}) \geq N(A, D/2^m)$, so we can simplify the log term to $\log(N(A, D/2^{m+1})^2) = 2\log(N(A, D/2^{m+1}))$. The sum becomes:

$$\sum_{m=0}^{\infty} \frac{3D}{2^{m+1}\sqrt{n}} \sqrt{4\log N(A, D/2^{m+1})}$$

This sum can be interpreted as a Riemann sum approximation of an integral. The term $D/2^{m+1}$ is the width of an interval, and the square root term is the height. Therefore, noting that the integrand is increasing with respect to δ, we have

$$\sum_{m=0}^{\infty} \frac{3D}{2^{m+1}\sqrt{n}} \sqrt{4\log N(A, D/2^{m+1})} \leq \frac{6}{\sqrt{n}} \int_0^D \sqrt{\log N(A, \delta)} \mathrm{d}\delta.$$

□

The theorem can be translated back from the vector set A to the original function class $\mathcal{F}$.

Theorem 2.2.5. Consider a function class $\mathcal{F}$ with an envelope function G, such that $|f(x)| \leq G(x)$ for all $f \in \mathcal{F}$ and for all x. Then there exists a constant c such that:

$$\mathbb{E}\left[\sup_{f \in \mathcal{F}} |P_n f - Pf|\right] \leq c\sqrt{\frac{\mathbb{E}[G(X)^2]}{n}} \int_0^1 \sqrt{\log \sup_Q N(\mathcal{F}, L^2(Q), \delta\|G\|_{L^2(Q)})}\, d\delta$$

Here, $L^2(Q)$ is the norm $\|f\|_{L^2(Q)} = (\int f^2 dQ)^{1/2}$, and the supremum $\sup_Q$ is taken over all discrete probability measures on at most n points. The term $N(\mathcal{F}, L^2(Q), \epsilon)$ is the "worst-case L^2 covering number".

Proof. Let $A_X = \{(f(X_i))_{i=1}^n \in \mathbb{R}^n : f \in \mathcal{F}\}$ be the random realization of the sets A above. Then, by Dudley's entropy integral and scaling, we have

$$\begin{aligned}
\mathbb{E}\left[\sup_{f \in \mathcal{F}} |P_n f - Pf|\right] &\lesssim \mathbb{E}_X\left[\frac{1}{\sqrt{n}} \int_0^{\mathrm{diam}(A_X)} \sqrt{\log N(A_X, \|\cdot\|_n, \delta)}\, d\delta\right] \\
&\lesssim \frac{1}{\sqrt{n}} \mathbb{E}_X\left[\|G\|_{L^2(Q_X)} \int_0^{\mathrm{diam}(A_X)/\|G\|_{L^2(Q_X)}} \sqrt{\log N(A_X, \|\cdot\|_n, \delta\|G\|_{L^2(Q_X)})}\, d\delta\right]
\end{aligned}$$

$$= \frac{1}{\sqrt{n}}\mathbb{E}_X\left[\|G\|_{L^2(Q_X)}\int_0^{\mathrm{diam}(A_X)/\|G\|_{L^2(Q_X)}}\sqrt{\log N(\mathcal{F}, L^2(Q_X), \delta\|G\|_{L^2(Q_X)})}\,d\delta\right]$$
$$\leq \frac{1}{\sqrt{n}}\mathbb{E}_X\left[\|G\|_{L^2(Q_X)}\int_0^{1}\sqrt{\log\sup_Q N(\mathcal{F}, L^2(Q), \delta\|G\|_{L^2(Q)})}\,d\delta\right]$$
$$= \sqrt{\frac{\mathbb{E}[G(X)^2]}{n}}\int_0^{1}\sqrt{\log\sup_Q N(\mathcal{F}, L^2(Q), \delta\|G\|_{L^2(Q)})}\,d\delta.$$

which completes the proof. □

Remark.
- The Dudley integral may diverge if the upper bound used for the covering number is too loose. In such cases, one must find tighter bounds on $N(\mathcal{F}, \epsilon)$.
- A more advanced technique called **"generic chaining"** provides a tighter bound that is known to be optimal in many settings, though it is often less convenient to apply than Dudley's integral.
- The problem of bounding the expected supremum deviation is thus reduced to the geometric problem of bounding the covering numbers of the function class $\mathcal{F}$.

2.3 Lecture 7: VC Dimension and Applications

2.3.1 VC Dimension of Sets

2.3.2 VC Dimension of Binary Functions

The primary quantity of interest in statistical learning theory is often the uniform deviation of the empirical mean from the true expectation over a class of functions $\mathcal{F}$:

$$\sup_{f\in\mathcal{F}} |P_n f - Pf|$$

This quantity can be bounded using techniques like **symmetrization** (leading to Rademacher complexity) and **Dudley chaining**, which involves the (log) covering numbers of the function class $\mathcal{F}$. The Vapnik-Chervonenkis (VC) theory provides a different, combinatorial approach to understanding the "complexity" of a function class, particularly for binary functions.

We now focus on **binary functions**, where $f : \mathcal{X} \to \{0, 1\}$.

Definition 2.3.1 (Shattering)**.** We say a set of n points, $S = \{x_1, x_2, \ldots, x_n\}$, is **shattered** by a function class $\mathcal{F}$ if $\mathcal{F}$ can generate every possible binary labeling on these n points. That is, the set of projected vectors is the entire hypercube:

$$\{(f(x_1), f(x_2), \ldots, f(x_n)) : f \in \mathcal{F}\} = \{0, 1\}^n$$

The size of this set of labelings is 2^n.

Definition 2.3.2 (VC Dimension)**.** The **Vapnik-Chervonenkis (VC) dimension** of a function class $\mathcal{F}$, denoted $\mathrm{VC}(\mathcal{F})$, is the largest integer D such that there exists *some* set of D points $\{x_1, \ldots, x_D\} \subseteq \mathcal{X}$ that is shattered by $\mathcal{F}$.
If $\mathcal{F}$ can shatter sets of arbitrarily large size, we say $\mathrm{VC}(\mathcal{F}) = \infty$.

Example (Indicator of line segments). Let $\mathcal{X} = \mathbb{R}$ and let $\mathcal{F}$ be the class of indicator functions of intervals:

$$\mathcal{F} = \{1_{x\in[\theta_1,\theta_2]} : \theta_1, \theta_2 \in \mathbb{R}\}$$

- **D=1:** We can shatter one point x_1.
  - To get $\{1\}$: Choose $[\theta_1, \theta_2] = [x_1, x_1]$.
  - To get $\{0\}$: Choose $[\theta_1, \theta_2] = [x_1 + 1, x_1 + 2]$.

  So, $\text{VC}(\mathcal{F}) \geq 1$.
- **D=2:** We can shatter two points $x_1 < x_2$.
  - To get $\{0, 0\}$: Choose an interval to the right of x_2.
  - To get $\{1, 0\}$: Choose $[\theta_1, \theta_2] = [x_1, x_1]$.
  - To get $\{0, 1\}$: Choose $[\theta_1, \theta_2] = [x_2, x_2]$.
  - To get $\{1, 1\}$: Choose $[\theta_1, \theta_2] = [x_1, x_2]$.

  All 4 combinations are possible, so $\text{VC}(\mathcal{F}) \geq 2$.
- **D=3:** We **cannot** shatter any set of three points $x_1 < x_2 < x_3$. Consider the labeling $(1, 0, 1)$. This would require an interval $[\theta_1, \theta_2]$ that contains x_1 and x_3 but *not* x_2. This is impossible for a single contiguous interval.

Since $\mathcal{F}$ can shatter 2 points but not 3 points, we conclude $\text{VC}(\mathcal{F}) = 2$.

Example (Union of K intervals). Let $\mathcal{F}$ be the class of indicator functions of unions of K intervals:

$$\mathcal{F} = \left\{1_{x\in\bigcup_{j=1}^{K}[\theta_j,\psi_j]}\right\}$$

This class is parameterized by $2K$ variables. It can be shown that $\text{VC}(\mathcal{F}) = 2K$. We can see that $\text{VC}(\mathcal{F}) < 2K + 2$. Consider $2K + 2$ points in order, $x_1, \ldots, x_{2K+2}$. The alternating labeling $(1, 0, 1, 0, \ldots, 1, 0)$ has $K + 1$ ones. This labeling would require $K + 1$ distinct intervals to achieve, which is not possible with a function from $\mathcal{F}$.

Sauer's Lemma (also known as the Sauer-Shelah Lemma) provides a crucial bound on the number of distinct labelings a function class $\mathcal{F}$ can generate on n points, based on its VC dimension. This number is often called the **growth function**.

Lemma 2.3.3 (Sauer's Lemma). Let $\mathcal{F}$ be a function class with $\text{VC}(\mathcal{F}) = D < \infty$. Let $\Pi_{\mathcal{F}}(S) = |\{(f(x_1), \ldots, f(x_n)) : f \in \mathcal{F}\}|$ be the number of distinct labelings $\mathcal{F}$ induces on a specific set $S = \{x_1, \ldots, x_n\}$.
Then for any $n \geq D$:

$$\Pi_{\mathcal{F}}(S) \leq \sum_{i=0}^{D} \binom{n}{i}$$

Remark. Letting $\Phi_n(D) = \sum_{i=0}^{D} \binom{n}{i}$, the lemma states $\Pi_{\mathcal{F}}(S) \leq \Phi_n(D)$. This bound holds regardless of the choice of S, so it also bounds the growth function $\Pi_{\mathcal{F}}(n) = \max_{|S|=n} \Pi_{\mathcal{F}}(S)$.
For large n, this bound is approximately $\left(\frac{en}{D}\right)^D$, which is polynomial in n, whereas 2^n is exponential.

Example (Linear Classifiers (Half-Spaces)). Let $\mathcal{F}$ be the class of linear threshold functions (half-spaces through the origin) in $\mathbb{R}^d$:

$$\mathcal{F} = \{1_{\{\theta^T x \geq 0\}} : \theta \in \mathbb{R}^d\}$$

Then $\text{VC}(\mathcal{F}) = d$.
To show this, it is equivalent to showing that no set of $d+1$ points $\{x_1, \ldots, x_{d+1}\}$ in $\mathbb{R}^d$ can be shattered.

(a). Since we have $d+1$ vectors in $\mathbb{R}^d$, they must be linearly dependent. Thus, $\exists \lambda_1, \ldots, \lambda_{d+1}$, not all zero, such that $\sum_{i=1}^{d+1} \lambda_i x_i = 0$.

(b). Let's construct a labeling that $\mathcal{F}$ cannot achieve. Define the labeling $b = (b_1, \ldots, b_{d+1})$ by:

$$b_i = \begin{cases} 1 & \text{if } \lambda_i > 0 \\ 0 & \text{if } \lambda_i \leq 0 \end{cases}$$

(c). Assume for contradiction that this labeling is achievable. Then $\exists \theta \in \mathbb{R}^d$ such that:

- If $\lambda_i > 0$, then $b_i = 1 \implies \theta^T x_i \geq 0$.
- If $\lambda_i \leq 0$, then $b_i = 0 \implies \theta^T x_i < 0$. (Unless $\lambda_i = 0$, in which case $\theta^T x_i$ can be anything, but $\lambda_i \theta^T x_i = 0$).

(d). Let's analyze the sign of $\lambda_i(\theta^T x_i)$:

- If $\lambda_i > 0$: $\lambda_i(\theta^T x_i) \geq 0$.
- If $\lambda_i < 0$: $\theta^T x_i < 0$, so $\lambda_i(\theta^T x_i) > 0$.
- If $\lambda_i = 0$: $\lambda_i(\theta^T x_i) = 0$.

(e). Since not all λ_i are zero, at least one of these terms must be strictly positive (from a non-zero λ_i).

(f). Therefore, $\sum_{i=1}^{d+1} \lambda_i(\theta^T x_i) > 0$.

(g). But we can rewrite the sum: $\sum_{i=1}^{d+1} \lambda_i(\theta^T x_i) = \theta^T \left(\sum_{i=1}^{d+1} \lambda_i x_i\right) = \theta^T(0) = 0$.

(h). We have the contradiction $0 > 0$. Thus, the labeling b is not achievable, and $\mathcal{F}$ cannot shatter any set of $d+1$ points.

(It can also be shown that $\text{VC}(\mathcal{F}) \geq d$, for example by picking the standard basis vectors plus the origin, so $\text{VC}(\mathcal{F}) = d$. For general half-spaces $1_{\{\theta^T x + b \geq 0\}}$, $\text{VC} = d+1$.)

A "naive" application of Sauer's Lemma to bound the generalization error gives:

$$\mathbb{E}[\sup_{f \in \mathcal{F}} |P_n f - Pf|] \leq C \sqrt{\frac{\log \Pi_{\mathcal{F}}(n)}{n}} \leq C \sqrt{\frac{\log((en/D)^D)}{n}} = C' \sqrt{\frac{D \log(n/D)}{n}}$$

This bound is "suboptimal" due to the $\log n$ term. A more powerful result connects VC dimension directly to covering numbers, independent of n.

Theorem 2.3.4 (VC Bound on Covering Numbers)**.** Let $\mathcal{F}$ be a class of binary functions with $\text{VC}(\mathcal{F}) = D < \infty$. Then for any probability measure Q and any $\varepsilon \in (0,1)$, the $L_2(Q)$ covering number is bounded by:

$$\log N(\varepsilon, \mathcal{F}, L_2(Q)) \leq C \cdot D \cdot \log\left(\frac{1}{\varepsilon}\right)$$

for some universal constant C.

Proof. Let $\{f_1, \dots, f_N\}$ be a maximal ε-packing of $\mathcal{F}$ under $L_2(Q)$. This means $N(\varepsilon, \mathcal{F}, L_2(Q)) \approx N$ and for all $i \neq j$, $\|f_i - f_j\|_{L_2(Q)} \geq \varepsilon$. Note that for binary functions, the $L_2(Q)$ distance squared is the probability of disagreement:

$$\|f_i - f_j\|^2_{L_2(Q)} = \int (f_i(x) - f_j(x))^2 dQ(x) = \int 1_{\{f_i(x) \neq f_j(x)\}} dQ(x) = Q(f_i \neq f_j)$$

So, for our packing, $Q(f_i \neq f_j) \geq \varepsilon^2$ for all $i \neq j$.
Now pick n i.i.d. samples $S = \{x_1, \dots, x_n\} \sim Q$ and consider the probability that a specific pair f_i, f_j are *not* distinguishable on S:

$$\begin{aligned}
\mathbb{P}((f_i(x_1), \dots, f_i(x_n)) = (f_j(x_1), \dots, f_j(x_n))) &= \prod_{k=1}^{n} \mathbb{P}(f_i(x_k) = f_j(x_k)) \\
&= \prod_{k=1}^{n} (1 - Q(f_i \neq f_j)) \\
&\leq (1 - \varepsilon^2)^n \leq \exp(-n\varepsilon^2)
\end{aligned}$$

By a union bound over all $\binom{N}{2} < N^2/2$ pairs, we have

$$\mathbb{P}(\exists i \neq j \text{ not distinguishable on } S) \leq \binom{N}{2} \exp(-n\varepsilon^2) \leq N^2 \exp(-n\varepsilon^2)$$

We want to choose n such that this probability is < 1. If we set $n > \frac{2 \log N}{\varepsilon^2}$, then the probability is $< N^2 \exp(-\frac{2 \log N}{\varepsilon^2} \varepsilon^2) = N^2 \cdot N^{-2} = 1$. This means there *exists* a set S of size $n = O(\frac{\log N}{\varepsilon^2})$ such that all N functions $f_1, \dots, f_N$ produce distinct labeling vectors on S. Let $A = \{(f_i(x_1), \dots, f_i(x_n)) : i \in [N]\}$ be this set of labelings. We have $|A| = N$. By Sauer's Lemma, $|A| \leq \Pi_{\mathcal{F}}(S) \leq \Phi_n(D) \leq \left(\frac{en}{D}\right)^D$. We now have a system of two inequalities:

(a). $N \leq \left(\frac{en}{D}\right)^D \implies \log N \leq D \log(en/D)$
(b). $n \approx O\left(\frac{\log N}{\varepsilon^2}\right)$

Plugging (a) into (b) gives $n \approx O\left(\frac{D \log(en/D)}{\varepsilon^2}\right)$. Solving this for n shows that n is bounded by something proportional to $\frac{D}{\varepsilon^2} \log(1/\varepsilon)$. Plugging this back into (a), we have

$$\log N \leq D \log \left(\frac{e}{D} \cdot O\left(\frac{D}{\varepsilon^2} \log(1/\varepsilon) \right) \right) \approx O(D \log(1/\varepsilon))$$

which gives the desired result. □

Remark. We can now plug this covering number bound into the Dudley integral to get the "optimal" VC bound.

$$\begin{aligned}
\mathbb{E}\left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum f(X_i) - \mathbb{E}[f(X)] \right| \right] &\leq \frac{C}{\sqrt{n}} \int_0^1 \sqrt{\log \sup_Q \mathbb{N}(\varepsilon, \mathcal{F}, L_2(Q))} d\varepsilon \\
&\leq \frac{C}{\sqrt{n}} \int_0^1 \sqrt{C' \cdot D \cdot \log(1/\varepsilon)} d\varepsilon \\
&= \frac{C'' \sqrt{D}}{\sqrt{n}} \int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon
\end{aligned}$$

The integral $\int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon$ is a finite constant (it's $\sqrt{\pi}/2$). Therefore, we get the classic VC generalization bound:

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum f(X_i)-\mathbb{E}[f(X)]\right|\right]\le O\left(\sqrt{\frac{D}{n}}\right)$$

where $D=\mathrm{VC}(\mathcal{F})$. For linear classifiers in $\mathbb{R}^d$, $D=d+1$, so the bound is $O\left(\sqrt{\frac{d+1}{n}}\right)$. This bound is a significant improvement over the $O\left(\sqrt{\frac{D\log n}{n}}\right)$ naive bound.

2.3.3 VC Subgraph Dimension and Generalizations

The standard Vapnik-Chervonenkis (VC) dimension applies to classes of binary functions ($\mathcal{F}:\mathcal{X}\to\{0,1\}$). To analyze real-valued function classes ($\mathcal{F}:\mathcal{X}\to\mathbb{R}$), we introduce the concept of the VC subgraph dimension.

Definition 2.3.5 (VC Subgraph Dimension). Let $\mathcal{F}$ be a class of real-valued functions $f:\mathcal{X}\to\mathbb{R}$. The **subgraph** of a function f is the set $\{(x,t)\in\mathcal{X}\times\mathbb{R}:t\le f(x)\}$. The **VC subgraph dimension** of $\mathcal{F}$, denoted $\mathrm{VC}(\mathcal{F})$, is defined as the standard VC dimension of the associated class of binary functions $\mathcal{G}$ defined on $\mathcal{X}\times\mathbb{R}$:

$$\mathcal{G}=\{(x,t)\mapsto 1_{\{t\le f(x)\}}:f\in\mathcal{F}\}$$

Equivalently, $\mathrm{VC}(\mathcal{F})$ is the largest integer D such that there exists a set of D points $\{(x_1,t_1),\dots,(x_D,t_D)\}$ that is shattered by $\mathcal{G}$. That is, for every binary vector $(b_1,\dots,b_D)\in\{0,1\}^D$, there exists a function $f\in\mathcal{F}$ such that:

$$1_{\{t_i\le f(x_i)\}}=b_i\quad\text{for all } i=1,\dots,D$$

This means $\mathcal{F}$ can "separate" the points, with $f(x_i)\ge t_i$ if $b_i=1$ and $f(x_i)<t_i$ if $b_i=0$.

Example (Affine Functions). Let

$$\mathcal{F}=\{f:x\mapsto\theta^Tx-\theta_0:\theta\in\mathbb{R}^d,\theta_0\in\mathbb{R}\}$$

be the class of affine functions on $\mathbb{R}^d$. The subgraph class is

$$\mathcal{G}=\{(x,t)\mapsto 1_{\{t\le\theta^Tx-\theta_0\}}\}=\{(x,t)\mapsto 1_{\{\theta^Tx-\theta_0-t\ge 0\}}\}.$$

This is a class of homogeneous half-spaces on the input $(x,1,t)\in\mathbb{R}^{d+2}$. The VC dimension of homogeneous half-spaces in $\mathbb{R}^k$ is k. Therefore, $\mathrm{VC}(\mathcal{F})=d+2$. (The note states $\mathrm{VC}(\mathcal{F})\le d+1$, which corresponds to linear functions through the origin, $f(x)=\theta^Tx$).

The VC subgraph dimension provides a powerful, distribution-free way to bound the covering numbers of real-valued function classes.

Theorem 2.3.6 (Covering Numbers via VC Dimension). Let $\mathcal{F}$ be a class of real-valued functions with an **envelope** $F(x)$. Let $D=\mathrm{VC}(\mathcal{F})$ be the VC subgraph dimension. Then, for any probability measure Q and any $\varepsilon > 0$, the $L_2(Q)$ covering number is bounded by:

$$\sup_Q N(\mathcal{F}, \varepsilon\|F\|_{L_2(Q)}, L_2(Q)) \leq \left(\frac{C}{\varepsilon}\right)^{C_2 D}$$

for universal constants C, C_2.

Proof Sketch. The key idea is to relate the $L_2(Q)$ distance between two functions $f, g \in \mathcal{F}$ to the $L_2(\tilde{Q})$ distance between their corresponding subgraph indicators $h_f, h_g \in \mathcal{G}$.

(a). We can write $f(x) - g(x) = \int_{-F(x)}^{F(x)} (1_{\{t \leq f(x)\}} - 1_{\{t \leq g(x)\}}) dt$.

(b). Applying Cauchy-Schwarz and several inequalities leads to a bound of the form:

$$\|f - g\|_{L_2(Q)}^2 \leq C\|F\|_{L_2(Q)}^2 \cdot \|h_f - h_g\|_{L_2(\tilde{Q})}^2$$

where $d\tilde{Q}(x,t) = \frac{F(x)dtdQ(x)}{\iint F(x)dtdQ(x)}$ is a probability measure on the subgraph domain $\{(x,t) : |t| \leq F(x)\}$.

(c). This implies that an ε'-covering of the binary class $\mathcal{G}$ (under $L_2(\tilde{Q})$) provides an ε-covering of $\mathcal{F}$ (under $L_2(Q)$), where $\varepsilon \approx C'\|F\|_{L_2(Q)}\varepsilon'$.

(d). Since $\mathcal{G}$ is a binary class with VC dimension D, its covering number is bounded by $\mathbb{N}(\varepsilon', \mathcal{G}, L_2(\tilde{Q})) \leq (C''/\varepsilon')^{C_3 D}$.

(e). Substituting $\varepsilon' \approx \varepsilon/(C'\|F\|_{L_2(Q)})$ gives the final result.

□

For many non-parametric classes, the VC subgraph dimension is infinite. This motivates a scale-dependent measure of complexity.

Definition 2.3.7 (ε-Fat Shattering Dimension)**.** A set of points $\{x_1, \ldots, x_D\}$ is **ε-shattered** by $\mathcal{F}$ if there exist "thresholds" $t_1, \ldots, t_D$ such that for any binary sequence $(b_1, \ldots, b_D) \in \{0,1\}^D$, there exists a function $f \in \mathcal{F}$ that satisfies:

- $f(x_i) \geq t_i + \varepsilon$ if $b_i = 1$
- $f(x_i) < t_i$ if $b_i = 0$

(Note: Definitions may vary, e.g., $f(x_i) \leq t_i - \varepsilon$ for $b_i = 0$). The **ε-fat shattering dimension**, $\text{fat}_\varepsilon(\mathcal{F})$, is the largest integer D for which such an ε-shattered set exists.

This dimension measures the complexity of $\mathcal{F}$ at a specific scale ε.

Theorem 2.3.8 (Mendelson-Vershynin Theorem)**.** If $\mathcal{F}$ is uniformly bounded by 1:

$$\sup_Q M(\epsilon; \mathcal{F}, L^2(Q)) \leq \left(\frac{1}{\epsilon}\right)^{C \cdot fat_{c\epsilon}(\mathcal{F})}$$

By Dudley's chaining integral, we get:

$$\sup_{f \in \mathcal{F}} |(P_n - P)f| \leq \frac{C}{\sqrt{n}} \int_0^1 \sqrt{fat_{c\epsilon}(\mathcal{F}) \cdot \log(1/\epsilon)} d\epsilon$$

Theorem 2.3.9 (Rudelson-Vershynin Theorem)**.** Under mild assumptions, we have

$$\sup_Q M(\epsilon; \mathcal{F}, L^2(Q)) \le \exp\left(C \cdot fat_{c\epsilon}(\mathcal{F})\right)$$

This implies a tighter bound:

$$\sup_{f \in \mathcal{F}} |(P_n - P)f| \le \frac{C}{\sqrt{n}} \int_0^1 \sqrt{fat_{c\epsilon}(\mathcal{F})} d\epsilon$$

2.4 Lecture 8: Localization and Bayesian Estimators

2.4.1 Localication Techniques

Consider the standard M-estimation problem:

$$\hat{\theta}_n = \text{argmin}_{\theta \in \Theta} L_n(\theta), \quad \theta^* = \text{argmin}_{\theta \in \Theta} L(\theta)$$

where $L_n(\theta) = \frac{1}{n} \sum_{i=1}^n l(x_i; \theta)$. The standard global bound on the excess risk is derived via uniform convergence over the entire parameter space Θ:

$$L(\hat{\theta}_n) - L(\theta^*) \le 2 \sup_{\theta \in \Theta} |L_n(\theta) - L(\theta)| \lesssim \sqrt{\frac{VC(\mathcal{F})}{n}}$$

Remark. Taking the supremum over the entire space Θ is often too conservative. Intuitively, consistency implies that $\hat{\theta}_n$ eventually falls within a small neighborhood of θ^*. If we knew this a priori, we could restrict the complexity analysis to a local neighborhood $\Theta \cap \{\theta : \|\theta - \theta^*\| \le r\}$. This creates a "Chicken and Egg" problem:

(a). To improve the bound, we need to know $\hat{\theta}_n$ is close to θ^*.

(b). To show $\hat{\theta}_n$ is close to θ^*, we need a tight bound.

The solution is the method of **Localization** (or Peeling): we can derive tighter bounds by analyzing the *local modulus of continuity* of the empirical process.

Theorem 2.4.1 (Localization)**.** Assume that

(a). **Curvature Assumption (Strong Convexity-type condition):** Assume the population risk behaves quadratically near the optimum:

$$L(\theta) - L(\theta^*) \ge \|\theta - \theta^*\|^2$$

(b). **Local Modulus of Continuity:** Let $\phi_n(u)$ be an upper bound on the expected supremum of the empirical process restricted to a ball of radius u:

$$\mathbb{E}\left[\sup_{\theta \in \Theta, \|\theta - \theta^*\| \le u} |(P_n - P)(l_\theta - l_{\theta^*})|\right] \le \phi_n(u)$$

(c). **Sub-quadratic Growth:** Assume $\phi_n(u)$ does not grow too fast. Specifically, for some $\alpha < 2$:

$$\phi_n(cx) \le c^\alpha \phi_n(x) \quad \forall c > 1, x > 0$$

The critical radius δ_n is defined as the fixed point solution (or the smallest value) satisfying:

$$\phi_n(\delta_n) \le \delta_n^2$$

Under the assumptions above, for any $\epsilon > 0$, there exists a constant C_ϵ such that with probability at least $1-\epsilon$:

$$\|\hat{\theta}_n - \theta^*\| \leq C_\epsilon \cdot \delta_n$$

Proof. Using the definition of $\hat{\theta}_n$ and the curvature assumption:

$$\begin{aligned}
\|\hat{\theta}_n - \theta^*\|^2 &\leq L(\hat{\theta}_n) - L(\theta^*) \\
&= (L(\hat{\theta}_n) - L_n(\hat{\theta}_n)) + (L_n(\hat{\theta}_n) - L_n(\theta^*)) + (L_n(\theta^*) - L(\theta^*)) \\
&\leq (L(\hat{\theta}_n) - L_n(\hat{\theta}_n)) + 0 + (L_n(\theta^*) - L(\theta^*)) \\
&= (P_n - P)(l_{\theta^*} - l_{\hat{\theta}_n})
\end{aligned}$$

We wish to bound the probability $\mathbb{P}(\|\hat{\theta}_n - \theta^*\| \geq 2^M \delta_n)$ by decomposing the event into disjoint "layers":

$$\mathbb{P}(\|\hat{\theta}_n - \theta^*\| \geq 2^M \delta_n) = \sum_{j \geq M} \mathbb{P}\left(2^{j-1}\delta_n < \|\hat{\theta}_n - \theta^*\| \leq 2^j \delta_n\right)$$

For a specific shell where $\|\hat{\theta}_n - \theta^*\| \leq 2^j \delta_n$, the error is bounded by the supremum over the ball of radius $u = 2^j \delta_n$. Also, from the first inequality, if $\|\hat{\theta}_n - \theta^*\| > 2^{j-1}\delta_n$, then the process value must be large:

$$(P_n - P)(l_{\theta^*} - l_{\hat{\theta}_n}) \geq \|\hat{\theta}_n - \theta^*\|^2 > (2^{j-1}\delta_n)^2$$

By Markov's inequality and the definition of ϕ_n:

$$\begin{aligned}
\text{Prob of shell } j &\leq \mathbb{P}\left(\sup_{\|\theta - \theta^*\| \leq 2^j \delta_n} |(P_n - P)(l_{\theta^*} - l_\theta)| \geq 2^{2j-2}\delta_n^2\right) \\
&\leq \frac{\phi_n(2^j \delta_n)}{2^{2j-2}\delta_n^2}
\end{aligned}$$

Using the sub-quadratic growth assumption $\phi_n(cx) \leq c^\alpha \phi_n(x)$:

$$\phi_n(2^j \delta_n) \leq 2^{j\alpha}\phi_n(\delta_n) \leq 2^{j\alpha}\delta_n^2 \quad (\text{since } \phi_n(\delta_n) \leq \delta_n^2)$$

Substituting back:

$$\text{Prob} \leq \frac{2^{j\alpha}\delta_n^2}{2^{2j-2}\delta_n^2} = 4 \cdot 2^{(\alpha-2)j}$$

Since $\alpha < 2$, the term $(\alpha - 2)$ is negative, and the sum over j forms a convergent geometric series. By choosing M large enough, we can make the total probability arbitrarily small ($\leq \epsilon$). □

Example. Let's apply this to a standard parametric class $\Theta = \mathbb{R}^d$. Assume the loss function is Lipschitz with respect to θ:

$$|l_{\theta_1}(x) - l_{\theta_2}(x)| \leq M(x)\|\theta_1 - \theta_2\|_2$$

where $M(x)$ is an envelope function with finite second moment $\|M\|_{L^2(P)}$. We analyze the local class $\mathcal{F}_u = \{l_\theta - l_{\theta^*} : \|\theta - \theta^*\| \leq u\}$.
Note that the envelope of $\mathcal{F}_u$ is $F_u(x) = u \cdot M(x)$. The covering number of the parameter space (a ball of radius u in $\mathbb{R}^d$) scales with dimension d:

$$\log N(\epsilon \|F_u\|; \mathcal{F}_u, \|\cdot\|) \approx \log\left(\frac{u}{\epsilon u}\right)^d = d \log\left(\frac{1}{\epsilon}\right)$$

By Dudley's entropy integral, the local modulus $\phi_n(u)$ is bounded by:

$$\phi_n(u) \leq C\frac{\|F_u\|_{L^2}}{\sqrt{n}} \int_0^1 \sqrt{\log N(\dots)} d\epsilon$$
$$= C\frac{u\|M\|_{L^2}}{\sqrt{n}} \int_0^1 \sqrt{d \log(1/\epsilon)} d\epsilon$$

The integral $\int_0^1 \sqrt{\log(1/\epsilon)} d\epsilon$ is a constant. Thus,

$$\phi_n(u) \leq C' \sqrt{\frac{d}{n}} \|M\|_{L^2(P)} \cdot u$$

We then solve for δ_n such that $\phi_n(\delta_n) = \delta_n^2$:

$$C' \sqrt{\frac{d}{n}} \delta_n = \delta_n^2 \implies \delta_n = C' \sqrt{\frac{d}{n}}$$

The estimation error bound is then

$$\|\hat{\theta}_n - \theta^*\| = O_p\left(\sqrt{\frac{d}{n}}\right)$$

This recovers the standard parametric rate of convergence, which is significantly faster than the slow rates (e.g., $n^{-1/4}$ or slower) that might be implied by loose global bounds without curvature assumptions.

Example. We now consider a scenario where the Lipschitz condition fails or the geometry of the problem changes, leading to a slower rate of convergence ($n^{-1/3}$).
Consider the estimator defined by maximizing the probability mass captured by a fixed-width interval. This corresponds to minimizing the loss:

$$f_\theta(x) = -\mathbb{1}_{\{x \in [\theta - 1, \theta + 1]\}}$$

- Population Risk: $Pf_\theta = -\mathbb{P}(X \in [\theta - 1, \theta + 1])$.
- Empirical Risk: $P_n f_\theta = -\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \in [\theta - 1, \theta + 1]\}$.

Assume the density $p(x)$ is continuously differentiable. We analyze the second derivative of the population risk at the optimum θ^*:

$$\left.\frac{d^2}{d\theta^2} Pf_\theta \right|_{\theta^*} = p'(\theta^* - 1) - p'(\theta^* + 1)$$

Assuming this quantity is strictly positive, the risk is locally quadratic:

$$Pf_\theta - Pf_{\theta^*} \geq C_0 |\theta - \theta^*|^2 \quad \text{for } \theta \in (\theta^* - \delta_0, \theta^* + \delta_0)$$

By consistency, for large n, we can restrict our attention to this local interval. We must bound the expectation of the supremum over the local class $\mathcal{F}_r = \{f_\theta - f_{\theta^*} : |\theta - \theta^*| \leq r\}$.
1. Envelope Function: The difference $|f_\theta(x) - f_{\theta^*}(x)|$ is non-zero only when x falls in the symmetric difference of the intervals $[\theta - 1, \theta + 1]$ and $[\theta^* - 1, \theta^* + 1]$. This region consists of two small strips of width at most r near the endpoints. We define the envelope $G(x)$:

$$G(x) = \mathbb{1}\{x \in [\theta^* + 1 - r, \theta^* + 1 + r] \cup [\theta^* - 1 - r, \theta^* - 1 + r]\}$$

2. Second Moment of the Envelope: Crucially, the variance scales linearly with r, not r^2 (unlike the Lipschitz case):

$$\mathbb{E}[G(x)^2] = \mathbb{P}(X \in \text{small strips}) \leq 4p_{\max}r$$

3. VC Dimension: The class of intervals has VC dimension ≤ 2. The local class $\mathcal{F}_r$ has VC dimension ≤ 8. Thus, the entropy integral is bounded by a constant.

4. Modulus of Continuity: Using Dudley's entropy bound:

$$\mathbb{E}\left[\sup_{\mathcal{F}_r} |(P_n - P)(f_\theta - f_{\theta^*})|\right] \leq C\sqrt{\frac{\mathbb{E}[G^2]}{n}} \int_0^1 \sqrt{\log N(\dots)}d\delta$$

Substituting the bound for $\mathbb{E}[G^2]$:

$$\phi_n(r) \leq C\sqrt{\frac{p_{\max}r}{n}}$$

By solving the fixed point equation $\delta_n^2 = \phi_n(\delta_n)$, we have $\delta_n \approx n^{-1/3}$.

Theorem 2.4.2 (Cube Root Rate)**.** For the interval location estimator, under the assumed curvature conditions:

$$|\hat{\theta}_n - \theta^*| \leq O_p\left(n^{-1/3}\right)$$

Remark. Since the convergence rate is $n^{-1/3}$, we look for a limiting distribution of $n^{1/3}(\hat{\theta}_n - \theta^*)$. Indeed, it is given by:

$$n^{1/3}(\hat{\theta}_n - \theta^*) \xrightarrow{d} \operatorname{argmax}_{t\in\mathbb{R}}\{\mathbb{G}(t) - ct^2\}$$

- $\mathbb{G}$ is some Gaussian Process originating from zero.
- ct^2 is a drift term representing the quadratic curvature of the population risk.
- The limit is **non-Gaussian**. It is the location of the maximum of a Brownian motion with quadratic drift.

2.4.2 Convergence of Bayesian Estimators

In the Bayesian framework, we observe data $X_1, \dots, X_n \overset{i.i.d.}{\sim} P_{\theta^*}$ given a prior distribution π on the parameter space Θ. The inference is based on the posterior distribution:

$$\pi(\theta|X^n) = \frac{\pi(\theta)\prod_{i=1}^n p_\theta(X_i)}{\int_\Theta \pi(\vartheta)\prod_{i=1}^n p_\vartheta(X_i)d\vartheta}$$

We are interested in three main asymptotic properties:

(a). **Posterior Consistency:** Does the posterior mass concentrate around the true parameter θ^*?

$$\forall \epsilon > 0, \quad \pi(\{\theta : \|\theta - \theta^*\| > \epsilon\} \mid X^n) \xrightarrow{P} 0$$

(b). **Contraction Rate:** How fast does the mass concentrate? We look for a sequence $\epsilon_n \to 0$ such that with high probability:

$$\pi(\{\theta : \|\theta - \theta^*\| \geq \epsilon_n\} \mid X^n) \leq \delta$$

(c). **Asymptotic Posterior Shape:** Does the posterior converge to a specific distribution?

$$d_{TV}(\pi(\cdot|X^n), \mathcal{D}) \xrightarrow{P} 0$$

Schwartz's Theorem provides sufficient conditions for posterior consistency.

Theorem 2.4.3 (Schwartz). Posterior consistency holds if the following two conditions are met:

(a). **KL Support (Prior Condition):** The prior places positive mass on any Kullback-Leibler (KL) neighborhood of the true parameter θ^*.

$$\forall \epsilon > 0, \quad \pi(\{\theta : D_{KL}(P_{\theta^*}||P_\theta) \leq \epsilon\}) > 0$$

(b). **Existence of Tests (Testing Condition):** For every $\epsilon > 0$, there exists a sequence of tests ϕ_n (functions of $X^n \to \{0,1\}$) that can distinguish θ^* from the complement of the neighborhood $\{\theta : \|\theta - \theta^*\| \geq \epsilon\}$:

$$\mathbb{E}_{\theta^*}[\phi_n] \to 0 \quad \text{and} \quad \sup_{\|\theta-\theta^*\|\geq\epsilon} \mathbb{E}_\theta[1-\phi_n] \to 0$$

Remark. The sequence of tests ϕ_n is purely a theoretical device used to prove the mass in the complement vanishes. The actual Bayesian algorithm does not need to know these tests.

Proof. The proof relies on bounding the posterior probability of the complement set $U^c = \{\theta : \|\theta - \theta^*\| > \epsilon\}$.

$$\pi(U^c|X^n) = \frac{\int_{U^c} \prod \frac{p_\theta}{p_{\theta^*}}(X_i)\pi(\theta)d\theta}{\int_\Theta \prod \frac{p_\theta}{p_{\theta^*}}(X_i)\pi(\theta)d\theta} = \frac{\text{Numerator}}{\text{Denominator}}$$

We first construct tests with exponentially decaying error probabilities:

(a). Assume there exists a test ϕ_{n_0} for a fixed sample size n_0 with Type I and Type II errors bounded by $1/4$.

(b). For large n, divide the data into $K = n/n_0$ independent blocks.

(c). Apply the base test to each block to get outcomes $Y_1, \ldots, Y_K$.

(d). Define the boosted test $\tilde{\phi}_n$ by a majority vote:

$$\tilde{\phi}_n(X^n) = \mathbb{1}\left\{\frac{1}{K}\sum_{j=1}^{K} Y_j > \frac{1}{2}\right\}$$

(e). By Hoeffding's inequality, the errors of $\tilde{\phi}_n$ decay exponentially:

$$\mathbb{E}_{\theta^*}[\tilde{\phi}_n] \leq e^{-cn}, \quad \sup_{\theta\in U^c} \mathbb{E}_\theta[1-\tilde{\phi}_n] \leq e^{-cn}$$

We use the tests $\tilde{\phi}_n$ to control the numerator. Let $R_n(\theta) = \prod \frac{p_\theta(X_i)}{p_{\theta^*}(X_i)}$, then

$$\text{Numerator} = \int_{U^c} R_n(\theta)\pi(\theta)d\theta$$

Since $\mathbb{E}_\theta^*[\phi_n]$ converges to zero, we can incorporate the test into the posterior probability:

$$\begin{aligned}
\mathbb{E}_{\theta^*}\left[(1-\tilde{\phi}_n)\int_{U^c} R_n(\theta)\pi(\theta)d\theta\right] &= \int_{U^c} \mathbb{E}_{\theta^*}[(1-\tilde{\phi}_n)R_n(\theta)]\pi(\theta)d\theta \\
&= \int_{U^c} \mathbb{E}_\theta[1-\tilde{\phi}_n(X^n)]\pi(\theta)d\theta \\
&\leq \sup_{\theta\in U^c} \mathbb{E}_\theta[1-\tilde{\phi}_n]\cdot\pi(U^c) \\
&\leq \exp(-cn)
\end{aligned}$$

Thus, the numerator is exponentially small with high probability.

To lower-bound the denominator, it suffices to focus on a KL-neighborhood $\Theta_0 = \{\theta : D_{KL}(P_{\theta^*}||P_\theta) \le \epsilon\}$. The log-likelihood ratio behaves according to the Law of Large Numbers:

$$\frac{1}{n}\sum_{i=1}^{n}\log\frac{p_{\theta^*}(X_i)}{p_\theta(X_i)} \xrightarrow{P} \mathbb{E}_{\theta^*}\left[\log\frac{p_{\theta^*}}{p_\theta}\right] = D_{KL}(P_{\theta^*}||P_\theta)$$

Therefore,

$$\frac{1}{n}\log\prod_{i=1}^{n}\frac{p_\theta(X_i)}{p_{\theta^*}(X_i)} \approx -D_{KL}(P_{\theta^*}||P_\theta) \ge -\epsilon \quad \text{for } \theta \in \Theta_0$$

Using the assumption on the prior, we have

$$\text{Denominator} \ge \int_{\Theta_0} e^{-n(D_{KL}+\text{small})}\pi(\theta)d\theta \ge e^{-2n\epsilon}\pi(\Theta_0)$$

Combining the bounds on the numerator and denominator, we have

$$\pi(U^c|X^n) \le \frac{\exp(-cn)}{\pi(\Theta_0)\exp(-2n\epsilon)}$$

By choosing ϵ small enough such that $2\epsilon < c$, the ratio goes to 0 as $n \to \infty$. □

The Bernstein-von Mises (BvM) theorem describes the asymptotic shape of the posterior distribution.

Theorem 2.4.4 (Bernstein-von Mises). Assuming sufficient regularity conditions (similar to those required for the asymptotic normality of the MLE), the posterior distribution converges in total variation to a normal distribution centered at the MLE $\hat{\theta}_n$ with covariance equal to the inverse Fisher information:

$$d_{TV}\left(\pi(\cdot|X^n), \mathcal{N}(\hat{\theta}_n, (nI(\theta^*))^{-1})\right) \xrightarrow{P} 0$$

Remark. This theorem justifies the use of Bayesian credible intervals as frequentist confidence intervals in large samples for regular models. In irregular cases (e.g., boundaries), the limit may be different (e.g., functionals of Brownian motion).

Chapter 3 Nonparametric Estimation

3.1 Lecture 9: Gaussian Complexity and its (Sub)optimality

3.1.1 Nonparametric Estimation and Gaussian Complexity

Given samples $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} p^* \in \mathcal{P}$, the goal is to estimate the true density p^* under a specific metric based on the observed pairs $(X_i, Y_i)_{i=1}^n$. The target function is the conditional expectation:

$$f^*(x) = \mathbb{E}[Y_i|X_i = x], \quad \text{where } f^* \in \mathcal{F}.$$

To simplify, we consider the **fixed design** setting where $x_1, \ldots, x_n$ are deterministic. The model is:

$$Y_i = f^*(x_i) + \varepsilon_i, \quad \varepsilon_i \overset{i.i.d.}{\sim} \mathcal{N}(0, 1).$$

This is essentially an n-dimensional vector recovery problem.

The Maximum Likelihood Estimator (MLE) under Gaussian noise is equivalent to the Least Squares estimator:

$$\hat{f}_n := \underset{f \in \mathcal{F}}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - f(x_i))^2 \right\}$$

> **Remark.** This optimization is efficiently computable when the function class $\mathcal{F}$ is convex. In literature, convexity can sometimes be relaxed to star-convexity.

Let $Y = (Y_i)_{i=1}^n$ be the observation vector in $\mathbb{R}^n$. The set of possible values for f on the design points is $\mathcal{F}([x_i])$. The estimator $\hat{f}_n$ is the projection of Y onto this convex set. The **First-order optimality condition** states:

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{f}_n(x_i))(f'(x_i) - \hat{f}_n(x_i)) \leq 0, \quad \forall f' \in \mathcal{F}.$$

Geometrically, the angle between the residual vector and any vector pointing into the set from $\hat{f}_n$ is obtuse.

Substituting f' with f^* in the optimality condition and rearranging terms, we derive the "Basic Inequality". Starting from:

$$\frac{1}{n} \left\langle Y - \hat{f}_n, f^* - \hat{f}_n \right\rangle \leq 0$$

Substituting $Y_i = f^*(x_i) + \varepsilon_i$, we get:

$$\frac{1}{n} \sum_{i=1}^n \left[(f^* - \hat{f}_n)(x_i) \right]^2 \leq \frac{1}{n} \sum_{i=1}^n (\hat{f}_n - f^*)(x_i) \cdot \varepsilon_i$$

Let $\hat{\Delta}_n = \hat{f}_n - f^*$ and define the empirical norm $\|f\|_n = \sqrt{\frac{1}{n} \sum f(x_i)^2}$. We have:

$$\|\hat{\Delta}_n\|_n^2 \leq \langle \hat{\Delta}_n, \varepsilon \rangle_n$$

Remark. Applying Cauchy-Schwarz ($\langle \hat{\Delta}_n, \varepsilon \rangle_n \le \|\hat{\Delta}_n\|_n \|\varepsilon\|_n$) leads to a naive bound because $\|\varepsilon\|_n = O(1)$, which does not vanish as $n \to \infty$.

To improve the bound, we utilize the fact that $\hat{\Delta}_n \in \mathcal{F}^*$, where $\mathcal{F}^* = \{f - f^* : f \in \mathcal{F}\}$. We bound the inner product uniformly over the class:

$$\langle \hat{\Delta}_n, \varepsilon \rangle_n \le \sup_{h \in \mathcal{F}^*, \|h\|_n \le \|\hat{\Delta}_n\|_n} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(x_i)$$

Definition 3.1.1 (Gaussian Complexity). Let $r > 0$. The **Gaussian complexity** of a the function class $\mathcal{F}^*$ is defined as

$$G_n(r) := \mathbb{E}\left[\sup_{h \in \mathcal{F}^*, \|h\|_n \le r} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(x_i) \right| \right]$$

where ε_i are i.i.d. standard normal random variables.

We face a "chicken-and-egg" problem: we are bounding $\|\hat{\Delta}_n\|_n$ using a quantity that depends on $\|\hat{\Delta}_n\|_n$. Similar to what we have done in the Rademader case, we also have the following localization bound on the error:

Theorem 3.1.2 (Critical Radius Bound). Suppose $G_n(r) \le \phi_n(r)$ such that the growth condition holds:

$$\phi_n(cr) \le c^\alpha \phi_n(r) \quad \text{for some } \alpha < 2.$$

If δ_n solves the fixed point equation $\delta_n^2 = \phi_n(\delta_n)$, then with high probability:

$$\|\hat{f}_n - f^*\|_n \le C \cdot \delta_n$$

Proof. We start from the bounds

$$\|\hat{\Delta}_n\|_n^2 \le \langle \hat{\Delta}_n, \varepsilon \rangle_n \le \sup_{h \in \mathcal{F}^*, \|h\|_n \le \|\hat{\Delta}_n\|_n} \frac{1}{n} \sum_{i=1}^n \varepsilon_i h(x_i)$$

and apply a peeling argument. Note that for any M,

$$\mathbb{P}\left(\|\hat{\Delta}_n\|_n > 2^M \delta_n\right) = \sum_{j \ge M} \mathbb{P}\left(2^{j+1}\delta_n \ge \|\hat{\Delta}_n\|_n > 2^j \delta_n\right) \le \sum_{j \ge M} \frac{G_n(2^{j+1}\delta_n)}{2^{2j}\delta_n^2} \le \delta^{\alpha-2} \sum_{j \ge M} 2^{(\alpha-2)j}$$

Note that $\alpha < 2$, we can pick M large to make this probability sufficiently small and conclude. $\square$

Remark. We can easily verify the growth condition on local Gaussian complexity if we assume that the function class $\mathcal{F}$ is convex (can be weakened to star-convexity), which is usually the case in nonparametric estimation. By convexity (or star-convexity), for $f \in \mathcal{F}^* \cap B(cr)$, we have $\frac{f}{c} \in \mathcal{F}^* \cap B(r)$. This implies:

$$\mathbb{E}\left[\sup_{f \in \mathcal{F}^* \cap B(cr)} \langle \varepsilon, f \rangle_n \right] \le c\mathbb{E}\left[\sup_{f \in \mathcal{F}^* \cap B(r)} \langle \varepsilon, f \rangle_n \right]$$

Thus, Gaussian complexity grows sub-linearly ($\alpha \le 1$), satisfying the growth condition.

To bound the Gaussian complexity, note that the moment generating function of a Rademacher variable is bounded by the MGF of $\mathcal{N}(0,1)$. Therefore, we can follow the same lines of previous proofs for Rademacher complexity to prove Dudley's entropy integral bound on Gaussian complexity.

Theorem 3.1.3 (Dudley's entropy integral, the Gaussian case)**.** Let $r > 0$, then we have

$$G_n(r) \leq \frac{C}{\sqrt{n}} \int_0^r \sqrt{\log N(\delta; \mathcal{F}^* \cap B_n(r), \|\cdot\|_n)}\, d\delta$$

Example. Consider the class of one-dimensional Lipschitz functions:

$$\mathcal{F} = \{f : [0,1] \to [0,1] \mid |f(x) - f(y)| \leq |x - y|, \forall x, y\}.$$

Using Rudelson-Vershynin results involving the fat-shattering dimension ($fat_{c\delta}$):

$$N(\delta; \mathcal{F}^* \cap B_n(r), \|\cdot\|_n) \leq N(\delta; \tilde{\mathcal{F}}, \|\cdot\|_n) \leq \exp\left(c \cdot fat_{c\delta}(\tilde{\mathcal{F}})\right).$$

From previous results, we know $fat_{c\delta}(\tilde{\mathcal{F}}) \leq c'/\delta$. Applying this to the integral:

$$G_n(r) \leq \frac{c}{\sqrt{n}} \int_0^r \sqrt{\frac{c'}{\delta}}\, d\delta \leq c\sqrt{\frac{r}{n}}.$$

To find the critical radius r_n, we solve for the fixed point $r^2 \sim G_n(r)$:

$$r^2 = c\sqrt{\frac{r}{n}} \implies r^{3/2} \sim n^{-1/2} \implies r_n \asymp n^{-1/3}.$$

Thus, with high probability (w.h.p.):

$$\|\hat{f}_n - f^*\|_n \lesssim n^{-1/3}.$$

Definition 3.1.4. Let $\beta > 0$ with $\beta = k + \gamma$, where $k \in \mathbb{N}$ and $\gamma \in (0,1]$. The Hölder class $\Sigma(\beta, L)$ is defined as:

$$\Sigma(\beta, L) := \left\{ f : [0,1]^d \to [0,1] \;\middle|\; \begin{array}{l} \forall \alpha \text{ (multi-index) s.t. } |\alpha| = k, \\ |\partial^\alpha f(x) - \partial^\alpha f(y)| \leq L\|x - y\|^\gamma, \\ \forall x, y \in [0,1]^d \end{array} \right\}$$

where a multi-index $\alpha = (\alpha_1, \ldots, \alpha_d)$ has magnitude $|\alpha| = \sum \alpha_i$, and $\partial^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \ldots \partial x_d^{\alpha_d}}$.

Theorem 3.1.5 (Metric Entropy of Hölder Classes)**.** For any finitely-supported measure Q, we have

$$\log N(\epsilon; \Sigma(\beta, L), \|\cdot\|_{L_2(Q)}) \leq c\left(\frac{1}{\epsilon}\right)^{d/\beta}.$$

where the constants c_1, c_2 may depend on (d, β).

Remark. Substituting the entropy bound into the complexity integral:

$$G_n(r) \leq \frac{c}{\sqrt{n}} \int_0^r \left(\frac{1}{\delta}\right)^{\frac{d}{2\beta}} d\delta.$$

The behavior of this integral depends heavily on the exponent $\frac{d}{2\beta}$:

Case 1: $\beta > d/2$ **(The "Donsker" Regime)** The integral converges at 0. Picking $r_0 = 0$:

$$G_n(r) \leq \frac{c}{\sqrt{n}} \int_0^r \delta^{-\frac{d}{2\beta}} \, d\delta \leq \frac{c'}{\sqrt{n}} r^{1-\frac{d}{2\beta}}.$$

Solving for the rate $r_n^2 = G_n(r_n)$:

$$r_n^2 = \frac{r_n^{1-\frac{d}{2\beta}}}{\sqrt{n}} \implies r_n^{1+\frac{d}{2\beta}} = n^{-1/2} \implies r_n \asymp n^{-\frac{\beta}{d+2\beta}}.$$

This is the **optimal rate**.

Case 2: $\beta < d/2$ **(The "Non-Donsker" Regime)** The integral $\int_0^r \delta^{-d/2\beta} d\delta$ diverges as $\delta \to 0$. To avoid divergence, we use the trivial bound $N(\delta) \leq (c/\delta)^n$ for very small δ. We split the integral at a truncation point r_0:

$$G_n(r) \leq r_0 + \frac{c}{\sqrt{n}} \int_{r_0}^r \delta^{-\frac{d}{2\beta}} \, d\delta \leq r_0 + \frac{c}{\sqrt{n}} \left(r_0^{1-\frac{d}{2\beta}} - r^{1-\frac{d}{2\beta}} \right).$$

Dominant term approximation:

$$G_n(r) \leq r_0 + \frac{c}{\sqrt{n}} r_0^{1-\frac{d}{2\beta}}.$$

We choose $r_0 = n^{-\beta/d}$ to balance terms, yielding:

$$G_n(r) \leq c \cdot n^{-\beta/d}.$$

Solving for the rate $r_n^2 = G_n(r_n)$:

$$r_n^2 = n^{-\beta/d} \implies r_n = n^{-\frac{\beta}{2d}}.$$

Remark. This rate is **suboptimal**.

Case 3: $\beta = d/2$ **(Boundary Case)** Using the split as in Case 2, we have

$$G_n(r) \leq r_0 + \frac{c}{\sqrt{n}} \int_{r_0}^r \delta^{-\frac{d}{2\beta}} \, d\delta \leq r_0 + \frac{c}{\sqrt{n}} (\log r - \log r_0).$$

Choosing $r_0 = n^{-1/2}$ gives

$$G_n(r) \leq c \cdot \frac{r \log n}{\sqrt{n}}$$

When n is large, we solve for the rate $r_n^2 = G_n(r_n)$ and obtain logarithmic factors:

$$r_n = (\log n)^{1/2} \cdot n^{-1/4}.$$

3.1.2 Sub-optimality of the Entropy Bound

If we plot the rate exponent versus β:

- For $\beta > d/2$: The rate is $n^{-\frac{\beta}{d+2\beta}}$, which is optimal.
- For $\beta \leq d/2$: The rate achieved by standard methods derived here is $n^{-\frac{\beta}{2d}}$ that is suboptimal.

Remark. The regime $\beta \leq d/2$ is often referred to as the **Non-Donsker** regime. Divergent entropy integrals are usually a strong indicator of the suboptimality of the least squares method.

Remark.
- The **optimal rate** for Hölder classes is *always* $n^{-\frac{\beta}{d+2\beta}}$.
- For Hölder classes, (constrained) least squares methods often fail to achieve the optimal rate in the Non-Donsker regime. Conversely, methods such as projection or local polynomials lead to optimal rates.
- The construction of optimal estimators in the Non-Donsker regime can be ad-hoc.

We consider the estimation of a function f^* belonging to a Sobolev class $\mathcal{F} = S(\beta, L)$ with smoothness parameter $\beta \in \mathbb{N}^+$ and radius $L > 0$:

Definition 3.1.6. The Sobolev class $S(\beta, L)$ is defined by the bound on the L^2 norm of the derivatives:

$$S(\beta, L) = \left\{ f : [0,1]^d \to \mathbb{R} \;\middle|\; \sum_{|\alpha| \le \beta} \int_{[0,1]^d} |\partial^\alpha f(x)|^2 dx \le L^2 \right\}$$

Remark. Let $\{\varphi_j\}$ be a Fourier basis in $[0,1]^d$. The smoothness condition can be characterized by the decay of Fourier coefficients:

$$\sum_{z \in \mathbb{Z}^d} \langle f, \varphi_z \rangle_{L^2}^2 \cdot |z|^{2\beta} \le L^2$$

We observe the function on a discrete grid: the design points $x_\alpha = \frac{\alpha}{m}$, where $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d) \in [m]^d$ and the total sample size is $n = m^d$.

Let $\varphi_1, \varphi_2, \dots, \varphi_n$ denote the "Discrete Fourier Basis," which forms an orthonormal basis of $(\mathbb{R}^n, \|\cdot\|_n)$. The target function class $\mathcal{F}$ is defined in terms of discrete coefficients c_j:

$$\mathcal{F} = \left\{ f = \sum_{j=1}^n c_j \varphi_j \;\middle|\; \sum_{j=1}^n c_j^2 \cdot j^{2\beta/d} \le L^2 \right\}$$

We know that the constrained least squares is suboptimal for $\mathcal{F}$ when $\beta < d/2$. Instead, we use a truncation strategy in the Fourier domain.

(a). Observe data vector $Y \in \mathbb{R}^n$.

(b). Compute empirical Fourier coefficients:

$$\hat{c}_j = \langle Y, \varphi_j \rangle_n \quad \text{for } j = 1, \dots, n$$

(c). Construct the estimator by keeping only the first N frequencies ($N < n$):

$$\hat{f}_n = \sum_{j=1}^N \hat{c}_j \varphi_j$$

We analyze the Mean Squared Error (MSE) using the bias-variance decomposition. Let c_j^* be the true coefficients of f^*.

$$\mathbb{E}\left[\|\hat{f}_n - f^*\|_n^2\right] = \frac{1}{n}\sum_{j=1}^N \mathbb{E}\left[|\hat{c}_j - c_j^*|^2\right] + \frac{1}{n}\sum_{j=N+1}^n (c_j^*)^2 = \underbrace{\frac{N}{n}}_{\text{Variance}} + \underbrace{\sum_{j=N+1}^n (c_j^*)^2}_{\text{Squared Bias}}$$

We bound the bias term using the Sobolev condition :

$$\sum_{j=N+1}^n (c_j^*)^2 = \sum_{j=N+1}^n \frac{j^{2\beta/d}}{j^{2\beta/d}} (c_j^*)^2 \le \frac{1}{(N+1)^{2\beta/d}} \underbrace{\sum_{j=N+1}^n j^{2\beta/d} (c_j^*)^2}_{\le L^2} \le \frac{L^2}{N^{2\beta/d}}$$

Thus, the total risk bound is:

$$\mathbb{E}\left[\|\hat{f}_n - f^*\|_n^2\right] \leq \frac{N}{n} + \frac{L^2}{N^{2\beta/d}}$$

To find the optimal cutoff N, we balance the variance and bias terms:

$$\frac{N}{n} \approx \frac{1}{N^{2\beta/d}} \implies N^{1+\frac{2\beta}{d}} \approx n$$

Solving for N, we have $N = n^{\frac{d}{2\beta+d}}$. Plugging this choice of N back into the risk bound yields the convergence rate r_n:

$$\text{MSE Rate} \leq \frac{n^{\frac{d}{2\beta+d}}}{n} = n^{\frac{d-(2\beta+d)}{2\beta+d}} = n^{-\frac{2\beta}{2\beta+d}}$$

Therefore, the rate of convergence (for the squared error) is: $r_n = n^{-\frac{2\beta}{2\beta+d}}$.

Question: How do we relate the empirical risk $\frac{1}{n}\sum(\hat{f}_n - f^*)(X_i)^2$ to the population risk $\left\|\hat{f}_n - f^*\right\|_{L^2(\mathbb{P})}^2$?

We define a local process modulus $Z_n(r)$ to bound the difference between the empirical and population norms:

$$Z_n(r) := \mathbb{E}\left[\sup_{\substack{h\in\mathcal{F}^* \\ \|h\|_{L^2(\mathbb{P})}\leq r}} \left|(P_n - P)h^2\right|\right]$$

where $\mathcal{F}^* = \{f - f^* : f \in \mathcal{F}\}$ is the error class (shifted function class).

Expanding this definition:

$$\begin{aligned} Z_n(r) &= \mathbb{E}\left[\sup\left|\frac{1}{n}\sum_{i=1}^n h(X_i)^2 - \mathbb{E}[h(X)^2]\right|\right] \\ &\leq 2\cdot\mathbb{E}\left[\sup\left|\frac{1}{n}\sum_{i=1}^n \xi_i h(X_i)^2\right|\right] \quad \text{(via Symmetrization)} \end{aligned}$$

where ξ_i are independent Rademacher random variables.

Once we can bound $Z_n(r)$, we have:

$$r_n^2 = \left\|\hat{f}_n - f^*\right\|_{L^2(\mathbb{P})}^2 \leq \frac{1}{n}\sum_{i=1}^n \varepsilon_i(\hat{f}_n - f^*)(X_i) + Z_n(\left\|\hat{f}_n - f^*\right\|_{L^2})$$

Bounding the noise term using the suprema over the class:

$$\leq \mathbb{E}\left[\sup_{\substack{h\in\mathcal{F} \\ \|h\|_2\leq r_n}} \left|\frac{1}{n}\sum \varepsilon_i h(X_i)\right|\right] + Z_n(r_n)$$

This holds with high probability (w.h.p.).

To handle the h^2 term inside the supremum, we utilize the Lipschitz property of the square function. Assume that the function class is uniformly bounded:

$$\forall f \in \mathcal{F}, \quad \|f\|_\infty \leq M$$

The mapping $x \mapsto x^2$ is $2M$-Lipschitz on the domain $[-M, M]$. Using the **Contraction Principle** (Ledoux-Talagrand), we have:

$$\mathbb{E}\left[\sup_{h\in\mathcal{F}^*(r)}\left|\frac{1}{n}\sum_{i=1}^n \xi_i h^2(X_i)\right|\right] \leq 2M\cdot\mathbb{E}\left[\sup_{h\in\mathcal{F}^*(r)}\left|\frac{1}{n}\sum_{i=1}^n \xi_i h(X_i)\right|\right]$$

Applying previous arguments yields specific bounds for the estimator.

Remark.
- **Concentration:** To improve the failure probability bounds, one should use concentration inequalities for the supremum of empirical processes (e.g., Talagrand's concentration inequality).
- **Loose Bounds:** The contraction-based bound derived here is **not tight**.
- **Small Noise Regime:** Consider the case where:

$$Y_i = f^*(X_i) + \varepsilon_i, \quad \varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

  When σ is small (high signal, low noise), or even when $\sigma = 0$, we expect *faster rates* of convergence. However, the contraction arguments above do not achieve these faster rates.
- **Reference:** To achieve tight bounds in these regimes, one must use geometry-dependent arguments, such as the **"Small Ball Method"** (cf. Mendelson, 2012).

3.2 Lecture 10: Kernel Density Estimation

3.2.1 Standard Kerner Density Estimators

Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} p^*$ on $\mathbb{R}$. We wish to estimate the density p^*. Common approaches include:

- **MLE:** $\hat{p} = \text{argmax}_{p \in \mathcal{P}} \frac{1}{n} \sum \log p(X_i)$. (Requires empirical process theory).
- **Skeleton Estimators:** Theoretically nice (Scheffé sets), but computationally infeasible.
- **Local Methods:** Smoothing via kernels.

The Kernel Density Estimator is defined as:

$$\hat{p}_n(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)$$

where K is a kernel function satisfying $\int_{\mathbb{R}} K(x)dx = 1$, and $h > 0$ is the bandwidth.

Consider a fixed point x_0 and a standard kernel (e.g., boxcar $K = \frac{1}{2}\mathbb{1}_{[-1,1]}$).

$$\begin{aligned}
\text{Var}(\hat{p}_n(x_0)) &= \frac{1}{h^2 n} \text{Var}\left(K\left(\frac{X - x_0}{h}\right)\right) \\
&\leq \frac{1}{h^2 n} \int K^2\left(\frac{y - x_0}{h}\right) p(y)dy \\
&\approx \frac{p(x_0)}{nh} \int K^2(u)du
\end{aligned}$$

Thus, the variance is of order $O(\frac{1}{nh})$.

Assume p is Hölder continuous with exponent $\beta \in (0, 1]$ and constant L: $|p(x) - p(y)| \leq L|x - y|^\beta$.

$$\mathbb{E}[\hat{p}_n(x_0)] - p(x_0) = \frac{1}{h} \int K\left(\frac{y - x_0}{h}\right) (p(y) - p(x_0))dy$$

By change of variables $y = x_0 + uh$:

$$\text{Bias}(x_0) = \int K(u)(p(x_0 + uh) - p(x_0))du \leq Lh^\beta \int |K(u)| \cdot |u|^\beta du$$

Thus, $|\text{Bias}(x_0)| \leq C \cdot h^\beta$.

The Mean Squared Error (MSE) is then bounded by:

$$\text{MSE}(x_0) \leq \frac{C_1}{nh} + C_2 h^{2\beta}$$

Optimizing with respect to h, we know that **optimal bandwidth** $= h_n \asymp n^{-\frac{1}{2\beta+1}}$ and the **optimal MSE** $\asymp n^{-\frac{2\beta}{2\beta+1}}$. For dimension d, the rate becomes $n^{-\frac{2\beta}{2\beta+d}}$.

If the density p is smoother (e.g., $\beta = \ell + \gamma$, with $\ell \in \mathbb{N}$), we use Taylor expansion:

$$p(x_0 + uh) - p(x_0) = \sum_{j=1}^{\ell} \frac{p^{(j)}(x_0)}{j!}(uh)^j + \frac{p^{(\ell)}(x_0 + \tau uh)}{\ell!}(uh)^\ell$$

To achieve better bias rates, we need the lower order terms in the integral $\int K(u)(\dots)du$ to vanish.

Definition 3.2.1. A kernel K is of order ℓ if:

$$\int u^j K(u)du = 0 \quad \text{for } j = 1, 2, \dots, \ell - 1.$$

Remark. To satisfy this for $\ell \geq 2$, K must take negative values. We can construct such kernels using an orthonormal basis on $L^2([-1,1])$, such as Legendre polynomials $(\varphi_k)_{k=0}^{\infty}$.

$$K(u) = \sum_{m=0}^{\ell} \varphi_m(0)\varphi_m(u) \cdot \mathbb{1}_{u\in[-1,1]}$$

Using an ℓ-th order kernel, the bias improves to $|\text{Bias}(x_0)| \leq Ch^\beta$, recovering the optimal rate $n^{-\frac{2\beta}{2\beta+1}}$ for $\beta > 1$.

We now analyze the Mean Integrated Squared Error (MISE): $\mathbb{E}\int(\hat{p}_n(x) - p(x))^2 dx$. Assume p belongs to a Sobolev class $W^{\beta,2}$, such that $\int(p^{(\beta)}(x))^2 dx \leq L^2$. The integrated variance is estimated as

$$\int \text{Var}(\hat{p}_n(x))dx = \frac{1}{nh}\int \text{Var}\left(K\left(\frac{X-x}{h}\right)\right)dx \leq \frac{1}{nh}\int K^2(u)du$$

Using a kernel of order $\beta - 1$, the bias term $b(x)$ is given by

$$b(x) = \int K(u)\frac{(uh)^\beta}{\beta!}\int_0^1 (1-\tau)^{\beta-1} p^{(\beta)}(x + \tau uh)d\tau du$$

and can be bounded via the **generalized Minkowski inequality**:

$$\left(\int \left|\int g(x,u)du\right|^2 dx\right)^{1/2} \leq \int \left(\int |g(x,u)|^2 dx\right)^{1/2} du$$

Applying this to the bias expression yields:

$$\int b(x)^2 dx \leq h^{2\beta}\left[\int |K(u)||u|^\beta \left(\int_{\mathbb{R}}\int_0^1 (p^{(\beta)}(x+\tau uh))^2 d\tau dx\right)^{1/2} du\right]^2$$

Since $\int(p^{(\beta)}(z))^2 dz \leq L^2$, this simplifies to:

$$\int b(x)^2 dx \leq C \cdot L^2 \cdot h^{2\beta}$$

By optimizing over h, we obtain the rate $n^{-\frac{2\beta}{2\beta+1}}$.

3.2.2 The Nadaraya-Watson Estimator

We consider the fixed design regression problem:

$$Y_i = f^*(x_i) + \varepsilon_i, \quad i = 1, \ldots, n$$

where x_i are deterministic design points (e.g., equispaced on $[0, 1]$), and ε_i are independent random errors satisfying:

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathrm{Var}(\varepsilon_i) \le \sigma^2$$

Our goal is to estimate the regression function $f^*(x)$. We made the following assumption:

Assumption: Let $f^* \in \Sigma(\beta, L)$ for $0 < \beta \le 1$. That is:

$$|f^*(x) - f^*(y)| \le L|x - y|^\beta$$

and the kernel K is non-negative and supported on $[-1, 1]$.

We start with the Nadaraya-Watson (NW) estimator, which is a "local" estimator defined as:

$$\hat{f}_n(x) = \frac{\sum_{i=1}^n Y_i K\left(\frac{x_i - x}{h}\right)}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}$$

where K is a kernel function and h is the bandwidth. For analysis, it is convenient to define the weights $W_{n,i}(x)$:

$$W_{n,i}(x) = \frac{K\left(\frac{x_i - x}{h}\right)}{\sum_{j=1}^n K\left(\frac{x_j - x}{h}\right)}$$

Thus, the estimator can be written as a linear function of the responses Y:

$$\hat{f}_n(x) = \sum_{i=1}^n W_{n,i}(x) Y_i$$

Note that by definition, $\sum_{i=1}^n W_{n,i}(x) = 1$. We analyze the Mean Squared Error (MSE) at a point x_0:

$$\mathrm{MSE}(x_0) = \mathbb{E}[(\hat{f}_n(x_0) - f^*(x_0))^2] = \mathrm{Bias}^2(x_0) + \mathrm{Var}(\hat{f}_n(x_0))$$

Proof of Bias Bound: The bias is given by $b(x_0) = \mathbb{E}[\hat{f}_n(x_0)] - f^*(x_0)$.

$$\begin{aligned}
b(x_0) &= \sum_{i=1}^n W_{n,i}(x_0) f^*(x_i) - f^*(x_0) \\
&= \sum_{i=1}^n W_{n,i}(x_0) f^*(x_i) - \sum_{i=1}^n W_{n,i}(x_0) f^*(x_0) \\
&= \sum_{i=1}^n W_{n,i}(x_0)(f^*(x_i) - f^*(x_0))
\end{aligned}$$

Taking the absolute value:

$$|b(x_0)| \le \sum_{i=1}^n |W_{n,i}(x_0)| \cdot |f^*(x_i) - f^*(x_0)|$$

Since K is supported on $[-1, 1]$, $W_{n,i}(x_0)$ is non-zero only if $|\frac{x_i - x_0}{h}| \le 1 \implies |x_i - x_0| \le h$. Using the Hölder condition:

$$|b(x_0)| \le \sum_{i=1}^n W_{n,i}(x_0) \cdot L|x_i - x_0|^\beta \le Lh^\beta \sum_{i=1}^n W_{n,i}(x_0) = Lh^\beta$$

Thus, the squared bias is bounded by $L^2 h^{2\beta}$.

The variance at x_0 is:

$$\sigma^2(x_0) = \mathrm{Var}\left(\sum_{i=1}^n W_{n,i}(x_0) Y_i\right) = \sum_{i=1}^n W_{n,i}^2(x_0)\mathrm{Var}(Y_i) \le \sigma^2 \sum_{i=1}^n W_{n,i}^2(x_0)$$

We can bound the sum of squares using the maximum weight:

$$\sum_{i=1}^n W_{n,i}^2(x_0) \le \left(\max_i |W_{n,i}(x_0)|\right) \underbrace{\sum_{i=1}^n W_{n,i}(x_0)}_{=1} = \max_i |W_{n,i}(x_0)|$$

> **Example** (Box Kernel with Equispaced Design). Let $K(u) = \frac{1}{2}\mathbb{I}(|u| \le 1)$ and $x_i = i/n$. The denominator of the weights approximates the density of points:
>
> $$\sum_{j=1}^n K\left(\frac{x_j - x_0}{h}\right) = \frac{1}{2} \cdot \#\{j : |x_j - x_0| \le h\}$$
>
> For equispaced points, the number of points in $[x_0 - h, x_0 + h]$ is approximately $2nh$. Thus, for any i in the window:
>
> $$W_{n,i}(x_0) \approx \frac{1/2}{1/2 \cdot 2nh} = \frac{1}{2nh}$$
>
> Therefore:
>
> $$\sigma^2(x_0) \le \frac{\sigma^2}{nh}$$
>
> Combining bias and variance:
>
> $$\mathrm{MSE}(x_0) \le L^2 h^{2\beta} + \frac{C\sigma^2}{nh}$$
>
> To find the optimal bandwidth h_n, we balance the two terms:
>
> $$h^{2\beta} \asymp \frac{1}{nh} \implies h^{2\beta+1} \asymp n^{-1} \implies h_n \asymp n^{-\frac{1}{2\beta+1}}$$
>
> Substituting this back gives the optimal minimax rate:
>
> $$\mathrm{MSE}(x_0) \asymp n^{-\frac{2\beta}{2\beta+1}}$$

3.2.3 Generalization to Local Polynomial Estimators

The Nadaraya-Watson estimator corresponds to locally fitting a constant. For smoother functions ($\beta > 1$), this suffers from boundary bias and cannot adapt to higher-order curvature. We generalize this to Local Polynomial Regression:

Instead of finding a constant θ, we approximate $f(x_i)$ near x using a Taylor expansion of order ℓ (where $\ell \ge \lfloor \beta \rfloor$):

$$f(x_i) \approx f(x) + f'(x)(x_i - x) + \frac{f''(x)}{2!}(x_i - x)^2 + \cdots + \frac{f^{(\ell)}(x)}{\ell!}(x_i - x)^\ell$$

We estimate the parameter vector $\boldsymbol{\theta} = [f(x), f'(x), \ldots, f^{(\ell)}(x)]^\top$ by minimizing the weighted least squares:

$$\hat{\boldsymbol{\theta}}(x) = \operatorname*{argmin}_{\boldsymbol{\theta} \in \mathbb{R}^{\ell+1}} \sum_{i=1}^n \left(Y_i - \sum_{j=0}^{\ell} \theta_j \frac{(x_i - x)^j}{j!}\right)^2 K\left(\frac{x_i - x}{h}\right)$$

The solution is a weighted least squares estimator. Define the matrix $B_{n,x}$:

$$B_{n,x} = \frac{1}{nh} \sum_{i=1}^{n} U\left(\frac{x_i - x}{h}\right) U\left(\frac{x_i - x}{h}\right)^{\top} K\left(\frac{x_i - x}{h}\right)$$

The weights for LPR are explicitly given by:

$$W_{n,i}(x) = \frac{1}{nh} e_1^{\top} B_{n,x}^{-1} U\left(\frac{x_i - x}{h}\right) K\left(\frac{x_i - x}{h}\right)$$

The variance is $\sigma^2 \sum W_{n,i}^2(x) \le \sigma^2 \max |W_{n,i}(x)|$. We bound the max weight. Assuming the design is regular enough such that the discrete sum converges to the integral, $B_{n,x} \succeq \lambda_0 I$ for large n.

$$|W_{n,i}(x)| \le \frac{1}{nh} \|e_1\| \cdot \|B_{n,x}^{-1}\|_{op} \cdot \left\| U\left(\frac{x_i - x}{h}\right) \right\|_2 \cdot |K(\cdot)|$$

Since $U(t)$ contains terms bounded on the compact support of K, $\|U\|_2 \le C$. Thus:

$$\max_i |W_{n,i}(x)| \le \frac{C}{\lambda_0 nh} \implies \mathrm{Var}(\hat{f}_n(x)) = O\left(\frac{1}{nh}\right)$$

We need the following lemma for analysis of the bias term:

Lemma 3.2.2 (Polynomial Reproduction)**.** For any polynomial Q of degree ℓ, the LPR weights satisfy:

$$\sum_{i=1}^{n} W_{n,i}(x) Q(x_i) = Q(x)$$

Consequently, $\sum_{i=1}^{n} W_{n,i}(x)(x_i - x)^k = 0$ for $k = 1, \ldots, \ell$.

Bias Proof for $\beta > 1$: Using the Taylor expansion with Lagrange remainder for f^*:

$$f^*(x_i) = f^*(x) + \sum_{k=1}^{\ell} \frac{f^{(k)}(x)}{k!}(x_i - x)^k + \frac{f^{(\ell)}(x_i^*)}{\ell!}(x_i - x)^{\ell}$$

Wait, strict expansion to order ℓ leaves a remainder related to Hölder class. Let's express $f^*(x_i)$ as the polynomial part $P_x(x_i)$ plus residual $R(x_i)$ where $|R(x_i)| \le L|x_i - x|^{\beta}$:

$$b(x) = \sum_{i=1}^{n} W_{n,i}(x) f^*(x_i) - f^*(x)$$

Substitute $f^*(x_i) = P_x(x_i) + R(x_i)$. By the Lemma, $\sum W_{n,i}(x) P_x(x_i) = P_x(x) = f^*(x)$.

$$b(x) = \underbrace{\left(\sum W_{n,i}(x) P_x(x_i) - f^*(x)\right)}_{0} + \sum_{i=1}^{n} W_{n,i}(x) R(x_i)$$

Thus, the bias is determined solely by the remainder term:

$$|b(x)| \le \sum_{i=1}^{n} |W_{n,i}(x)| \cdot L|x_i - x|^{\beta} \le Lh^{\beta} \sum |W_{n,i}(x)|$$

Since $\sum |W_{n,i}|$ is bounded (by properties of the inverse matrix $B_{n,x}^{-1}$), we have:

$$\mathrm{Bias}^2(x) = O(h^{2\beta})$$

Combining the variance $O((nh)^{-1})$ and squared bias $O(h^{2\beta})$, LPR achieves the rate $n^{-\frac{2\beta}{2\beta+1}}$ for any $\beta > 0$.

3.3 Lecture 11: Minimax Lower bounds

3.3.1 Sub-optimality of KDE

In parametric estimation, MLE is often asymptotically efficient (achieving the Cramér-Rao lower bound). For KDE, can we achieve a specific asymptotic constant?

Remark. For a class like $S(2, L)$ (Sobolev class with second derivatives), the minimax rate is $n^{-4/5}$.

- For a *fixed* density p as $n \to \infty$, the error is asymptotically dominated by the variance term if h is chosen small enough.
- However, for the **minimax risk** (worst case over the class), the "hardest" density depends on n.
- Every specific density is asymptotically "easier" than the worst-case density in the class (super-efficiency phenomenon).

We now justify the minimax rate. Recall that the variance can be expressed as

$$\int \mathrm{Var}(\hat{p}_n(x))dx = \frac{1}{nh}R(K) + o\left(\frac{1}{nh}\right)$$

The bias is $b(x) = \mathbb{E}[\hat{p}_n(x)] - p(x)$.

$$b(x) = \int \frac{1}{h}K\left(\frac{x-y}{h}\right)p(y)dy - p(x)$$

Let $y = x - uh$, so $dy = -hdu$.

$$b(x) = \int K(u)p(x-uh)du - p(x) = \int K(u)[p(x-uh) - p(x)]du$$

To analyze the second-order behavior, we use the Taylor expansion of $p(x - uh)$ around x with the integral form of the remainder:

$$p(x-uh) = p(x) - uhp'(x) + \int_0^1 (1-\tau)\frac{d^2}{d\tau^2}p(x-\tau uh)d\tau$$

Using the chain rule, $\frac{d^2}{d\tau^2}p(x-\tau uh) = (-uh)^2 p''(x-\tau uh) = u^2h^2p''(x-\tau uh)$. Thus:

$$p(x-uh) - p(x) = -uhp'(x) + h^2u^2\int_0^1 (1-\tau)p''(x-\tau uh)d\tau$$

Substituting this back into the bias equation:

$$b(x) = -hp'(x)\int uK(u)du + h^2\int u^2K(u)\left[\int_0^1(1-\tau)p''(x-\tau uh)d\tau\right]du$$

The linear term vanishes because K is a second-order kernel. We define an approximation $\tilde{b}(x)$ by replacing the shifted second derivative with $p''(x)$:

$$\tilde{b}(x) = h^2\left(\int u^2K(u)du\right)\left(\int_0^1(1-\tau)d\tau\right)p''(x)$$

Since $\int_0^1(1-\tau)d\tau = \frac{1}{2}$, we have:

$$\tilde{b}(x) = \frac{1}{2}h^2\kappa_2 p''(x)$$

We aim to show that $\int b^2(x)dx \approx \int \tilde{b}^2(x)dx$. First, the squared norm of the approximation is:

$$\int \tilde{b}^2(x)dx = \frac{h^4}{4}\kappa_2^2 \int (p''(x))^2 dx = \frac{h^4}{4}\kappa_2^2 R(p'')$$

To prove rigor, we must show the error goes to zero: $\int |b(x) - \tilde{b}(x)|^2 dx \to 0$ faster than h^4 (or specifically that the ratio goes to 1).

$$b(x) - \tilde{b}(x) = h^2 \int u^2 K(u) \int_0^1 (1-\tau)[p''(x - \tau uh) - p''(x)]d\tau du$$

By the Generalized Minkowski Inequality (integral form), the L^2 norm satisfies:

$$\|b - \tilde{b}\|_2 \leq h^2 \int |u^2 K(u)| \int_0^1 (1-\tau)\|p''(\cdot - \tau uh) - p''(\cdot)\|_2 d\tau du$$

For any function $g \in L^2$, the translation map is continuous in L^2. That is, $\|g(\cdot + \delta) - g(\cdot)\|_2 \to 0$ as $\delta \to 0$. Since $\tau \in [0,1]$ and u is fixed in the integration (weighted by $K(u)$ which decays), the shift $\tau uh \to 0$ as $h \to 0$. Therefore, $\|p''(\cdot - \tau uh) - p''(\cdot)\|_2 = o(1)$. This implies $\|b - \tilde{b}\|_2 = o(h^2)$, and thus:

$$\int b^2(x)dx = \frac{h^4}{4}\kappa_2^2 R(p'') + o(h^4)$$

Combining the bias and variance:

$$\text{AMISE}(h) = \frac{R(K)}{nh} + \frac{h^4}{4}\kappa_2^2 R(p'')$$

Minimizing this with respect to h yields the optimal bandwidth:

$$h_{opt} = \left[\frac{R(K)}{n\kappa_2^2 R(p'')}\right]^{1/5} \sim n^{-1/5}$$

Substituting h_{opt} back gives the optimal rate $O(n^{-4/5})$.

3.3.2 Minimax Lower Bound for Hölder Class

We prove that no estimator can achieve a faster rate than $n^{-\frac{2\beta}{2\beta+1}}$ over $\Sigma(\beta, L)$.

Proof. We use Le Cam's two-point method.
We construct two hypotheses $f_0, f_1 \in \Sigma(\beta, L)$:

- $H_0 : f_0(x) = 0$
- $H_1 : f_1(x) = \psi \cdot K_0\left(\frac{x - x_0}{h}\right)$

where K_0 is a smooth bump function supported on $[-1, 1]$, and ψ is a scaling factor. To ensure $f_1 \in \Sigma(\beta, L)$, we require the ℓ-th derivative to satisfy the Hölder condition. The derivatives scale as:

$$|f_1^{(\ell)}(x)| \propto \psi h^{-\ell}$$

The Hölder condition requires $\psi h^{-\ell} \cdot h^{\ell - \beta} \lesssim L \implies \psi \lesssim Lh^\beta$. We choose $\psi = cLh^\beta$ for a small constant c.
By Le Cam's Lemma, the minimax risk is lower bounded by:

$$\inf_{\hat{f}} \sup_{f \in \Sigma} \mathbb{E}[(\hat{f}(x_0) - f(x_0))^2] \geq \frac{(f_0(x_0) - f_1(x_0))^2}{8}(1 - D_{TV}(P_0, P_1))$$

The squared separation is $(f_1(x_0) - f_0(x_0))^2 = \psi^2 \asymp h^{2\beta}$. We bound the TV distance using KL divergence:

$$D_{KL}(P_0||P_1) = \frac{1}{2\sigma^2}\sum_{i=1}^{n}(f_1(x_i) - 0)^2 \approx \frac{n}{2\sigma^2}\int f_1^2(x)dx$$

$$\int f_1^2(x)dx = \psi^2 \int K_0^2\left(\frac{x-x_0}{h}\right)dx = \psi^2 h\|K_0\|_2^2 \asymp h^{2\beta}\cdot h = h^{2\beta+1}$$

Thus, $D_{KL} \asymp nh^{2\beta+1}$. To make the errors indistinguishable (e.g., $D_{TV} \leq 1/2$), we require $D_{KL} \leq C$, which implies:

$$nh^{2\beta+1} \asymp 1 \implies h \asymp n^{-\frac{1}{2\beta+1}}$$

Substituting this h into the separation ψ^2:

$$\text{Minimax Risk} \gtrsim h^{2\beta} \asymp n^{-\frac{2\beta}{2\beta+1}}$$

□

3.3.3 Lepski's Method

We aim to construct a single estimator $\hat{f}_n$ that is simultaneously optimal for all smoothness parameters β. Ideally, for all β and all $f^* \in \Sigma(\beta, L)$, we want the Mean Squared Error (MSE) at a point x_0 to satisfy:

$$MSE(x_0) = \mathbb{E}[(\hat{f}_n(x_0) - f^*(x_0))^2] \lesssim n^{-\frac{2\beta}{2\beta+1}}$$

This capability is desirable because f^* may exhibit different smoothness levels at different locations x_0, allowing the estimator to adapt to local regularity.

Surprisingly, perfect adaptation is not possible without an additional cost. The following theorem demonstrates that a logarithmic penalty is unavoidable.

Theorem 3.3.1 (Lepski's Lower Bound)**.** Suppose $0 < \beta_1 < \alpha < \beta_2 \leq 1$. Define the rates for the rougher class (β_1) and the smoother class (α) as follows:

$$r_1^2(n) = \left(\frac{\log n}{n}\right)^{\frac{2\beta_1}{2\beta_1+1}}, \quad r_2^2(n) = \left(\frac{1}{n}\right)^{\frac{2\alpha}{2\alpha+1}}$$

Then, for any estimator $\hat{f}_n$, the following minimax lower bound holds for some constant $C > 0$:

$$\inf_{\hat{f}_n}\sup_{i\in\{1,2\}}\sup_{f^*\in\Sigma(\beta_i,L)}\frac{\mathbb{E}[(\hat{f}_n(x_0) - f^*(x_0))^2]}{r_i^2(n)} \geq C$$

Interpretation: This is an *asymmetric lower bound*. It implies that if we wish to estimate the easier (smoother) class at a rate faster than the minimax rate of the harder (rougher) class, we must "pay" a logarithmic factor in the rate for the harder class.

Proof. We define two specific functions centered at the point of interest x_0:

(a). **Hypothesis 2 (Smooth):** Let $f_2^* \in \Sigma(\beta_2, L)$. We choose the zero function:

$$f_2^*(x) = 0$$

(b). **Hypothesis 1 (Rough):** Let $f_1^* \in \Sigma(\beta_1, L)$. We construct a "bump" function using a kernel

K:

$$f_1^*(x) = \psi \cdot K\left(\frac{x - x_0}{h}\right)$$

where h is the bandwidth and ψ is the amplitude. From the asymmetric two-point lemma, the minimax risk is lower bounded by the separation distance weighted by the error probability. Specifically, we examine the ratio:

$$\frac{\mathbb{E}[|\hat{f}_n(x_0) - f_i^*(x_0)|^2]}{r_i(n)^2}$$

The lemma implies a bound related to the separation at x_0:

$$\text{Risk} \gtrsim \frac{(f_1^*(x_0) - f_2^*(x_0))^2}{8r_1(n)^2}\left\{1 - \frac{r_2^2(n)}{r_1^2(n)}\left(1 + \chi^2(P_{f_1}||P_{f_2})\right)\right\}$$

Here, the separation is determined by the amplitude ψ at x_0 (since $K(0) = 1$ typically, $f_1^*(x_0) = \psi$). Thus $(f_1^*(x_0) - f_2^*(x_0))^2 = \psi^2$.

To obtain a contradiction or a valid bound, we need the term in the curly braces to be strictly positive (e.g., $< 1/2$). This requires:

$$\frac{r_2^2(n)}{r_1^2(n)}\left(1 + \chi^2(P_{f_1}||P_{f_2})\right) < \frac{1}{2}$$

Since r_2 (smooth rate) decays faster than r_1 (rough rate), the ratio $\frac{r_2^2(n)}{r_1^2(n)}$ is very small ($1/\text{poly}(n)$). Therefore, we only need the divergence $\chi^2(P_{f_1}||P_{f_2})$ to be bounded by a polynomial in n (rather than exponential). For the symmetric version, we would typically require $D_{KL}(P_{f_1}||P_{f_2}) \leq \frac{1}{2}$, but here the asymmetry allows for larger divergence.

We choose the bandwidth h and amplitude ψ to satisfy the smoothness condition $f_1^* \in \Sigma(\beta_1, L)$ and the divergence constraint:

$$h = C_0\left(\frac{\log n}{n}\right)^{\frac{1}{2\beta_1+1}}, \quad \psi = C'h^{\beta_1}$$

With these choices, the separation is $\psi^2 \asymp h^{2\beta_1} \asymp \left(\frac{\log n}{n}\right)^{\frac{2\beta_1}{2\beta_1+1}}$, which matches $r_1^2(n)$. Substituting these into the divergence calculation ensures $\chi^2(\cdot) \leq \text{poly}(n)$, completing the proof that the rate cannot be faster than $r_1^2(n)$ (which includes the log factor) when we also demand high performance on the smooth class. □

The lower bound proves we *must* pay a log factor. To actually achieve this adaptive rate, we use **Lepski's Method**.

- **Key Observation:** For each candidate β, let h_β be the optimal bandwidth.
- Let $\hat{f}_{h_\beta}(x_0)$ be the estimator using bandwidth h_β.
- The method involves comparing estimators across a grid of bandwidths and selecting the largest bandwidth where the estimators do not differ significantly from those with smaller bandwidths.

Lepski's method provides a selection rule to choose an estimator from a family of estimators ordered by their bandwidths (or assumed smoothness).

Step I: Discretization Since we cannot search over continuous β, we discretize the search space $[\beta_{\min}, \beta_{\max}]$.

Definition 3.3.2 (The Grid $\mathcal{B}$). Let $\mathcal{B} = \{\beta_1, \beta_2, \ldots, \beta_N\}$ be a grid of candidate smoothness parameters such that:

$$\beta_{\min} = \beta_1 < \beta_2 < \cdots < \beta_N = \beta_{\max}$$

where the grid size is $N \approx \log n$. The spacing is chosen to be uniform, typically $\beta_j - \beta_{j-1} = \frac{1}{\log n}$.

Justification: This discretization is sufficiently fine because the convergence rates do not change distinctively within intervals of size $1/\log n$. Specifically:

$$n^{-\frac{2\beta_j}{2\beta_j+1}} \approx n^{-\frac{2\beta_{j-1}}{2\beta_{j-1}+1}}$$

within polynomial factors relevant to the sample size.

Step II: The Selection Rule For each $\beta \in \mathcal{B}$, let $\hat{f}_{h_\beta}$ be the estimator constructed using bandwidth $h_\beta \asymp (\frac{\log n}{n})^{\frac{1}{2\beta+1}}$.

We define the adaptive estimator $\hat{\beta}$ by selecting the largest smoothness (largest β, smallest variance, largest bias) that is not statistically distinguishable from the rougher estimates (smaller β, larger variance, smaller bias).

Definition 3.3.3 (Lepski's Estimator). The selected smoothness $\hat{\beta}$ is:

$$\hat{\beta} = \max\left\{\beta \in \mathcal{B} : \forall \beta' \in \mathcal{B}, \beta' < \beta, \quad |\hat{f}_{h_{\beta'}}(x_0) - \hat{f}_{h_\beta}(x_0)| \leq C_0 \psi_n(\beta')\right\}$$

where $\psi_n(\beta')$ is the threshold associated with the standard deviation of the rougher estimator:

$$\psi_n(\beta') \asymp \left(\frac{\log n}{n}\right)^{\frac{\beta'}{2\beta'+1}} \approx \sqrt{\frac{\log n}{n h_{\beta'}}}$$

The final estimator is $\hat{f}_{\text{Lepski}}(x_0) := \hat{f}_{h_{\hat{\beta}}}(x_0)$.

The adaptive estimator achieves the optimal rate for the unknown β^*, paying only a logarithmic penalty factor for the adaptation.

Theorem 3.3.4 (Oracle Inequality). Let f^* have true smoothness $\beta^* \in [\beta_{\min}, \beta_{\max}]$. Then:

$$\mathbb{E}\left[|\hat{f}_{\text{Lepski}}(x_0) - f^*(x_0)|^2\right] \leq C\left(\frac{\log n}{n}\right)^{\frac{2\beta^*}{2\beta^*+1}}$$

Proof. The risk is decomposed based on the event where the selected $\hat{\beta}$ falls relative to the true β^*

$$\mathbb{E}|\hat{f}_{\hat{\beta}} - f^*|^2 = \sum_{j=1}^{N} \mathbb{E}\left[|\hat{f}_{\beta_j} - f^*|^2 \mathbb{I}(\hat{\beta} = \beta_j)\right]$$

We split the sum into two regimes:

Case 1: $j \geq j^*$ **(Over-smoothing)** Here, we selected a smoothness β_j greater than or equal to the truth β^*. By the construction of the set in Lepski's rule, if β_j was selected, it implies that it satisfied the intersection condition with all $\beta' < \beta_j$. Specifically, it satisfied the condition with β_{j^*}

(the truth).

$$|\hat{f}_{\beta_j} - \hat{f}_{\beta_{j^*}}| \leq C\psi_n(\beta_{j^*})$$

Using the triangle inequality:

$$|\hat{f}_{\beta_j} - f^*| \leq \underbrace{|\hat{f}_{\beta_j} - \hat{f}_{\beta_{j^*}}|}_{\leq \text{Threshold}} + \underbrace{|\hat{f}_{\beta_{j^*}} - f^*|}_{\text{Oracle Risk}}$$

The first term is dominated by $\psi_n(\beta')$ by the definition of $\hat{\beta}$, and the second term is bounded by the risk of the true parameter.

Case 2: $j < j^*$ **(Under-smoothing)** Here, we selected a smoothness lower than the truth (high variance, low bias). This event $\hat{\beta} = \beta_j < \beta^*$ occurs only if the selection rule rejected the true β^*. This implies that for some comparison, the noise term was large enough to violate the threshold. Since the estimators are based on sums of independent variables, we can control this probability using **concentration inequalities** (e.g., Bernstein or Gaussian tail bounds).

The probability of choosing $j < j^*$ decays polynomially in n, making the expected risk contribution from this regime negligible compared to the oracle rate. $\square$

Chapter 4 Exponential Family

4.1 Lecture 12: Definitions and Basic Properties

Consider a random variable $X = (X_1, \dots, X_m) \in \mathcal{X} \subseteq \mathbb{R}^m$ equipped with a σ-finite measure μ (typically a counting measure or the Lebesgue measure).

Definition 4.1.1. A parametric statistical model for X is defined as an exponential family with canonical parameter vector $\Theta = (\theta_1, \dots, \theta_d)$ and canonical statistics $t(x) = (t_1(x), \dots, t_d(x))$ if it admits a density with respect to μ of the form:

$$f(x; \theta) = h(x) \exp\{\langle \theta, t(x) \rangle - A(\theta)\}$$

Remark. Formally, $\mathcal{X}$ is defined as the smallest closed set such that $P_\theta(x \in \mathcal{X}) = 1$, and it is independent of θ.

Definition 4.1.2. The term $A(\theta)$ acts as a normalization constant (log-partition function) and is related to the partition function $Z(\theta)$ as follows:

$$Z(\theta) := \int_{\mathcal{X}} h(x) \exp\{\langle \theta, t(x) \rangle\} \mu(dx), \quad A(\theta) = \log Z(\theta)$$

Consequently, the integral of the unnormalized density is $e^{A(\theta)}$.
The space of the canonical parameter Θ is defined as the set where the partition function is finite:

$$\Theta := \{\theta \in \mathbb{R}^d : Z(\theta) < +\infty\}$$

Example (Bernoulli Distribution)**.** Consider $X \sim \text{Bern}(p)$ where $p \in (0, 1)$. The probability mass function is:

$$f(x; p) = p^x (1-p)^{1-x} = \exp\left\{x \log \frac{p}{1-p} + \log(1-p)\right\}$$

Mapping this to the exponential family form:

- **Canonical Parameter:** $\theta = \log \frac{p}{1-p}$ (Log-odds).
- **Inverse Mapping:** $p = \frac{e^\theta}{1+e^\theta}$.
- **Log-Partition Function:** $A(\theta) = \log(1 + e^\theta)$ (derived from $\log(1-p)$).
- **Canonical Statistic:** $t(x) = x$.

Example (Univariate Gaussian)**.** Consider $X \sim N(\mu, \sigma^2)$. The density is expanded as:

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2\sigma^2} + \frac{x\mu}{\sigma^2} + \frac{1}{2}\left(\log\left(\frac{1}{\sigma^2}\right) - \frac{\mu^2}{\sigma^2}\right)\right\}$$

Identifying the components:

- **Canonical Statistic:** $t(x) = (x, -\frac{x^2}{2}) \in \mathbb{R}^2$.
- **Canonical Parameter:** $\theta = (\frac{\mu}{\sigma^2}, \frac{1}{\sigma^2}) \in \mathbb{R}^2$.
- **Base Function:** $h(x) = \frac{1}{\sqrt{2\pi}}$.
- **Log-Partition Function:** $A(\theta) = -\frac{1}{2}(\log(\theta_2) - \frac{\theta_1^2}{\theta_2})$.

Example (Centered Multivariate Gaussian)**.** Consider $X \sim N_m(0, \Sigma)$ with covariance Σ and precision matrix $K := \Sigma^{-1}$. The density is:

$$f(x; K) = \frac{1}{(2\pi)^{m/2}} \sqrt{\det(K)} \exp\left(-\frac{1}{2}x^T K x\right)$$

Using the trace trick $x^T K x = \operatorname{tr}(K x x^T) = \langle K, x x^T \rangle$, we find:

- **Canonical Parameter:** $\theta = K$.
- **Base Function:** $h(x) = (2\pi)^{-m/2}$.
- **Log-Partition Function:** $A(K) = -\frac{1}{2} \log \det(K)$.
- **Canonical Statistic:** $t(x) = -\frac{1}{2} x x^T$.

Remark (Connection with Statistical Inference)**.** Given a sample $x_{1:n} = (x_1, \ldots, x_n)$ i.i.d. from $\mathbb{P}$, the log-likelihood function is defined as:

$$l_n(\theta) := \frac{1}{n} \sum_{i=1}^{n} \log f(x_i; \theta)$$

Substituting the exponential family form, the log-likelihood becomes:

$$l_n(\theta) = \left\langle \theta, \frac{1}{n} \sum_{i=1}^{n} t(x_i) \right\rangle - A(\theta) + \text{const}$$

This demonstrates that inference depends on the data only through the empirical mean of the canonical statistics $\bar{\mu}_n$.

Definition 4.1.3. An exponential family representation is considered **minimal** if the canonical statistics and parameters satisfy the following conditions(to ensure the representation is unique and efficient):

(a). **No redundancy on statistics:** There is no hyperplane $H(\alpha, c) = \{x : \langle \alpha, t(x) \rangle = c\}$ such that $P_\theta(x \in H) = 1$.

(b). **No redundancy on parameters:** There is no hyperplane containing the parameter space Θ.

Example (Bernoulli Distribution)**. Non-Minimal Representation:** Consider the Bernoulli distribution modeled with $t(x) = (\mathbb{I}\{x = 0\}, \mathbb{I}\{x = 1\})$. Since $\mathbb{I}\{x = 0\} + \mathbb{I}\{x = 1\} = 1$, the statistics satisfy a linear constraint $\langle \mathbf{1}, t(x) \rangle = 1$. Because of this linear dependency, this parametrization is **not minimal**.
Minimal Representation: By recognizing $\mathbb{I}\{x = 1\} = 1 - \mathbb{I}\{x = 0\}$, we can reduce the statistic to $t(x) = \mathbb{I}\{x = 1\}$ (or simply x). This yields the standard minimal parametrization.

Definition 4.1.4. A minimal exponential family is called "full" if its parameter space Θ is maximal (i.e., Θ contains all θ for which the partition function $Z(\theta)$ is finite). Otherwise it is called curved.

In a full exponential family, the canonical statistic $t(x)$ is **minimally sufficient** for θ. Under regularity conditions, the distribution of the statistic $T = t(X)$ itself forms an exponential family:

$$f(t;\theta) = g(t)\exp\{\langle\theta, t\rangle - A(\theta)\} \tag{4.1}$$

Here, the base measure $g(t)$ is derived by integrating (or summing) $h(x)$ over the level set $\{x : t(x) = t\}$.

Definition 4.1.5. Let $t(x)$ be a sufficient statistic. It is minimal if, for every other sufficient statistic $t'(x)$, there exists a measurable function g such that:

$$t(x) = g\left(t'(x)\right)$$

Remark. This is equivalent to the statement that if $t'(x) = t'(y)$, then it must imply that $t(x) = t(y)$.

Proposition 4.1.6. In a full exponential family, $t(x)$ is minimally sufficient.

Remark. Recall the **Fisher-Neyman Factorization Theorem**: A statistic $t(x)$ is sufficient if and only if the density factorizes as:

$$f(x;\theta) = h(x)g_\theta(t(x)) \tag{4.2}$$

For exponential families, $t(x)$ naturally fits this form.

Intuition. If $t'(x)$ is another sufficient statistic, the ratio $f(x;\theta)/f(y;\theta)$ must be independent of θ whenever $t'(x) = t'(y)$. For exponential families, this ratio is proportional to $\exp\{\langle\theta, t(x) - t(y)\rangle\}$. For this to be independent of θ for all θ, we essentially need $t(x) = t(y)$ (assuming minimality), implying $t(x)$ is a function of any other sufficient statistic. □

Theorem 4.1.7. For every exponential family defined with parameter space Θ, the following properties hold:

(a). **Convex Set:** The parameter space Θ is a convex set.

(b). **Convex Function:** The log-partition function $A(\theta)$ is convex on Θ.

(c). **Strict Convexity & Identifiability:** If the exponential family is minimal, $A(\theta)$ is **strictly convex**. This implies that the mapping from parameters to distributions is one-to-one: $P_{\theta_1} = P_{\theta_2}$ if and only if $\theta_1 = \theta_2$.

(d). **Continuity:** $A(\theta)$ is lower semi-continuous on Θ and continuous in the interior of Θ.

Example (Implications for Maximum Likelihood Estimation (MLE))**.** These properties ensure that MLE is a well-posed convex optimization problem. The negative log-likelihood is given by:

$$-l_n(\theta) = -\langle\theta, \bar{\mu}_n\rangle + A(\theta) + \text{const} \tag{4.3}$$

Since $A(\theta)$ is convex and the term $-\langle\theta, \bar{\mu}_n\rangle$ is linear (and thus convex), the negative log-likelihood is a convex function over a convex set. Therefore, gradient descent (or Newton's method) is guaranteed to find the global optimal solution (the MLE) effectively.

Proof. Proof of Convexity: Let $\theta_1, \theta_2 \in \Theta$ and $\lambda \in (0, 1)$. Define

$$\theta_\lambda = (1 - \lambda)\theta_1 + \lambda\theta_2.$$

Then

$$Z(\theta_\lambda) = \int h(x)e^{\langle\theta_\lambda,t(x)\rangle}\mu(dx) = \int h(x)\left(e^{\langle\theta_1,t(x)\rangle}\right)^{1-\lambda}\left(e^{\langle\theta_2,t(x)\rangle}\right)^{\lambda}\mu(dx).$$

Applying Hölders inequality with exponents $\frac{1}{1-\lambda}$ and $\frac{1}{\lambda}$,

$$Z(\theta_\lambda) \le Z(\theta_1)^{1-\lambda}Z(\theta_2)^{\lambda}.$$

Taking logs,

$$A(\theta_\lambda) \le (1-\lambda)A(\theta_1) + \lambda A(\theta_2).$$

Thus Θ is convex and A is convex.

Proof of (ii): Equality in Hölder occurs iff the two functions are proportional a.e., i.e.

$$e^{\langle\theta_1,t(x)\rangle} = C\,e^{\langle\theta_2,t(x)\rangle} \quad \mu\text{-a.e.}$$

Taking logs gives

$$\langle\theta_1-\theta_2, t(x)\rangle = \text{const}.$$

Proof of Strict Convexity: If the family is minimal, the sufficient statistics are not linearly dependent a.s., so the equality condition cannot hold unless $\theta_1 = \theta_2$. Hence A is strictly convex. **Proof of Lower Semicontinuity:** Let $\theta_n \to \theta$. Define $g_n(x) = h(x)e^{\langle\theta_n,t(x)\rangle}$. Then

$$Z(\theta) = \int \liminf g_n(x)\,\mu(dx) \le \liminf \int g_n(x)\,\mu(dx) = \liminf Z(\theta_n)$$

by Fatous lemma. Taking logs gives lower semicontinuity. □

Definition 4.1.8. A minimal exponential family is **regular** if Θ is open.

Remark. If $\mathcal{X}$ is finite, $\Theta = \mathbb{R}^d$ and the family is regular.

Remark (Local moment condition)**.** Fix $\theta_0 \in \Theta$. Assume there exists $r > 0$ such that the closed ball $B(\theta_0, r) \subset \Theta$ and

$$\int_{\mathcal{X}} \|t(x)\|\, e^{\theta^\top t(x)}h(x)\,\nu(dx) < \infty, \qquad \int_{\mathcal{X}} \|t(x)\|^2\, e^{\theta^\top t(x)}h(x)\,\nu(dx) < \infty, \tag{4.4}$$

for all $\theta \in B(\theta_0, r)$. This is automatically satisfied in many common regular families; it ensures we may differentiate $Z(\theta)$ under the integral sign up to second order in a neighborhood of θ_0.

Lemma 4.1.9 (Interchange of differentiation and integration)**.** Assume (4.4) holds at some $\theta_0 \in \Theta$. Then Z is twice continuously differentiable in a neighborhood of θ_0, and for $i, j \in \{1, \dots, d\}$,

$$\frac{\partial Z}{\partial\theta_i}(\theta) = \int_{\mathcal{X}} t_i(x)\, e^{\theta^\top t(x)}h(x)\,\nu(dx), \tag{4.5}$$

$$\frac{\partial^2 Z}{\partial\theta_i\,\partial\theta_j}(\theta) = \int_{\mathcal{X}} t_i(x)t_j(x)\, e^{\theta^\top t(x)}h(x)\,\nu(dx), \tag{4.6}$$

for all θ in a (possibly smaller) neighborhood of θ_0.

Proof. We prove (4.5); the proof of (4.6) is analogous.

Fix i and consider $u \to 0$. For θ near θ_0 and u small, write

$$\frac{Z(\theta + ue_i) - Z(\theta)}{u} = \int_{\mathcal{X}} \frac{e^{(\theta+ue_i)^\top t(x)} - e^{\theta^\top t(x)}}{u} h(x)\, \nu(dx),$$

where e_i is the ith coordinate vector in $\mathbb{R}^d$.
For each fixed x, the function $s \mapsto e^{(\theta+se_i)^\top t(x)}$ is differentiable, and by the (one-dimensional) mean value theorem there exists $\xi = \xi(x,u) \in (0,1)$ such that

$$\frac{e^{(\theta+ue_i)^\top t(x)} - e^{\theta^\top t(x)}}{u} = t_i(x)\, e^{(\theta+\xi ue_i)^\top t(x)}.$$

Hence, for $|u| \le r$ and $\theta \in B(\theta_0, r/2)$ we have $\theta + \xi ue_i \in B(\theta_0, r)$, so

$$\left| \frac{e^{(\theta+ue_i)^\top t(x)} - e^{\theta^\top t(x)}}{u} \right| \le |t_i(x)| \sup_{\vartheta \in B(\theta_0, r)} e^{\vartheta^\top t(x)}.$$

By (4.4) (and shrinking neighborhoods if needed), the right-hand side is integrable w.r.t. $h(x)\nu(dx)$, so dominated convergence applies. Therefore, we may pass the limit inside the integral:

$$\frac{\partial Z}{\partial \theta_i}(\theta) = \int_{\mathcal{X}} \lim_{u\to 0} \frac{e^{(\theta+ue_i)^\top t(x)} - e^{\theta^\top t(x)}}{u} h(x)\, \nu(dx) = \int_{\mathcal{X}} t_i(x)\, e^{\theta^\top t(x)} h(x)\, \nu(dx).$$

Continuity of the derivatives in a neighborhood follows similarly. The second derivative formula (4.6) is obtained by repeating the same argument with $t_i(x)e^{\theta^\top t(x)}$ in place of $e^{\theta^\top t(x)}$ and using the $\|t(x)\|^2$ integrability in (4.4). $\square$

Proposition 4.1.10 (Derivatives of the log-partition function)**.** Let $\theta \in \Theta$ and assume (4.4) holds in a neighborhood of θ. Then $A(\theta) = \log Z(\theta)$ is twice continuously differentiable in a neighborhood of θ and

$$\nabla A(\theta) = \mathbb{E}_\theta[t(X)] =: \mu(\theta), \tag{4.7}$$
$$\nabla^2 A(\theta) = \mathrm{Var}_\theta(t(X)) =: V(\theta). \tag{4.8}$$

In components,

$$\frac{\partial^2 A}{\partial\theta_i \partial\theta_j}(\theta) = \mathrm{Cov}_\theta\left(t_i(X), t_j(X)\right).$$

Proof. Since $Z(\theta) > 0$ on Θ, by the chain rule,

$$\frac{\partial A}{\partial \theta_i}(\theta) = \frac{1}{Z(\theta)} \frac{\partial Z}{\partial \theta_i}(\theta).$$

Using Lemma 4.1.9 and the definition of p_θ,

$$\begin{aligned}\frac{\partial A}{\partial \theta_i}(\theta) &= \frac{1}{Z(\theta)} \int_{\mathcal{X}} t_i(x)\, e^{\theta^\top t(x)} h(x)\, \nu(dx) = \int_{\mathcal{X}} t_i(x) \frac{e^{\theta^\top t(x)} h(x)}{Z(\theta)} \nu(dx) \\ &= \int_{\mathcal{X}} t_i(x)\, p_\theta(x)\, \nu(dx) = \mathbb{E}_\theta[t_i(X)].\end{aligned}$$

Stacking coordinates yields (4.7).

For the Hessian, differentiate once more:

$$\frac{\partial^2 A}{\partial\theta_i\partial\theta_j}(\theta) = \frac{\partial}{\partial\theta_j}\left(\frac{1}{Z(\theta)}\frac{\partial Z}{\partial\theta_i}(\theta)\right) = \frac{1}{Z(\theta)}\frac{\partial^2 Z}{\partial\theta_i\partial\theta_j}(\theta) - \frac{1}{Z(\theta)^2}\frac{\partial Z}{\partial\theta_i}(\theta)\frac{\partial Z}{\partial\theta_j}(\theta).$$

Apply Lemma 4.1.9 again:

$$\frac{1}{Z(\theta)}\frac{\partial^2 Z}{\partial\theta_i\partial\theta_j}(\theta) = \mathbb{E}_\theta[t_i(X)t_j(X)], \qquad \frac{1}{Z(\theta)}\frac{\partial Z}{\partial\theta_i}(\theta) = \mathbb{E}_\theta[t_i(X)].$$

Thus

$$\frac{\partial^2 A}{\partial\theta_i\partial\theta_j}(\theta) = \mathbb{E}_\theta[t_i(X)t_j(X)] - \mathbb{E}_\theta[t_i(X)]\,\mathbb{E}_\theta[t_j(X)] = \mathrm{Cov}_\theta\left(t_i(X), t_j(X)\right).$$

In matrix form this is exactly (4.8). □

Remark (Smoothness and higher derivatives)**.** Under mild strengthened versions of (4.4) (existence of local exponential moments of all orders), one can iterate the above argument to show A is C^∞ on Θ, and in fact real-analytic. Moreover, higher derivatives of A correspond to higher-order cumulants of $t(X)$ under p_θ.

Remark. For i.i.d. data,

$$\ell_n(\theta) = \langle\theta, \bar{t}_n\rangle - A(\theta), \quad \bar{t}_n = \frac{1}{n}\sum t(X_i).$$

Observed and Fisher information:

$$J(\theta) = -\nabla^2\ell_n(\theta) = \nabla^2 A(\theta), \quad I(\theta) = \mathbb{E}_\theta[J(\theta)].$$

Recall that $\mathcal{X}$ is formally defined as the support of $\mathbb{P}_{\boldsymbol{\theta}}$. Define $\mathcal{K}$ to be the convex hull of the image $\mathbf{t}(\mathcal{X})$:

$$\mathcal{K} := \mathrm{conv}(\mathbf{t}(\mathcal{X})).$$

Example. For the multivariate Gaussian distribution, $\mathbf{x} \in \mathbb{R}^m$ and so $\mathbf{t}(\mathbf{x}) = -\frac{1}{2}\mathbf{x}\mathbf{x}^\top$ is a rank-one negative semi-definite matrix. In this case, the convex hull of all such matrices, namely $\mathcal{K}$, is the cone of all negative semidefinite matrices.

The next results offer an alternative parametrization for the exponential model.

Proposition 4.1.11. In a regular exponential family:

(a). The mapping $\mu : \mathbb{R}^d \to \mathbb{R}^d$ given by $\boldsymbol{\theta} \mapsto \nabla A(\boldsymbol{\theta})$ is one-to-one on Θ.

(b). The log-likelihood function with data $\bar{\mu} \in \mathcal{K}$ has a maximum in Θ if and only if $\overline{\boldsymbol{\mu}} \in \mu(\Theta)$ and then the maximum $\overline{\boldsymbol{\theta}}$ is given uniquely as $\overline{\boldsymbol{\theta}} = \mu^{-1}(\overline{\boldsymbol{\mu}})$.

(c). $\mu(\Theta) = \mathrm{int}(\mathcal{K})$ and so in particular $\mu(\Theta)$ is open.

Remark. A direct consequence of the result (a) is that we can alternatively parametrize the exponential family using the mean of the canonical statistics

$$\nabla A(\boldsymbol{\theta}) = \mathbb{E}_\theta[t(X)],$$

and from the result (c), it is a diffeomorphism between the natural parameter space Θ and the interior of the mean-parameter space $\mathrm{int}(\mathcal{K})$.

4.2 Lecture 13: Conditional Inference

We will consider partitioning of the sufficient statistics $\mathbf{t}$ into $\boldsymbol{u}$ and v, $\mathbf{t} = (\boldsymbol{u}, \boldsymbol{v})$ with the corresponding partition of $\boldsymbol{\theta} = (\boldsymbol{\theta}_u, \boldsymbol{\theta}_v)$ and $\mu = (\mu_u, \mu_v)$. We consider two basic examples.

Example. We have shown that $X \sim N(\mu, \sigma)$ forms an exponential family with $\mathbf{t}(x) = (x, -x^2/2)$, $\boldsymbol{\theta} = \left(\frac{\mu}{\sigma^2}, \frac{1}{\sigma^2}\right)$ and $\boldsymbol{\mu} = \left(\mu, -\frac{1}{2}\left(\mu^2 + \sigma^2\right)\right)$. Given a sample x_i for $i = 1, \ldots, n$, we get an exponential family with the same canonical parameter and the sufficient statistics $\mathbf{t}\left(\mathbf{x}_{1:n}\right) = \left(\sum_i x_i, -\frac{1}{2}\sum_i x_i^2\right)$ and the mean parameter $\left(n\mu, -\frac{n}{2}\left(\mu^2 + \sigma^2\right)\right)$. Here we could take $\boldsymbol{u}(x) = \sum_i x_i$ and $\boldsymbol{v}(x) = -\frac{1}{2}\sum_i x_i^2$ or the other way around.

Proposition 4.2.1 (Marginal distribution)**.** In a regular exponential family with $\mathbf{t} = (\boldsymbol{u}, \boldsymbol{v})$ and $\boldsymbol{\theta} = (\boldsymbol{\theta}_u, \boldsymbol{\theta}_v)$ the marginal model for $\boldsymbol{u}$ is a regular exponential family for each given $\boldsymbol{\theta}_v$, depending on $\boldsymbol{\theta}_v$ but with the same parameter space for its mean value parameter μ_u.

Proof. The marginal distribution for u is obtained by integrating v out:

$$\begin{aligned} f(\boldsymbol{u};\boldsymbol{\theta}) &= \int g(\boldsymbol{u},\boldsymbol{v}) \exp\left\{\langle\boldsymbol{\theta}_u,\boldsymbol{u}\rangle + \langle\boldsymbol{\theta}_v,\boldsymbol{v}\rangle - A\left(\boldsymbol{\theta}_u,\boldsymbol{\theta}_v\right)\right\} \mathrm{d}v \\ &= \exp\left\{\langle\boldsymbol{\theta}_u,\boldsymbol{u}\rangle - A\left(\boldsymbol{\theta}_u,\boldsymbol{\theta}_v\right)\right\} \left(\int g(\boldsymbol{u},\boldsymbol{v}) \exp\left\{\langle\boldsymbol{\theta}_v,\boldsymbol{v}\rangle\right\} \mathrm{d}v\right). \end{aligned}$$

For any fixed $\boldsymbol{\theta}_v$ this has the form of a regular exponential family. This exponential family has canonical parameter $\boldsymbol{\theta}_u$ but the space of canonical parameters will typically depend on $\boldsymbol{\theta}_v$ (it is an intersection of Θ with $\boldsymbol{\theta}_v = $ fixed). However, the mean parameter space is always the same and equal to the interior of the convex hull of $\boldsymbol{u}(\mathcal{X})$, which is equal to the projection of $\mu(\Theta)$ on the u coordinates (i.e. μ_u). □

Proposition 4.2.2 (Conditional distribution)**.** With the same setting as in the previous proposition, the conditional model for $\mathbf{x}$ given $\boldsymbol{u}$ (and thus also for $\boldsymbol{v}$ given $\boldsymbol{u}$) is a regular exponential family with canonical statistics $\boldsymbol{v}$. The conditional model depends on $\boldsymbol{u}$ but with one and the same canonical parameter $\boldsymbol{\theta}_v$ as in the joint model.

Proof. We have

$$f(\mathbf{x} \mid \boldsymbol{u};\boldsymbol{\theta}) = \frac{f(\boldsymbol{u},\mathbf{x};\boldsymbol{\theta})}{f(\boldsymbol{u};\boldsymbol{\theta})} = \frac{h(\mathbf{x}) \exp\left\{\langle\boldsymbol{\theta}_v, v(\mathbf{x})\right\}}{\int g(\boldsymbol{u},v) \exp\left\{\langle\boldsymbol{\theta}_v, v(\mathbf{x})\right\} \mathrm{d}v}.$$

For the fixed value of u, the expression in the denominator does not depend on $\mathbf{x}$ but only on $\boldsymbol{\theta}_v$ and so it represents the normalizing constant of this distribution. Note $f(\mathbf{x} \mid \boldsymbol{u};\boldsymbol{\theta})$ is defined only for those $\mathbf{x}$ for which $\mathbf{t}(\mathbf{x}) = (\boldsymbol{u}(\mathbf{x}), \boldsymbol{v}(\mathbf{x})) \in \mathbf{t}(\mathcal{X})$, and thus the space of sufficient statistics depends on $\boldsymbol{u}$. However, the canonical parameter is the same, $\boldsymbol{\theta}_v$ in the projection Θ_v of Θ on the coordinates $\boldsymbol{\theta}_v$. To get $f(\boldsymbol{v} \mid \boldsymbol{u};\boldsymbol{\theta})$ we only substitute $h(\mathbf{x})$ for $g(\boldsymbol{u},\boldsymbol{v})$ above but otherwise the argument is the same. □

Remark. Explicit calculations with these marginal and conditional distributions are typically hard but the two results above are important in guiding our analysis.

As a corollary of the above two results, we obtain a useful result on a range of possible alternative parametrizations.

For a given split $\mathbf{t} = (\boldsymbol{u}, \boldsymbol{v})$ consider the vector $(\boldsymbol{\mu}_u, \boldsymbol{\theta}_v)$ with $\boldsymbol{\mu}_u \in \mu_u(\Theta)$ and $\boldsymbol{\theta}_v \in \Theta_v$. Here by Θ_v we denote the projection of Θ on the coordinates $\boldsymbol{\theta}_v$ and by $\mu_u(\Theta)$ we mean the projection of $\mu(\Theta)$ on the coordinates μ_u.

Proposition 4.2.3. The mixed parametrization $(\boldsymbol{\mu}_u, \boldsymbol{\theta}_v)$ is a valid parametrization with the parameter space $\mu_u(\Theta) \times \Theta_v$ (variational independence!). The Fisher information for $(\boldsymbol{\mu}_u, \boldsymbol{\theta}_v)$ is

$$I\left(\boldsymbol{\mu}_{u'}, \boldsymbol{\theta}_v\right) = \begin{bmatrix} (\Sigma_{uu})^{-1} & 0 \\ 0 & \left((\Sigma^{-1})_{vv}\right)^{-1} \end{bmatrix},$$

where $\Sigma = \operatorname{var}(\mathbf{t})$ and $\Sigma_{uu} = \operatorname{var}(\boldsymbol{u})$. The same formula holds for the observed information in the MLE, $J\left(\bar{\mu}_u, \overline{\boldsymbol{\theta}}_v\right)$.

Proof. Fix an exponential family with canonical statistics $\mathbf{t}(\mathbf{x})$ and canonical parameter $\boldsymbol{\theta} \in \Theta$. Note that the distribution of $\mathbf{t}(\mathbf{X})$ is an exponential family with the same canonical parameter. This distribution is uniquely defined by the marginal distribution of u and the conditional distribution of v given u. The latter forms an exponential family with canonical parameter $\boldsymbol{\theta}_v \in \Theta_v$. Now fix $\boldsymbol{\theta}_v$, which corresponds to fixing the conditional distribution of v given u. Therefore, the marginal distribution of u is an exponential family with the mean parameter μ_u. Hence, the range of this mean parameter is the interior of the convex hull of $\boldsymbol{u}(\mathcal{X})$ (independent on $\boldsymbol{\theta}_v$). This is precisely the projection of $\boldsymbol{\mu}(\Theta)$ on $\boldsymbol{\mu}_u$ and this shows that the map $\boldsymbol{\theta} \mapsto (\boldsymbol{\mu}_{u'}, \boldsymbol{\theta}_v)$ is one-to-one with range $\mu_u(\Theta) \times \Theta_v$. □

Now suppose ψ is the parameter of interest, where (λ, ψ) is a transformation of $\boldsymbol{\theta}$. For simplicity we focus on the case when $\boldsymbol{\psi} = \boldsymbol{\theta}_v$ and $\lambda = \mu_u$ is regarded as nuisance parameter. Recall that $\lambda = \mu_u = \mathbb{E}_{\boldsymbol{\theta}}(u)$ is the preferable nuisance parameter (rather than $\boldsymbol{\theta}_u$), since $\boldsymbol{\theta}_v$ and $\boldsymbol{\mu}_u$ are variation independent and information orthogonal.

Proposition 4.2.4 (Conditionality principle for full families)**.** Statistical inference about the canonical parameter component $\boldsymbol{\theta}_v$ in presence of the nuisance parameter $\boldsymbol{\lambda} = \boldsymbol{\mu}_u = \mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{u})$ should be made conditional on $\boldsymbol{u}$, that is, the conditional model for $\mathbf{x}$ or $\boldsymbol{v}$ given $\boldsymbol{u}$.

Remark. In words, the conditional principle is exercised as follows:

(a). Observe data $X_1, \cdots, X_n$.

(b). Derive the marginal model of $\boldsymbol{\mu}_u$ assuming $\boldsymbol{\theta}_v$ is fixed and using the data to make inference on $\boldsymbol{\mu}_u$. (Under a regular exponential family, the *MLE* is independent of $\boldsymbol{\theta}_v$).

(c). Derive the conditional model $\boldsymbol{\theta}_v \mid \boldsymbol{\mu_u}$ and make inference on $\boldsymbol{\theta}_v$.

This is only a recommendation so rather than providing a formal proof we motivate this statement informally.

Example (Conditional MLE for the variance component in the normal family)**.** Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(m, \sigma^2),$$

and consider the mixed parametrization

$$\lambda = \mu_1 = m, \qquad \theta_2 = \frac{1}{\sigma^2}.$$

Then:

(a). the MLE of the nuisance mean-parameter is

$$\hat{\lambda} = \bar{X};$$

(b). the conditional likelihood of θ_2 given $\bar{X}$ is

$$L_c(\theta_2 \mid \bar{X} = \bar{x}) \propto \theta_2^{(n-1)/2} \exp\left\{ -\frac{\theta_2}{2} \sum_{i=1}^{n} (X_i - \bar{X})^2 \right\};$$

(c). hence the conditional maximum likelihood estimator of θ_2 is

$$\hat{\theta}_2^c = \frac{n-1}{\sum_{i=1}^{n} (X_i - \bar{X})^2},$$

or equivalently,

$$\hat{\sigma}_c^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

Proof. The joint density of $X_1, \ldots, X_n$ under the parametrization (λ, θ_2) is

$$f(x_1, \ldots, x_n; \lambda, \theta_2) = (2\pi)^{-n/2} \theta_2^{n/2} \exp\left\{ -\frac{\theta_2}{2} \sum_{i=1}^{n} (x_i - \lambda)^2 \right\}.$$

Let

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i, \qquad R = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Using the decomposition

$$\sum_{i=1}^{n} (x_i - \lambda)^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2 + n(\bar{x} - \lambda)^2 = R + n(\bar{x} - \lambda)^2,$$

we obtain

$$f(x; \lambda, \theta_2) = (2\pi)^{-n/2} \theta_2^{n/2} \exp\left\{ -\frac{\theta_2}{2} R \right\} \exp\left\{ -\frac{n\theta_2}{2} (\bar{x} - \lambda)^2 \right\}.$$

First, consider inference for the nuisance parameter λ. The marginal law of $\bar{X}$ is

$$\bar{X} \sim N\left(\lambda, \frac{1}{n\theta_2} \right),$$

so its log-likelihood is, up to an additive constant,

$$\ell_{\text{marg}}(\lambda, \theta_2; \bar{x}) = \frac{1}{2} \log \theta_2 - \frac{n\theta_2}{2} (\bar{x} - \lambda)^2.$$

For fixed θ_2, this is maximized at

$$\hat{\lambda} = \bar{x}.$$

Thus the MLE of the nuisance mean-parameter is

$$\hat{\lambda} = \bar{X}.$$

Now apply the mixed-parametrization principle: inference about the canonical component θ_2 in the presence of the nuisance parameter λ should be made conditionally on $U = \bar{X}$. The conditional density is

$$f(x \mid \bar{x}; \theta_2) = \frac{f(x; \lambda, \theta_2)}{f_{\bar{X}}(\bar{x}; \lambda, \theta_2)}.$$

Since
$$f_{\bar{X}}(\bar{x};\lambda,\theta_2) = \left(\frac{n\theta_2}{2\pi}\right)^{1/2} \exp\left\{-\frac{n\theta_2}{2}(\bar{x}-\lambda)^2\right\},$$
the terms involving λ cancel, leaving
$$f(x \mid \bar{x};\theta_2) \propto \theta_2^{(n-1)/2} \exp\left\{-\frac{\theta_2}{2}R\right\}.$$
Hence the conditional log-likelihood is
$$\ell_c(\theta_2 \mid \bar{x}) = \frac{n-1}{2}\log\theta_2 - \frac{\theta_2}{2}R + \text{const.}$$
Differentiating with respect to θ_2,
$$\frac{d}{d\theta_2}\ell_c(\theta_2 \mid \bar{x}) = \frac{n-1}{2\theta_2} - \frac{R}{2}.$$
Setting this equal to zero gives
$$\hat{\theta}_2^c = \frac{n-1}{R} = \frac{n-1}{\sum_{i=1}^n (X_i-\bar{X})^2}.$$
Since $\theta_2 = 1/\sigma^2$, the corresponding conditional estimator of σ^2 is
$$\hat{\sigma}_c^2 = \frac{1}{\hat{\theta}_2^c} = \frac{1}{n-1}\sum_{i=1}^n (X_i-\bar{X})^2.$$
This proves the result. □

Remark. This conditional estimator differs from the ordinary joint MLE of σ^2, which is
$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^n (X_i-\bar{X})^2.$$
The reason is that the conditional likelihood removes the nuisance location parameter by conditioning on $\bar{X}$, so the residual variation carries only $n-1$ degrees of freedom.

4.3 Lecture 14: Applications

4.3.1 KL Divergence

The Fenchel conjugate of the cumulant function A is the function
$$A^*(\mathbf{t}) = \sup\left\{\langle\boldsymbol{\theta},\mathbf{t}\rangle - A(\boldsymbol{\theta}) : \boldsymbol{\theta}\in\mathbb{R}^d\right\}.$$

Proposition 4.3.1. The function A^* admits the following properties:

- For regular exponential families $A^*(\mathbf{t}) < \infty$ if and only if $\mathbf{t} \in \mu(\Theta)$.
- The function A^* is convex as a supremum of linear functions and, in fact, strictly convex.
- If $\mathbf{t} \in \mu(\Theta)$, the unique optimizer of the log-likelihood is $\theta(\mathbf{t})$, where $\theta : M \to \Theta$ is the inverse of the map $\mu : \Theta \to M$. It follows that, for $\mathbf{t} \in \mu(\Theta)$,
$$A^*(\mathbf{t}) = \langle\theta(\mathbf{t}),\mathbf{t}\rangle - A(\theta(\mathbf{t}))$$

Remark. Alternatively, for $\boldsymbol{\theta} \in \Theta$,

$$A^*(\mu(\boldsymbol{\theta})) = \langle \boldsymbol{\theta}, \mu(\boldsymbol{\theta}) \rangle - A(\boldsymbol{\theta})$$

implying in particular that A^* is smooth, since μ and A are both smooth. Since $\mu(\theta(\mathbf{t})) = \mathbf{t}$, we obtain

$$\nabla A^*(\mathbf{t}) = \theta(\mathbf{t}) + \nabla\theta(\mathbf{t}) \cdot \mathbf{t} - \nabla\theta(\mathbf{t}) \cdot \mu(\theta(\mathbf{t})) = \theta(\mathbf{t}),$$

where the gradient is taken with respect to t.

Definition 4.3.2. For two distributions over some state-space $\mathcal{X}$ with densities p, q with respect to some measure μ, the Kullback-Leibler divergence $\mathrm{K}(p, q)$ is defined as

$$\mathrm{K}(p, q) := \int_{\mathcal{X}} p(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})} \mu(\mathrm{d}\mathbf{x}).$$

The following result is well-known.

Proposition 4.3.3. We have $K(p, q) \geq 0$ with equality if and only if $p = q$ almost surely.

Given two distributions $\mathbb{P}_{\theta_1}, \mathbb{P}_{\theta_2}$ in the given exponential family we write the corresponding Kullback-Leibler divergence as $K(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$. We easily check that

$$K(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = A(\boldsymbol{\theta}_2) - A(\boldsymbol{\theta}_1) - \langle \boldsymbol{\theta}_2 - \boldsymbol{\theta}_1, \boldsymbol{\mu}_1 \rangle .$$

Proposition 4.3.4. Consider two distributions in the exponential family (1.1), one with the mean parameter $\mu_1 \in \mu(\Theta)$ and the other with canonical parameter $\boldsymbol{\theta}_2 \in \Theta$. If this exponential family is regular, the Kullback-Leibler divergence between these two distributions is

$$\mathrm{K}(\mu_1, \theta_2) = -\langle \mu_1, \theta_2 \rangle + A^*(\mu_1) + A(\theta_2).$$

The Kullback-Leibler divergence is well defined and nonnegative over $\mu(\Theta) \times \Theta$. Moreover, $\boldsymbol{K}(\boldsymbol{\mu}_1, \boldsymbol{\theta}_2) = 0$ if and only if $\boldsymbol{\mu}_1 = \mu(\boldsymbol{\theta}_2)$.

Proof. It suffices to notice that $A^*(\mu_1) = \langle \boldsymbol{\theta}_1, \mu_1 \rangle - A(\boldsymbol{\theta}_1)$. □

The reason to express the Kullback-Leibler distance in terms of μ_1 and θ_2 rather than θ_1, θ_2 (as usually done in the literature) is that we wish to exploit the following basic result.

Proposition 4.3.5. The Kullback-Leibler divergence $\mathrm{K}(\boldsymbol{\mu}_1, \boldsymbol{\theta}_2)$ is strictly convex both in μ_1 and in $\boldsymbol{\theta}_2$.

Proof. This follows directly from

$$\mathrm{K}(\mu_1, \theta_2) = -\langle \mu_1, \theta_2 \rangle + A^*(\mu_1) + A(\theta_2).$$

and the fact that both $A(\boldsymbol{\theta})$ and $A^*(\boldsymbol{\mu})$ are strictly convex functions. □

We state now the most fundamental results motivating exponential families. The maximum entropy principle states that under uncertainty, one should take a model which maximizes the entropy subject to constraints on the known features about the system. We show that the exponential family arises naturally if the constraint is given by the expected value of some statistics.

Recall that for a distribution $\mathbb{P}$ that admits a density function $p(x)$ with respect to the base measure μ, the entropy $H_{\mathbb{P}}$ of $\mathbb{P}$ is

$$\mathrm{H}_{\mathbb{P}} = -\mathbb{E}_{\mathbb{P}} \log p(X) = -\int \log p(x) p(x) \mu(\mathrm{d}x).$$

For notational simplicity, in what follows consider the exponential family where the function $h(\mathbf{x})$ has been incorporated into the base measure μ. In this case each distribution in this family has the same support, which is equal to the support of μ. For such an exponential family $\mathbb{P}_\theta$ we have

$$\mathrm{H}_{\mathbb{P}_{\boldsymbol{\theta}}} = -\langle \boldsymbol{\theta}, \mu(\boldsymbol{\theta}) \rangle + A(\boldsymbol{\theta}) = -A^*(\mu(\boldsymbol{\theta})).$$

Consider the problem of maximizing the entropy for all distributions $\mathbb{P}$ that admit a density function absolutely continuous with respect to μ and $\mathbb{E}_{\mathbb{P}}(t(X)) = t_0$

$$\text{maximize}\, \mathrm{H}_{\mathrm{P}} \quad \text{s.t. } \mathbb{P} \sim \mu, \quad \mathbb{E}_{\mathbb{P}} t(X) = t_0.$$

The following result provides an important characterization of exponential families.

Theorem 4.3.6. Consider the exponential family and suppose there exists $\boldsymbol{\theta}_0 \in \Theta$ such that $\nabla A(\boldsymbol{\theta}_0) = \boldsymbol{t}_0$. Then for any distribution $\mathbb{P}$, which satisfies the mean condition $\mathbb{E}_{\mathbb{P}} t(X) = t_0$, we have

$$H_{\mathbb{P}_{\theta_0}} - H_{\mathbb{P}} = K\left(\mathbb{P}, \mathbb{P}_{\theta_0}\right).$$

Thus, $\mathbb{P}_{\theta_0}$ is the unique solution to the maximum entropy problem.

Proof. Note that $\nabla A(\boldsymbol{\theta}_0) = t_0$ is equivalent to $\mu(\boldsymbol{\theta}_0) = \mathbb{E}_{\boldsymbol{\theta}_0} \mathbf{t}(X) = \mathbf{t}_0$. Let $p(\mathbf{x})$ be the density of $\mathbb{P}$ with respect to μ. Consider

$$\begin{aligned}
\mathrm{K}\left(\mathbb{P}, \mathbb{P}_{\theta_0}\right) &= \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{f(\mathbf{x}; \boldsymbol{\theta}_0)} \mu(\mathrm{d}\mathbf{x}) \\
&= -H_{\mathbb{P}} - \int \left(\langle \boldsymbol{\theta}_0, \mathbf{t}(\mathbf{x}) \rangle - A(\boldsymbol{\theta}_0)\right) p(x) \mu(\mathrm{d}\mathbf{x}) \\
&= -H_{\mathbb{P}} - \langle \boldsymbol{\theta}_0, \mathbf{t}_0 \rangle + A(\boldsymbol{\theta}_0) = H_{\mathbb{P}_{\theta_0}} - H_{\mathbb{P}}.
\end{aligned}$$

Therefore, $H_{\mathbb{P}} \leq H_{\mathbb{P}_{\theta_0}}$ with equality if and only if $\mathbb{P} = \mathbb{P}_{\theta_0}$. This concludes the proof. □

4.3.2 Generalized Linear Models

The generalized linear models are formulated based on the construction of exponential families. Here we provide only a basic treatment that explains the origin of the construction and the most important examples.

Consider the pairs $(y_1, x_1), \ldots, (y_n, x_n)$, where the input $x_1, \ldots, x_n \in \mathbb{R}^d$ are considered fixed and the outputs $y_1, \ldots, y_n$ are independent observations each from density

$$f\left(y_i; \mathbf{x}_i, w, \sigma^2\right) = h\left(y_i, \sigma^2\right) \exp\left\{\frac{1}{\sigma^2}\left(y_i \theta_i(w) - A\left(\theta_i(w)\right)\right)\right\}.$$

where $\theta_i(w) = \mathbf{x}_i^\top w$ and σ^2 is called the dispersion term. From the standard theory, we get immediately that

$$\mu(\theta_i) := \mathbb{E}\left[Y_i \mid \mathbf{x}_i, w, \sigma^2\right] = A'(\theta_i).$$

Indeed, the standard result show that for a fixed $\sigma^2, \mathbb{E}\left(\frac{1}{\sigma^2}Y\right) = \frac{1}{\sigma^2}A'(\theta)$. In the same way, we argue that

$$V\left(\theta, \sigma^2\right) := \operatorname{var}_\theta\left(Y_i \mid \mathbf{x}_i, w, \sigma^2\right) = \sigma^2 A''\left(\theta_i\right).$$

From now on we fix σ^2 and with no loss of generality we take $\sigma^2 = 1$. The log-likelihood function is

$$\ell_n(w) = \frac{1}{n}\sum_{i=1}^n \log f\left(y_i; \mathbf{x}_i, w\right) = \left(\frac{1}{n}\sum_{i=1}^n y_i \mathbf{x}_i\right)^\top w - \frac{1}{n}\sum_{i=1}^n A\left(\mathbf{x}_i^\top w\right) + \text{ const.}$$

Since a composition of a convex and a linear function is convex, we conclude that $\ell_n(w)$ is a concave function.

Let $\mathbf{X} \in \mathbb{R}^{n\times d}$ be the data matrix with rows $\mathbf{x}_i$ and let $\mathbf{y}$ be the vectors with entries y_i. It can be shown that the unique optimizer of $\ell_n(w)$ (if it exists), must satisfy the likelihood equations

$$\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top A'(\mathbf{X}\widehat{w}),$$

where the function A' is applied elementwise to the vector $\mathbf{X}\widehat{w}$. In machine learning ,it is customary to call A' in this context an activation function.

Example (Linear regression)**.** Consider the univariate Gaussian distribution. Suppose now that σ^2 is fixed and the model is parametrized only by the mean μ. We can rewrite the density as

$$f\left(y; \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{y^2}{2\sigma^2}} \exp\left\{\frac{1}{\sigma^2}\left(y\mu - \frac{\mu^2}{2}\right)\right\} = h\left(y, \sigma^2\right)\exp\left\{\frac{1}{\sigma^2}(y\mu - A(\mu))\right\},$$

where $h\left(y, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{y^2}{2\sigma^2}}$ and $A(\mu) = \mu^2/2$. If we model the mean μ as a linear function of a vector $\mathbf{x}$, $\mu = w^\top \mathbf{x}$ we get the standard Gaussian linear regression

$$f\left(y \mid \mathbf{x}, w, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}\left(y - w^\top \mathbf{x}\right)^2\right\}.$$

We easily see that this is a generalized linear model as defined above. As $A'(\mu) = \mu$ and $A''(\mu) = 1$, we get $\mathbb{E}\left(Y \mid \mathbf{x}, w, \sigma^2\right) = \mu = w^\top \mathbf{x}$, $\operatorname{var}\left(Y \mid \mathbf{x}, w, \sigma^2\right) = \sigma^2$ and the MLE equations are $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X}$ w, which are identical to the OLS estimating equations.

Example (Binomial regression)**.** The binomial distribution for the number of successes in n trials, $y \in \{0, \ldots, n\}$ has the probability mass function

$$\operatorname{Bin}(y; n, p) = \binom{n}{y} p^y (1-p)^{n-y} = \binom{n}{y}\exp\left\{y \log\frac{p}{1-p} + n\log(1-p)\right\}.$$

If $\theta = \log\frac{p}{1-p}$ then $p = \frac{e^\theta}{1+e^\theta}$, which is typically called a sigmoid function and denoted by $\sigma(\theta)$. Moreover, $A(\theta) = n\log\left(1 + e^\theta\right)$ and so $A'(\theta) = n\sigma(\theta)$. Now we use this distribution for the GLM setup. If the response variable is the number of successes in n trials, $y \in \{0, \ldots, n\}$, we can use binomial regression, which is defined by

$$f(y; \mathbf{x}, N, w) = \operatorname{Bin}\left(y; n, \sigma\left(w^\top \mathbf{x}\right)\right),$$

where the logistic regression becomes a special case with $n = 1$. This clearly has the right form and $\mathbb{E}(Y) = n\theta, \operatorname{var}(Y) = np(1-p)$. The likelihood equations are

$$\mathbf{X}^\top \mathbf{y} = n\mathbf{X}^\top \sigma(\mathbf{X}\hat{w}).$$

Example (Poisson regression)**.** The Poisson distribution is a distribution over $\mathcal{X} = \{0, 1, 2, \ldots\}$ with the probability mass function

$$f(y; \mu) = \frac{\mu^y}{y!} e^{-\mu} = \frac{1}{y!} \exp\left(y \log \mu - \mu\right) \quad y \in \mathcal{X}.$$

This is an exponential family with canonical parameter $\theta = \log(\mu)$ and $A(\theta) = \mu = \exp(\theta)$. In Poisson regression, we take $\theta = w^\top x$.

Remark. Modelling the canonical parameters as a linear function of external variables is not the only choice. For example, the Bernoulli distribution gives the logistic regression. An alternative approach is to model $p = \Phi\left(w^\top x\right)$, which gives the probit regression. Generalized linear models with non-canonical link functions correspond to curved exponential families.

4.3.3 Conjugate Priors

An important advantage of exponential families over more general classes of distributions is that they admit an explicit conjugate prior for Bayesian computations. The conjugate prior measure for the exponential family is given by the density (w.r.t. the Lebesgue measure) of the form

$$\pi(\boldsymbol{\theta}) = C \exp\left\{\langle \boldsymbol{\theta}, \tau \rangle - n_0 A(\boldsymbol{\theta})\right\}, \quad \tau \in \mathbb{R}^d, n_0 \geq 0.$$

Note that $\pi(\boldsymbol{\theta}) \equiv 0$ outside of Θ because there $A(\boldsymbol{\theta}) \equiv +\infty$. It can also be shown that the distribution is normalizable if $n_0 > 0$ and $\tau / n_0 \in \mathcal{K} = \operatorname{conv}(T(\mathcal{X}))$.

Proposition 4.3.7. For a regular exponential family, consider the conjugate prior

$$\pi(\boldsymbol{\theta}) = C \exp\left\{\langle \boldsymbol{\theta}, \tau \rangle - n_0 A(\boldsymbol{\theta})\right\}, \quad \tau \in \mathbb{R}^d, n_0 \geq 0.$$

If $\boldsymbol{\mu}$ is the mean parameter then we have $\mathbb{E}_{\boldsymbol{\theta}}[\boldsymbol{\mu}] = \frac{\tau}{n_0}$.

Proof. We use the fact that $\boldsymbol{\mu} = \nabla A(\boldsymbol{\theta})$. We have

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{\theta}}\left[\tau - n_0 \nabla A(\boldsymbol{\theta})\right] &= \int_\Theta \left(\tau - n_0 \nabla A(\boldsymbol{\theta})\right) C \exp\left\{\langle \boldsymbol{\theta}, \tau \rangle - n_0 A(\boldsymbol{\theta})\right\} \mathrm{d}\boldsymbol{\theta} \\
&= \int_\Theta \nabla\left(C \exp\left\{\langle \boldsymbol{\theta}, \tau \rangle - n_0 A(\boldsymbol{\theta})\right\}\right) \mathrm{d}\boldsymbol{\theta} \\
&= \int_\Theta \nabla \pi(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta} = 0
\end{aligned}$$

□

Example. Consider the data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ from the distribution $\mathbb{P}_\theta$. If the prior is of the form above, then the posterior $\pi\left(\boldsymbol{\theta} \mid \mathbf{x}_{1:n}\right)$ satisfies

$$\begin{aligned}
\pi\left(\boldsymbol{\theta} \mid \mathbf{x}_{1:n}\right) &\propto \pi(\boldsymbol{\theta}) \prod_{i=1}^n f\left(\mathbf{x}_i; \boldsymbol{\theta}\right) \\
&= C \prod_{i=1}^n h\left(\mathbf{x}_i\right) \exp\left\{\left\langle \boldsymbol{\theta}, \tau + \sum_{i=1}^n \mathbf{t}\left(\mathbf{x}_i\right)\right\rangle - \left(n + n_0\right) A(\boldsymbol{\theta})\right\}
\end{aligned}$$

And so it has the same general form as the prior, where (τ, n_0) is replaced with $(\tau + n\bar{\mu}_n, n_0 + n)$. The Bayes estimator of the mean parameter is then

$$\mathbb{E}\left[\boldsymbol{\mu} \mid \mathbf{x}_{1:n}\right] = \frac{\tau + n\bar{\mu}_n}{n_0 + n} = \lambda \frac{\tau}{n_0} + (1 - \lambda)\bar{\mu}_{n'}$$

where $\lambda = \frac{n_0}{n_0+n}$ and so the Bayes estimator is a convex combination of the MLE $\bar{\mu}_n$ and the prior mean τ/n_0.

Remark. We can easily define conjugate priors over non-canonical parameters. For example, to get a conjugate prior over the mean parameter we simply change $\boldsymbol{\theta}$ with $\theta(\boldsymbol{\mu})$ and change the definition of C so that the corresponding expression integrates to 1. Note that this is not the same as the density obtained through the change of variable formula.

Chapter 5 Multiple Testing

5.1 Lecture 17: Testing using Family-wise Error Rate

5.1.1 Testing on Submodels

Consider an exponential family and a p-dimensional linear space $\mathcal{L} \subseteq \mathbb{R}^d$ and the corresponding $\Theta_0 = \Theta \cap \mathcal{L}$. There exists a simple affine change of coordinates form θ to (λ, ψ) such that $\Theta_0 = \{(\boldsymbol{\lambda}, \boldsymbol{\psi}) : \boldsymbol{\psi} = \mathbf{0}\}$. If the basis of $\mathcal{L}$ is given by the columns of $\mathbb{R}^{d\times p}$ then $\boldsymbol{\theta} = A\boldsymbol{\lambda}$ parametrizes Θ_0. Thus, for $\boldsymbol{\theta} \in \Theta_0$ we have

$$\langle \boldsymbol{\theta}, t\rangle = \langle A\boldsymbol{\lambda}, t\rangle = \left\langle \boldsymbol{\lambda}, A^\top t\right\rangle = \langle \boldsymbol{\lambda}, u\rangle,$$

where $u = A^\top t$ is the sufficient statistics of the p-dimensional model parametrized by Θ_0.

Thus, without loss of generality we decompose $\mathbf{t} = (u, v)$ with $\boldsymbol{\lambda} = \mu_u$ and $\boldsymbol{\psi} = \boldsymbol{\theta}_v$. We want to test the q-dimensional hypothesis $\psi_0 = 0$, where $(q = d - p)$:

$$H_0: \quad \boldsymbol{\psi} = 0, \quad \text{versus} \quad \boldsymbol{\psi} \neq 0.$$

We first consider a model reduction of the canonical statistics from $t = (u, v)$ to u. Recall that any inference on ψ should be done conditionally on u. All information provided by u is consumed in estimating μ_u and this parameter has no information about ψ by variational inference.

Example. Let $X_1, \ldots, X_n \in \mathbb{R}^2$ be i.i.d.

$$X_i \sim \mathcal{N}_2(\mu, I_2), \qquad \mu = (\mu_1, \mu_2)^\top.$$

This is a regular exponential family with natural parameter $\theta = \mu \in \mathbb{R}^d$ $(d = 2)$ and sufficient statistic

$$t = \sum_{i=1}^n X_i \in \mathbb{R}^2.$$

Up to an additive constant, the log-likelihood is

$$\ell(\theta) = \langle \theta, t\rangle - \frac{n}{2}\|\theta\|^2.$$

Consider the p-dimensional linear space $(p = 1)$

$$\mathcal{L} = \{(\alpha, \alpha) : \alpha \in \mathbb{R}\} = \operatorname{span}\{(1, 1)\}.$$

Choose the basis matrix $A \in \mathbb{R}^{2\times 1}$ as

$$A = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad \Theta_0 = \Theta \cap \mathcal{L} = \{\theta : \theta_1 = \theta_2\}.$$

Then $\theta = A\lambda$ parametrizes Θ_0, and for $\theta \in \Theta_0$,

$$\langle \theta, t \rangle = \langle A\lambda, t \rangle = \langle \lambda, A^\top t \rangle = \langle \lambda, u \rangle, \qquad u := A^\top t = \frac{t_1 + t_2}{\sqrt{2}}.$$

Hence the reduced model has 1-dimensional sufficient statistic u. Recall that the desired parametrization of $(\boldsymbol{\lambda}, \boldsymbol{\psi})$ satisfies

$$\Theta_0 = \{(\lambda, \psi) : \psi = 0\}.$$

Let B be an orthonormal basis of $\mathcal{L}^\perp$:

$$B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The corresponding decomposition of the sufficient statistic (under an affine transformation) is

$$u = A^\top t = \frac{t_1 + t_2}{\sqrt{2}}, \qquad v = B^\top t = \frac{t_1 - t_2}{\sqrt{2}}.$$

The corresponding parameters that are variational independent are then given by

$$\boldsymbol{\lambda} = \mu_u = \frac{1}{\sqrt{2}}(\mu_1 + \mu_2), \quad \boldsymbol{\psi} = \boldsymbol{\theta}_v = \frac{1}{\sqrt{2}}(\mu_1 - \mu_2)$$

Here $q = d - p = 1$, and the hypothesis

$$H_0 : \ \psi = 0 \quad \text{vs} \quad H_1 : \ \psi \neq 0$$

is equivalent to

$$H_0 : \ \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \ \mu_1 \neq \mu_2.$$

A convenient (Wald/LR/score-equivalent) test statistic is

$$W = \frac{v^2}{n} = \frac{1}{n}\left(\frac{t_1 - t_2}{\sqrt{2}}\right)^2 = \frac{n}{2}\,(\bar{X}_1 - \bar{X}_2)^2,$$

and under H_0,

$$W \ \sim \ \chi^2_1.$$

Consider the conditional distribution for v given u. Inserting $\psi = 0$ simplifies this to

$$f_0(v \mid u) = \frac{g(u, v)}{g_0(u)},$$

where index 0 indicates distribution under H_0, and in particular $g_0(u) = \int g(u, v) \mathrm{d}v$ is the structure function in the marginal exponential family for u under H_0. Note that, under the null, the conditional distribution of v given u is parameter-free.

We propose the following test for

$$H_0 : \boldsymbol{\psi} = \mathbf{0} \quad \text{v.s.} \quad H_1 : \boldsymbol{\psi} \neq \mathbf{0}$$

(a). Use $f_0(v \mid \bar{u})$ as the test statistic, which, under the null, is equal to the conditional density of v given $\bar{u}$.

(b). Reject H_0 if $f_0(\bar{v} \mid \bar{u})$ is too small, and calculate the p-value as

$$\mathbb{P}\left(f_0(v \mid \bar{u}) \leq f_0(\bar{v} \mid \bar{u})\right) = \int_{f_0(v|\pi) \leq f_0(\bar{v}|\pi)} f_0(v \mid \bar{u}) \mathrm{d}v,$$

where $\bar{u}, \bar{v}$ denote the observed quantities.

Remark. These test follow the Fisher's principle of exact tests. "Exactness" comes from the fact that the test achieves exactly the desired size α (as opposed to approximate tests). One obvious question is about the power of such a procedure.

Consider a special case of the above setting when $\mathbf{t} = (u, v)$ with v one-dimensional. In this case θ_v is one-dimensional too and we can obtain a uniform most powerful test for $H_0 : \psi = 0$ versus $H_1 : \psi \neq 0$. Recall that, under the null, the conditional distribution of v given u is parameter-free. We will then use v directly as the test statistic with u fixed and consider the test $\varphi^*(x)$ defined as

$$\varphi^*(x) = \begin{cases} 1 & \text{if } v > c^*(u), \\ 1 & \text{if } v < c_*(u), \\ \gamma^*(u) & \text{if } v = c^*(u), \\ \gamma_*(u) & \text{if } v = c_*(u), \\ 0 & \text{if } v \in \left(c_*(u), c^*(u)\right), \end{cases}$$

with $c(\cdot)$ and $\gamma(\cdot)$ adjusted so that the test has exactly level α.

Theorem 5.1.1. If the exponential family is regular and θ_v is one-dimensional, then φ^* is a uniformly most powerful unbiased test of $H_0 : \theta_v = 0$ versus $H_1 : \theta_v \neq 0$.

Example (Mean value test for a normal distribution)**.** Consider testing $H_0 : \mu = 0$ for a sample of size n from $N\left(\mu, \sigma^2\right)$. Note that the canonical parameters of this family are $\left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$ with sufficient statistics $\left(\sum x_i, \sum x_i^2\right)$, so equivalently we would like to test $\theta_1 = 0$ with $v = \sum_i x_i$. The exact test should be conditional, given $u = \sum_i x_i^2 = \|x\|^2$.
Indeed, from the lemma below, $u \perp\!\!\!\perp v/\sqrt{u}$ and the vector $\frac{1}{\sqrt{u}}\left(x_1, \ldots, x_n\right)$ is uniformly distributed on the $(n-1)$-dimensional unit sphere. Therefore, under $H_0 : \mu = 0$, define

$$Z := \frac{V}{\sqrt{nU}}.$$

Then given $U = u$,

$$Z \mid (U = u) \stackrel{d}{=} S_1,$$

where S_1 is the first coordinate of a point uniform on the unit sphere S^{n-1}. That coordinate has density on $[-1, 1]$

$$f_Z(z) = c_n \left(1 - z^2\right)^{\frac{n-3}{2}}, \quad c_n = \frac{\Gamma(n/2)}{\sqrt{\pi}\Gamma((n-1)/2)}.$$

Therefore, the conditional density of $V \mid U = u$ is

$$f_0(v \mid u) = \frac{1}{\sqrt{nu}} c_n \left(1 - \frac{v^2}{nu}\right)^{\frac{n-3}{2}}, \quad |v| \leq \sqrt{nu},$$

and 0 otherwise. The test proposed above is then equivalent to a test on the ratio $v/\sqrt{u}$.

The last step is to show that this test is equivalent to the t-test. Let $\tau = \sqrt{n}x/s$ be the t-test statistic. Let us rewrite $v/\sqrt{u}$ as

$$\frac{n\bar{x}}{\sqrt{\sum_i x_i^2}} = \frac{n\bar{x}}{\sqrt{(n-1)s^2 + nx^2}} = \frac{n\bar{x}/s}{\sqrt{(n-1) + (\sqrt{n}x/s)^2}} = \frac{\sqrt{n}\tau}{\sqrt{(n-1)+\tau^2}}.$$

The right-hand side is seen to be an odd function and a monotone function of τ. Thus, a single tail or a symmetric pair of tails in $u/\sqrt{v}$ is equivalent to a single or symmetric pair of tails in the usual t-test.

5.1.2 The Bonferroni Procedure

As a general motivation, consider the problem of simultaneously testing a finite number of null hypotheses H_0^i for $i = 1, \dots, m$.

Example. Suppose that we have m genes and data about expression levels for each gene among healthy individuals and those with lung cancer.

	Healthy (k patients)	Lung cancer (l patients)
Expression Level of Gene i	$x_{ij}^{(0)}, 1 \le j \le k$	$x_{ij}^{(1)}, 1 \le j \le l$

The i-th null hypothesis, denoted H_0^i, would state that the mean expression level of the i-th gene is the same in both groups of patients.

We assume that the tests for the individual hypotheses are available (T_i test statistic, R_i rejection region, and the test rejects if $T_i \in R_i$, P_i the associated p-value, i.e. the smallest α leading to rejection). And the problem is how to combine them into a simultaneous test procedure. The easiest yet extremely naive approach is to disregard the multiplicity and simply test each hypothesis at level α. However, with such a procedure, the probability of one or more false rejections rapidly increases with n. For example, if all tests are independent, of size α, and all null hypotheses are true, then

$$\mathbb{P}_0\left(\bigcap \{T_i \notin R_i\}\right) = \prod_{i=1}^{m} \mathbb{P}_0\left(T_i \notin R_i\right) = (1-\alpha)^m.$$

In this sense, the claim that the procedure controls the probability of false rejections at level α is clearly misleading.

Definition 5.1.2. Let $\mathcal{H}_0 \subseteq \{1, \dots, m\}$ be the index set of the true hypotheses and let $\mathcal{R} \subseteq \{1, \dots, m\}$ be the (random) set of rejected hypotheses. Denote $m_0 = |\mathcal{H}_0|$. The family-wise error rate (FWER) is

$$\text{FWER } = \mathbb{P}\left(|\mathcal{H}_0 \cap \mathcal{R}| \ge 1\right).$$

A natural approach is to replace the usual condition for testing a single hypothesis, that the probability of false rejection not exceed α, by the requirement

$$\text{FWER } \le \alpha$$

for all possible combinations of true and false hypotheses. Methods that control the FWER are often described by the p-values of individual tests.

For simplicity, we first restrict our discussion to the important example given by the Gaussian sequence model. Consider a model $Y_i = \mu_i + \varepsilon_i$ for $i = 1, \dots, m$, where $\epsilon_i \sim N(0,1)$. For now, we will not assume that ε_i are independent. Also, the case when the variance of the noise is a general

$\sigma^2 > 0$ but known can be easily covered. The typical question that is asked about this model is how we can test, which elements of $\mu = (\mu_1, \ldots, \mu_m)$ are zero, or perhaps, which elements of μ are equal to each other.

In the question of testing which μ_i are zero, we already mentioned the naive approach: take $z_{\alpha/2} = \Phi^{-1}(1-\alpha/2)$. Test $H_0^i : \mu_i = 0$ using the test statistic $T_i = |Y_i|$ and rejection rule $1\{|Y_i| > z_{\alpha/2}\}$. One classical fix is given by the Bonferroni correction, which is to use α/m instead of α. Now, we test $H_0^i : \mu_i = 0$ with the test $\mathbb{1}\{|Y_i| > z_{\alpha/2m}\}$. Assuming all the nulls are true, we obtain

$$\text{FWER } = \mathbb{P}_0\left(\exists i\, |\varepsilon_i| > z_{\alpha/2m}\right) \leq \sum_{i=1}^m \mathbb{P}_0\left(|\varepsilon_i| > z_{\alpha/2m}\right) = m\frac{\alpha}{m} = \alpha,$$

where in the inequality we used the union bound.

Remark. The union bound can be conservative, which implies that the Bonferroni bound can be conservative too. In the special case when all ε_i are independent, the events $\{|\varepsilon_i| > z_{\alpha/2m}\}$ are independent and

$$\begin{aligned}\text{FWER } &= \mathbb{P}_0\left(\exists i\, |\varepsilon_i| > z_{\alpha/2m}\right) = 1 - \prod_{i=1}^m \mathbb{P}_0\left(|\varepsilon_i| \leq z_{\alpha/2m}\right)\\ &= 1 - \left(1 - \frac{\alpha}{m}\right)^m \approx 1 - e^{-\alpha} \approx \alpha\end{aligned}$$

Therefore, in this case, the Bonferroni procedure provides a good control over the family-wise error rate. This also tells us that if we have many hypotheses (e.g. $m = 10,000$ genes in the biological example) then Bonferroni's test has size approximately $1 - e^{-\alpha}$, which for small α is approximately α. For example, if $\alpha = 0.05$, then $1 - e^{-\alpha} = 0.04877\ldots$. So to get a test of size 0.05 , we could test each hypothesis at level $0.0512/m$.

Remark. (idák correction). The above calculation shows that we could use $z_{\tilde{\alpha}/2}$ with $\tilde{\alpha} = 1 - (1 - \alpha)^{1/m}$ in order to get the error rate precisely under independence.

The Bonferroni correction can become overly conservative in the case when ε_i are dependent. In the extreme situation, when they are all equal, we get

$$\text{FWER } = \mathbb{P}\left(\exists i\, |\varepsilon_i| > z_{\alpha/2m}\right) = \mathbb{P}\left(|\varepsilon_1| > z_{\alpha/2m}\right) = \frac{\alpha}{m}.$$

Remark. In this section, we always assumed that $\sigma^2 = 1$ or equivalently that σ^2 is known. If it is unknown, it can still often be estimated using replicates. So suppose $Y_{ij} = \mu_i + \varepsilon_{ij}$, where $\varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} N\left(0, \sigma^2\right)$ for $i = 1, \ldots, m$ and $j = 1, \ldots, n$. Let $\bar{Y}_i = \frac{1}{n}\sum_j Y_{ij}, \bar{\varepsilon}_i = \frac{1}{n}\sum_j \varepsilon_{ij}$. Then $\bar{Y}_i = \mu_i + \bar{\varepsilon}_i$ with $\bar{\varepsilon}_i \overset{\text{i.i.d.}}{\sim} N\left(0, \sigma^2/n\right)$. Using ideas in Example 1.4.5, we can then simply construct an estimate $\hat{\sigma}^2$ of σ^2, where $\hat{\sigma}^2 \perp\!\!\!\perp \bar{\varepsilon}_1, \ldots, \bar{\varepsilon}_m$. Moreover,

$$\frac{\hat{\sigma}^2}{\sigma^2}(n-1)m \sim \chi^2_{(n-1)m}$$

Now the correct modification of Bonferroni bounds is to use the quantiles of $t_{(n-1)m}$ instead of $N(0,1)$.

First, recall a basic fact of statistical tests. Suppose a null hypothesis H_0 is true, and we perform a statistical test of H_0 and obtain a p-value P. What is the distribution of P? We have the following standard lemma:

Lemma 5.1.3. Let P be the p-value of a statistical testing problem with rejection for small T or large T, or both large and small T. Then, under the null hypothesis H_0,

$$\mathbb{P}(P \leq u) \leq u \quad \text{for all } u \in (0,1).$$

Moreover, if our test statistic T has a continuous distribution under H_0, then for any $u \in (0,1)$

$$\mathbb{P}_0(P \leq u) = u.$$

So $P \sim U(0,1)$ under the null.

Theorem 5.1.4 (Bonferroni Procedure). If, for $i = 1, \ldots, m$, hypothesis H_0^i is rejected when $P_i \leq \alpha/m$, then the FWER for the simultaneous testing of $H_0^1, \ldots, H_0^m$ satisfies FWER $\leq \alpha$.

Proof. Suppose hypotheses H_0^i with $i \in \mathcal{H}_0$ are true, and the others are false. From the union bound, it follows that

$$\begin{aligned}\text{FWER} &= \mathbb{P}\left(\text{reject any } H_0^i \text{ with } i \in \mathcal{H}_0\right) \leq \sum_{i \in \mathcal{H}_0} \mathbb{P}\left(\text{reject } H_0^i\right) \\ &= \sum_{i \in \mathcal{H}_0} \mathbb{P}\left(P_i \leq \frac{\alpha}{m}\right) \overset{(3.16)}{\leq} \sum_{i \in \mathcal{H}_0} \frac{\alpha}{m} \leq \frac{m_0}{m}\alpha \leq \alpha\end{aligned}$$

□

Remark. From the proof, it is clear that, although this procedure controls FWER, it is generally too conservative to be useful in detecting H_0^i that are false. If m_0 is smaller than m, using α/m_0 would be preferable. The problem, of course, is that m_0 is not known. Note, however, that if one H_0^i is false, then $m_0 \leq m-1$ and we could use $\alpha/(m-1)$ as the threshold. If this H_0^i was true and we rejected it, then we already made a mistake, so we can do whatever we want, and there is no harm in using $\alpha/(m-1)$ for the other hypotheses (in our situation, if we make a mistake, it does not matter how many).

5.1.3 The Holm Procedure

The Holm procedure tries to make use of the above observations. It can conveniently be stated in terms of the p -values $P_1, \ldots, P_n$ of the n individual tests. The procedure starts by sorting the p -values.

Definition 5.1.5 (The Holm Procedure). Call them $P_{(1)} \leq \cdots \leq P_{(m)}$. Then The Holm Procedure goes as follows:

(a). If $P_{(1)} \leq \frac{\alpha}{m}$, reject $H_0^{(1)}$ and continue. Else stop.

(b). If $P_{(2)} \leq \frac{\alpha}{m-1}$, reject $H_0^{(2)}$ and continue. Else stop.

(c). ... ...

(d). If $P_{(m)} \leq \alpha$, reject $H_0^{(m)}$.

In other words, the procedure finds the smallest r such that $P_{(r)} > \frac{1}{m-(r-1)}\alpha$ and it rejects the null hypotheses $H_0^{(1)}, \ldots, H_0^{(r-1)}$.

Theorem 5.1.6. The Holm procedure satisfies FWER $\leq \alpha$.

Remark. The Holm procedure is an adapted version of the Bonferroni procedure: once we reject a hypothesis, we treat the problem with the remaining hypothesis as a new problem and apply the Bonferroni procedure. This gives the adapted threshold α/k.

Proof. Suppose $\mathcal{H}_0$ is the set of true hypotheses. Order the P-values as above $P_{(1)} \leq \cdots \leq P_{(m)}$. Ideally, if $i \in \mathcal{H}_0$ then P_i appears later in this order sequence. Let j be the smallest (random) index satisfying

$$P_{(j)} = \min_{i \in \mathcal{H}_0} P_i.$$

Note that, by construction, $j \leq m - m_0 + 1$ (there are at least $m_0 - 1$ indices following (j)). Now, the Holm procedure commits a false rejection if and only if $r \geq j + 1$, or in other words,

$$P_{(1)} \leq \frac{\alpha}{m}, \quad P_{(2)} \leq \frac{\alpha}{m-1}, \quad \ldots, \quad P_{(j)} \leq \frac{\alpha}{m-j+1}$$

which implies that

$$\min_{i \in \mathcal{H}_0} P_i = P_{(j)} \leq \frac{\alpha}{m-j+1} \leq \frac{\alpha}{m_0}.$$

We thus get

$$\begin{aligned} \mathbb{P}\left(|\mathcal{H}_0 \cap \mathcal{R}| \geq 1\right) &= \mathbb{P}\left(P_{(1)} \leq \frac{\alpha}{m}, \ldots, P_{(j)} \leq \frac{\alpha}{m-j+1}\right) \\ &\leq \mathbb{P}\left(\min_{i \in \mathcal{H}_0} P_i \leq \frac{\alpha}{m-j+1}\right) \leq \mathbb{P}\left(\min_{i \in \mathcal{H}_0} P_i \leq \frac{\alpha}{m_0}\right). \end{aligned}$$

By the union bound, the probability of a false rejection is bounded above by

$$\mathbb{P}\left(\min_{i \in \mathcal{H}_0} P_i \leq \frac{\alpha}{m_0}\right) \leq \sum_{i \in \mathcal{H}_0} \mathbb{P}\left(P_i \leq \frac{\alpha}{m_0}\right) \leq \alpha$$

This means that the Holm procedure is strictly more powerful than the Bonferroni procedure without any additional assumptions. □

Remark. However, when m is very large, the Holm procedure may not substantially differ from the Bonferroni correction.

5.2 Lecture 18: Testing using False Discovery Rate

Controlling the FWER may be too conservative and greatly reduce our power to detect real effects, especially when m is large.

Definition 5.2.1. In many modern "large-scale testing" applications, focus has shifted from FWER to the false-discovery proportion (FDP)

$$\text{FDP} = \frac{|\mathcal{H}_0 \cap \mathcal{R}|}{|\mathcal{R}| \vee 1}$$

and on procedures that control its expected value

$$\mathbb{E}(\text{FDP}) = \mathbb{E}\frac{|\mathcal{H}_0 \cap \mathcal{R}|}{|\mathcal{R}| \vee 1}$$

, called the false-discovery rate (FDR).

Remark. The FDR can be interpreted as follows: if FDR ≤ 0.1, we expect around 90% of the discoveries to be true.

Controlling FDR is a shift in paradigm - we are willing to tolerate some type I errors (false discoveries), as long as most of the discoveries we make are still true. It has been argued that in applications where the statistical test is thought of as providing a "definitive answer" for whether an effect is real, FWER control is still the correct objective. In contrast, for applications where the statistical test identifies candidate effects that are likely to be real and which merit further study, it may be better to target FDR control.

5.2.1 The Benjamini-Hochberg Procedure

The false discovery rate depends heavily on the number of true hypotheses. Thus, any procedure that controls FDR should be adaptive. As we showed earlier, if $i \in \mathcal{H}_0$ then $\mathbb{P}(P_i \leq u) \leq u$ for all $u \in (0,1)$. The Benjamini-Hochberg (BH) procedure compares the sorted p-values to a diagonal cutoff line, finds the largest p-value that still falls below this line, and rejects the null hypotheses for the p-values up to and including this one.

Definition 5.2.2 (The Benjamini-Hochberg Procedure)**.** The BH procedure at level α is defined as follows:

(a). Sort the p-values. Call them $P_{(1)} \leq \cdots \leq P_{(m)}$ as before.

(b). Find the largest r such that $P_{(r)} \leq \frac{r}{m}\alpha$.

(c). Reject the null hypotheses $H_0^{(1)}, \ldots, H_0^{(r)}$.

Remark. Just to avoid confusion with the Holm procedure, note that we do not require that $P_{(j)} \leq \frac{j}{m}\alpha$ for $j < r$.

Lemma 5.2.3. Suppose the B-H procedure rejects exactly r hypotheses. Then $i \in \mathcal{R}$ if and only if $P_i \leq \frac{r}{m}\alpha$.

Proof. If $i \in \mathcal{R}$ then $P_i \leq P_{(r)} \leq \frac{r}{m}\alpha$, which proves the right implication. For the left implication, we argue using a contrapositive statement. Suppose $i \notin \mathcal{R}$, let s be such that $P_i = P_{(s)}$. Then $s > r$ and $P_{(s)} > \frac{s}{m}\alpha > \frac{r}{m}\alpha$. □

Theorem 5.2.4 (Benjamini and Hochberg)**.** Consider tests of m null hypotheses. If the test statistics (or equivalently, p-values) of these tests are independent, then the FDR of the above procedure satisfies

$$\text{FDR} \leq \alpha \frac{m_0}{m} \leq \alpha.$$

Proof. We have

$$\text{FDR} = \mathbb{E}\left(\frac{|\mathcal{H}_0 \cap \mathcal{R}|}{|\mathcal{R}| \vee 1}\right) = \mathbb{E}\left(\frac{\sum_{i \in \mathcal{H}_0} \mathbb{1}\{i \in \mathcal{R}\}}{|\mathcal{R}| \vee 1}\right) = \sum_{i \in \mathcal{H}_0} \mathbb{E}\left(\frac{\mathbb{1}\{i \in \mathcal{R}\}}{|\mathcal{R}| \vee 1}\right).$$

Let $R = |\mathcal{R}|$ and let $R_{\setminus i}$ be the number of rejections if we replace P_i with zero and run BH$_\alpha$ on $P_1, \ldots, P_{i-1}, 0, P_{i+1}, \ldots, P_m$. Note that $P_i \perp\!\!\!\perp R_{\setminus i}$ and $R_{\setminus i} \geq 1$. Consider the following claim that

$$i \in \mathcal{R} \quad \Leftrightarrow \quad R = R_{\setminus i} \quad \Leftrightarrow \quad P_i \leq \frac{\alpha R_{-i}}{m}.$$

Before we prove the claim, we show how it helps us to prove the theorem. Assuming that the claim holds

$$\begin{aligned}
\mathrm{FDR} &= \sum_{i\in\mathcal{H}_0} \mathbb{E}\left(\frac{\mathbb{1}\left\{P_i \le \frac{\alpha R_{-i}}{m}\right\}}{R_{\setminus i}}\right) = \sum_{i\in\mathcal{H}_0} \mathbb{E}\left(\frac{\mathbb{P}\left(P_i \le \frac{\alpha R_{-i}}{m} \,\middle|\, R_{\setminus i}\right)}{R_{\setminus i}}\right) \\
&\overset{P_i \perp\!\!\!\perp R_{-i}}{\le} \sum_{i\in\mathcal{H}_0} \mathbb{E}\left(\frac{\frac{\alpha R_{-i}}{m}}{R_{\setminus i}}\right) = \alpha\frac{m_0}{m} \le \alpha.
\end{aligned}$$

It remains to prove the claim. We do it in several steps.
$\left[i \in \mathcal{R} \Rightarrow P_i \le \frac{\alpha R_i}{m}\right]$: First note that BH_α is monotone in the p-values: If $P_i \ge P_i'$ for all i then $R \le R'$. From this it follows that $R \le R_{\setminus i}$. Suppose $i \in \mathcal{R}$. Then

$$P_i \le \frac{\alpha R}{m} \le \frac{\alpha R_{\setminus i}}{m}.$$

$\left[P_i \le \frac{\alpha R_{-i}}{m} \Rightarrow R = R_{\setminus i}\right]$: By definition of $R_{\setminus i}$, we must have $R_{\setminus i}$ many values from $P_1, \dots, P_{i-1}, 0, P_{i+1}, \dots, P_m$ which are $\le \frac{\alpha R_{\setminus i}}{m}$. Since $P_i \le \frac{\alpha R_{-i}}{m}$, it follows that there are $R_{\setminus i}$ many values from $P_1, \dots, P_{i-1}, P_i, P_{i+1}, \dots, P_m$ which are $\le \frac{\alpha R_{Vi}}{m}$. In particular, $R \ge R_{\setminus i}$. The other inequality was shown above and so we get equality.
$\left[R = R_{\setminus i} \Rightarrow i \in \mathcal{R}\right]$: We prove this implication by contradiction. Suppose $R = R_{\setminus i}$ but $i \notin \mathcal{R}$. If $i \notin \mathcal{R}$ then $P_i > \frac{\alpha R}{m}$ and there are R many P_j 's with $P_j \le \frac{\alpha R}{m}$. Now run BH_α on $P_1 \dots, P_{i-1}, 0, P_{i+1}, \dots, P_m$. Then there are $R+1$ many values $\le \frac{\alpha R}{m} \le \frac{\alpha(R+1)}{m}$. Thus, we must have $R_{\setminus i} \ge R+1 > R$, which gives contradiction. □

5.2.2 The Benjamini-Yekutieli Justification

The main problem with the validation of the Benjamini-Hochberg procedure is that it requires that the hypotheses are independent. Handling all the dependence structure between the hypotheses is hard. One popular setting where the B-H procedure can still be validated is when the dependence between the p-values satisfies a form of positive dependence. We will now discuss this in more detail.

Definition 5.2.5. A set $S \subset \mathbb{R}^n$ is nondecreasing if $x \in S$ and $y \ge x$ implies that $y \in S$.

Definition 5.2.6. A random vector $X = (X_1, \dots, X_m)$ is positively regression dependent (PRDS) on $I \subseteq \{1, \dots, m\}$ if $\mathbb{P}\left(X \in S \mid X_i = x_i\right)$ is non-decreasing in x_i for every nondecreasing set S and any $i \in I$.

If $(X_1, \dots, X_m)$ is PDRS on I and $Y_i := f_i\left(X_i\right)$ for all $1 \le i \le m$ with f_i strictly increasing or decreasing, then $(Y_1, \dots, Y_m)$ is PRDS on I as well. Transformation of this form are called co-monotone transformations. Thus PRDS property is preserved under co-monotone transformations. It follows that $P_i = \mathbb{P}\left(T_i \le X_i\right)$ is PDRS as well (at least in the continuous case).

Theorem 5.2.7. If the joint distribution of $P = (P_1, \dots, P_m)$ is PRDS on the subset $\mathcal{H}_0$, the Benjamini-Hochberg procedure controls the FDR at level $\frac{m_0}{m}\alpha$.

The proof of this result relies on the following proposition.

Proposition 5.2.8. If P is PRDS on the set of true nulls, then the function $\mathbb{P}(P \in S \mid P_i \leq t)$ for $i \in \mathcal{H}_0$ is non-decreasing in t for S a non-decreasing set.

Proof. For any $t, \mathbb{P}(\boldsymbol{P} \in S \mid P_i \leq t) = \frac{\mathbb{P}(\boldsymbol{P} \in S, P_i \leq t)}{\mathbb{P}(P_i \leq t)}$. For $t > t'$, we have that

$$\mathbb{P}\left(\mathbb{P} \in S \mid P_i \leq t'\right) = \frac{\mathbb{P}(P \in S, P_i \leq t) + \mathbb{P}(P \in S, P_i \in (t, t'])}{\mathbb{P}(P_i \leq t) + \mathbb{P}(P_i \in (t, t'])}$$

To show that $\mathbb{P}(\boldsymbol{P} \in S \mid P_i \leq t') \leq \mathbb{P}(\boldsymbol{P} \in S \mid P_i \leq t)$, it suffices to show that

$$\frac{\mathbb{P}(\mathbb{P} \in S, P_i \leq t)}{\mathbb{P}(P_i \leq t)} \leq \frac{\mathbb{P}(\mathbb{P} \in S, P_i \in (t, t'])}{\mathbb{P}(P_i \in (t, t'])}.$$

The last statement is because for any positive number a, b, c, d, $\frac{a}{b} \leq \frac{a+c}{b+d}$ if and only if $\frac{a}{b} \leq \frac{c}{d}$. If F_i denotes the CDF of P_i then

$$\begin{aligned}
\mathbb{P}\left(\boldsymbol{P} \in S, P_i \in \left(t, t'\right]\right) &= \mathbb{E}\mathbb{1}\left\{\boldsymbol{P} \in S, P_i \in \left(t, t'\right]\right\} \\
&= \mathbb{E}\left[\mathbb{1}\left\{P_i \in \left(t, t'\right]\right\} \mathbb{E}\left(\mathbb{1}\{\boldsymbol{P} \in S\} \mid P_i\right)\right] \\
&= \mathbb{E}\left[\mathbb{1}\left\{P_i \in \left(t, t'\right]\right\} \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i\right)\right] \\
&= \int_t^{t'} \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = s\right) \mathrm{d}F_i(s) \\
&\overset{(PRDS)}{\geq} \int_t^{t'} \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = t\right) \mathrm{d}F_i(s) \\
&= \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = t\right) \mathbb{P}\left(P_i \in \left(t, t'\right]\right)
\end{aligned}$$

Similarly we have

$$\begin{aligned}
\mathbb{P}\left(\boldsymbol{P} \in S, P_i \leq t\right) &= \int_0^t \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = s\right) \mathrm{d}F_i(s) \\
&\overset{(PRDS)}{\leq} \int_0^t \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = t\right) \mathrm{d}F_i(s) \\
&= \quad \mathbb{P}\left(\boldsymbol{P} \in S \mid P_i = t\right) \mathbb{P}\left(P_i \leq t\right)
\end{aligned}$$

We have just shown that

$$\frac{\mathbb{P}(\boldsymbol{P} \in S, P_i \leq t)}{\mathbb{P}(P_i \leq t)} \leq \mathbb{P}(\boldsymbol{P} \in S \mid P_i = t) \leq \frac{\mathbb{P}(\boldsymbol{P} \in S, P_i \in (t, t'])}{\mathbb{P}(P_i \in (t, t'])}$$

as claimed. □

Proof of The Benjamini-Yekutieli justification. Since we have

$$\mathrm{FDR} = \sum_{i \in \mathcal{H}_0} \mathbb{E}\left(\frac{\mathbb{1}\{i \in \mathcal{R}\}}{|\mathcal{R}| \vee 1}\right)$$

Note that it is enough to show that $\mathbb{E}\left(\frac{\mathbb{1}\{i \in \mathcal{R}\}}{|\mathcal{R}| \vee 1}\right) \leq \alpha/m$ for $i \in \mathcal{H}_0$. By the lemma above, if r rejections are made, then H_0^i is rejected if and only if $P_i \leq \frac{\alpha r}{m}$. Hence, $\mathbb{1}\{i \in \mathcal{R}\} = \mathbb{1}\left\{P_i \leq \frac{\alpha r}{m}\right\}$. This gives

$$\mathbb{E}\left(\frac{\mathbb{1}\{i \in \mathcal{R}\}}{R \vee 1}\right) = \mathbb{E}\left(\mathbb{E}\left[\frac{\mathbb{1}\{i \in \mathcal{R}\}}{R \vee 1} \,\middle|\, R\right]\right) = \mathbb{E}\left(\frac{1}{R \vee 1} \mathbb{P}[i \in \mathcal{R} \mid R]\right) = \sum_{r=1}^m \frac{1}{r} \mathbb{P}\left(P_i \leq \frac{\alpha r}{m}, R = r\right).$$

For any true null, we now have

$$\begin{aligned}\sum_{r=1}^{m} \frac{1}{r}\mathbb{P}\left(P_i \leq \frac{\alpha r}{m}, R = r\right) &= \sum_{r=1}^{m} \frac{1}{r}\mathbb{P}\left(P_i \leq \frac{\alpha r}{m}\right)\mathbb{P}\left(|\mathcal{R}| = r \,\middle|\, P_i \leq \frac{\alpha r}{m}\right) \\ &\leq \frac{\alpha}{m}\sum_{r=1}^{m}\mathbb{P}\left(R = r \,\middle|\, P_i \leq \frac{\alpha r}{m}\right)\end{aligned}$$

Hence, it suffices to show that $\sum_{r=1}^{m} \mathbb{P}\left(R = r \,\middle|\, P_i \leq \frac{\alpha r}{m}\right) \leq 1$. Observe that $\{R \leq r\}$ is an increasing event, that is, it can be written as $\{\boldsymbol{P} \in S\}$ for some non-decreasing set S. This is because increasing all p-values increases the p-value at each rank. Hence, any ranked p-value above the threshold remains above its threshold, that is, we accept at least as many as before, and hence, do not reject more hypotheses. Using this, we get that

$$\begin{aligned}\sum_{r=1}^{m}\mathbb{P}\left(R = r \,\middle|\, P_i \leq \frac{\alpha r}{m}\right) &= \left(\mathbb{P}\left(R \leq m \mid P_i \leq \alpha\right) - \mathbb{P}\left(R \leq 0 \,\middle|\, P_i \leq \frac{\alpha}{m}\right)\right) \\ &\quad + \sum_{r=1}^{m-1}\left(\mathbb{P}\left(R \leq r \,\middle|\, P_i \leq \frac{\alpha r}{m}\right) - \mathbb{P}\left(R \leq r \,\middle|\, P_i \leq \frac{\alpha(r+1)}{m}\right)\right)\end{aligned}$$

Note that each summand in the second line is non-positive since $\mathbb{P}\left(R \leq r \mid P_i \leq x\right)$ is increasing in x. Also $\mathbb{P}\left(R \leq 0 \,\middle|\, P_i \leq \frac{\alpha}{m}\right) \geq 0$, which implies that

$$\sum_{k=1}^{m}\mathbb{P}\left(R = k \,\middle|\, P_i \leq \frac{\alpha k}{m}\right) \leq \mathbb{P}\left(R \leq m \mid P_i \leq \alpha\right) \leq 1$$

As stated before, this proves the upper bound on the FDR. □

Chapter 6 Concentration of Measures

6.1 Lecture 20: Generalization and Transformations

We first motivate why we care about the concentration inequalities:

The main themes are:

(1) concentration bounds imply consistency and stochastic rates;

(2) sub-Gaussian and sub-exponential classes encode useful tail behavior;

(3) Bernstein-type inequalities control sums and sample means;

(4) bounded differences give concentration for functions of independent samples.

Proposition 6.1.1 (Consistency and stochastic rates). Let $\hat{\theta}_n$ be an estimator of θ.

(a) Suppose that for every $t > 0$,

$$\mathbb{P}\big(|\hat{\theta}_n - \theta| > t\big) \leq a_n(t), \qquad a_n(t) \to 0 \text{ as } n \to \infty.$$

Then $\hat{\theta}_n \to \theta$ in probability.

(b) Suppose further that there exists a deterministic sequence $r_n \to \infty$ such that for every $M > 0$,

$$\mathbb{P}\big(r_n|\hat{\theta}_n - \theta| > M\big) \leq a_n(M/r_n) \to 0.$$

Then

$$r_n(\hat{\theta}_n - \theta) = O_p(1).$$

Proof. For (a), fix $t > 0$. By assumption,

$$\mathbb{P}\big(|\hat{\theta}_n - \theta| > t\big) \leq a_n(t) \to 0,$$

which is exactly convergence in probability.

For (b), fix $M > 0$. Then

$$\mathbb{P}\big(|r_n(\hat{\theta}_n - \theta)| > M\big) = \mathbb{P}\big(|\hat{\theta}_n - \theta| > M/r_n\big) \leq a_n(M/r_n).$$

By assumption, this tends to 0 for each fixed M, which is the definition of $O_p(1)$. □

Remark. A concentration inequality is a *non-asymptotic* form of error control. Consistency says the estimation error vanishes; a sharper concentration bound says *how fast* and with *what tail behavior*.

Proposition 6.1.2 (Equivalent characterizations of sub-Gaussianity). For a centered random variable X, the following are equivalent up to universal constants:

(i) There exists $K_1 > 0$ such that

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{K_1^2\lambda^2}{2}\right) \qquad \forall \lambda \in \mathbb{R}.$$

(ii) There exists $K_2 > 0$ such that

$$\mathbb{P}(|X| > t) \leq 2e^{-t^2/K_2^2} \qquad \forall t \geq 0.$$

(iii) There exists $K_3 > 0$ such that

$$(\mathbb{E}|X|^p)^{1/p} \leq K_3\sqrt{p} \qquad \forall p \geq 1.$$

(iv) There exists $K_4 > 0$ such that

$$\mathbb{E}\exp(X^2/K_4^2) \leq 2.$$

Proof. Assume throughout that $\mathbb{E}X = 0$ when discussing condition *(i)*. If X is not centred, then one should replace X by $X - \mathbb{E}X$ in *(i)*.

(i)⇒(ii). Fix $t > 0$ and $\lambda > 0$. By Markov's inequality and the condition *(i)*,

$$\mathbb{P}(X > t) = \mathbb{P}\left(e^{\lambda X} > e^{\lambda t}\right) \leq e^{-\lambda t}\mathbb{E}e^{\lambda X} \leq \exp\left(-\lambda t + \frac{K_1^2\lambda^2}{2}\right).$$

The right-hand side is minimized at

$$\lambda = \frac{t}{K_1^2},$$

and substituting this value gives

$$\mathbb{P}(X > t) \leq \exp\left(-\frac{t^2}{2K_1^2}\right).$$

Applying the same argument to $-X$ and the union bound gives

$$\mathbb{P}(|X| > t) \leq 2\exp\left(-\frac{t^2}{2K_1^2}\right).$$

Thus *(ii)* holds with $K_2 = \sqrt{2}\,K_1$.

(ii)⇒(iii). Let $p \geq 1$. By Tonelli's theorem,

$$\begin{aligned}\mathbb{E}|X|^p &= \mathbb{E}\int_0^\infty \mathbf{1}_{\{s<|X|^p\}}ds = \int_0^\infty \mathbb{P}(|X|^p > s)ds \\ &= \int_0^\infty \mathbb{P}\left(|X| > s^{1/p}\right)ds\end{aligned}$$

Now make the change of variables

$$s = t^p, \quad ds = pt^{p-1}dt$$

we then have

$$\mathbb{E}|X|^p = \int_0^\infty pt^{p-1}\mathbb{P}(|X| > t)dt$$

By *(ii)*,

$$\mathbb{E}|X|^p \le 2p \int_0^\infty t^{p-1} e^{-t^2/K_2^2}\, dt.$$

Make the change of variable

$$u = \frac{t^2}{K_2^2}, \qquad t = K_2\sqrt{u}, \qquad dt = \frac{K_2}{2} u^{-1/2}\, du.$$

Then

$$\begin{aligned}
\mathbb{E}|X|^p &\le 2p \int_0^\infty (K_2\sqrt{u})^{p-1} e^{-u} \frac{K_2}{2} u^{-1/2}\, du \\
&= pK_2^p \int_0^\infty u^{p/2-1} e^{-u}\, du \\
&= pK_2^p\, \Gamma\left(\frac{p}{2}\right) = 2K_2^p\, \Gamma\left(\frac{p}{2}+1\right).
\end{aligned}$$

Now use the standard bound

$$\Gamma(r+1) \le (Cr)^r, \qquad r \ge \frac{1}{2},$$

for some universal constant $C > 0$. With $r = p/2$ this gives

$$\mathbb{E}|X|^p \le 2K_2^p \left(C\frac{p}{2}\right)^{p/2} \le (C' K_2 \sqrt{p})^p$$

for another universal constant C'. Taking pth roots,

$$(\mathbb{E}|X|^p)^{1/p} \le C' K_2 \sqrt{p}.$$

Thus *(iii)* holds with $K_3 \le C' K_2$.

(iii)⇒(iv). Expand the exponential into its power series:

$$\mathbb{E} e^{X^2/L^2} = \sum_{m=0}^\infty \frac{\mathbb{E} X^{2m}}{m!\, L^{2m}}.$$

By *(iii)*, with $p = 2m$,

$$(\mathbb{E}|X|^{2m})^{1/(2m)} \le K_3 \sqrt{2m},$$

hence

$$\mathbb{E} X^{2m} \le (K_3\sqrt{2m})^{2m} = K_3^{2m} (2m)^m.$$

Therefore

$$\frac{\mathbb{E} X^{2m}}{m!\, L^{2m}} \le \frac{K_3^{2m}(2m)^m}{m!\, L^{2m}}.$$

Choose

$$L = 2\sqrt{e}\, K_3.$$

Then

$$L^{2m} = (4eK_3^2)^m,$$

so

$$\frac{\mathbb{E} X^{2m}}{m!\, L^{2m}} \le \frac{(2m)^m}{m!(4e)^m} = \frac{1}{m!} \left(\frac{m}{2e}\right)^m.$$

Using Stirling's lower bound

$$m! \ge \left(\frac{m}{e}\right)^m,$$

we obtain

$$\frac{\mathbb{E}X^{2m}}{m!\,L^{2m}} \le 2^{-m}.$$

Hence

$$\mathbb{E}e^{X^2/L^2} \le \sum_{m=0}^{\infty} 2^{-m} = 2.$$

So *(iv)* holds with

$$K_4 \le 2\sqrt{e}\,K_3.$$

(iv)⇒(ii). For any $t \ge 0$, Markov's inequality gives

$$\mathbb{P}(|X| > t) = \mathbb{P}\left(e^{X^2/K_4^2} > e^{t^2/K_4^2}\right) \le e^{-t^2/K_4^2}\mathbb{E}e^{X^2/K_4^2}.$$

By *(iv)*,

$$\mathbb{P}(|X| > t) \le 2e^{-t^2/K_4^2}.$$

Thus *(ii)* holds with $K_2 \le K_4$.

(iii)⇒(i). This is the step where centring matters. Let X' be an independent copy of X. Since $\mathbb{E}X = 0$, Jensen's inequality gives

$$e^{\lambda X} = e^{\lambda(X-\mathbb{E}X')} \le \mathbb{E}_{X'}e^{\lambda(X-X')}.$$

Taking expectation in X,

$$\mathbb{E}e^{\lambda X} \le \mathbb{E}e^{\lambda(X-X')}.$$

Set

$$Y := X - X'.$$

Then Y is symmetric, so all odd moments of Y vanish, and therefore

$$\mathbb{E}e^{\lambda Y} = \sum_{m=0}^{\infty} \frac{\lambda^{2m}\mathbb{E}Y^{2m}}{(2m)!}.$$

By Minkowski's inequality and *(iii)*,

$$(\mathbb{E}|Y|^{2m})^{1/(2m)} \le (\mathbb{E}|X|^{2m})^{1/(2m)} + (\mathbb{E}|X'|^{2m})^{1/(2m)} \le 2K_3\sqrt{2m}.$$

Hence

$$\mathbb{E}Y^{2m} \le (2K_3\sqrt{2m})^{2m} = (8K_3^2 m)^m.$$

Also,

$$(2m)! = (1\cdots m)(m+1\cdots 2m) \ge m!\,m^m.$$

Thus

$$\frac{\lambda^{2m}\mathbb{E}Y^{2m}}{(2m)!} \le \frac{\lambda^{2m}(8K_3^2 m)^m}{m!\,m^m} = \frac{(8K_3^2\lambda^2)^m}{m!}.$$

Summing over m,

$$\mathbb{E}e^{\lambda Y} \le \sum_{m=0}^{\infty} \frac{(8K_3^2\lambda^2)^m}{m!} = \exp(8K_3^2\lambda^2).$$

Therefore

$$\mathbb{E}e^{\lambda X} \le \exp(8K_3^2\lambda^2) = \exp\left(\frac{(4K_3)^2\lambda^2}{2}\right).$$

So *(i)* holds with

$$K_1 \le 4K_3.$$

Combining the implications above shows that the four conditions are equivalent up to universal constant factors. □

Remark. Sub-Gaussianity means that X behaves, from the point of view of tails and moments, like a Gaussian random variable. The four formulations say the same thing in different languages: mgf, tails, moments, and Orlicz norms.

We now state a one-sided version of the sub-exponential random variables.

Definition 6.1.3 (One-sided Bernstein condition / sub-exponential behavior)**.** A centered random variable X satisfies the *one-sided Bernstein condition* with parameters (σ^2, c) if

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\sigma^2\lambda^2}{2(1-c\lambda)}\right), \qquad 0 \leq \lambda < \frac{1}{c}.$$

When a corresponding two-sided version also holds for $-X$, one says that X is *sub-exponential*.

Proposition 6.1.4 (Equivalent characterizations of sub-exponentiality)**.** For a centered random variable X, the following are equivalent up to universal constants:

(i) X satisfies a two-sided Bernstein mgf bound.

(ii) There exists $K_1 > 0$ such that

$$\mathbb{P}(|X| > t) \leq 2e^{-t/K_1}, \qquad t \geq 0.$$

(iii) There exists $K_2 > 0$ such that

$$(\mathbb{E}|X|^p)^{1/p} \leq K_2 p, \qquad p \geq 1.$$

(iv) There exists $K_3 > 0$ such that

$$\mathbb{E}e^{|X|/K_3} \leq 2.$$

Proof sketch. The proof parallels the sub-Gaussian case.
(i)⇒(ii): use Chernoff with the Bernstein mgf control.
(ii)⇒(iii): integrate the tail:

$$\mathbb{E}|X|^p = \int_0^\infty pt^{p-1}\mathbb{P}(|X| > t)\,dt \leq 2p\int_0^\infty t^{p-1}e^{-t/K_1}\,dt = 2p!K_1^p.$$

Thus $(\mathbb{E}|X|^p)^{1/p} \leq CK_1 p$.
(iii)⇒(iv): expand $\mathbb{E}e^{|X|/L}$ into a power series and choose L large.
(iv)⇒(ii): apply Markov:

$$\mathbb{P}(|X| > t) \leq e^{-t/K_3}\mathbb{E}e^{|X|/K_3} \leq 2e^{-t/K_3}.$$

□

Corollary 6.1.5 (Square of a sub-Gaussian variable is sub-exponential)**.** If X is sub-Gaussian, then X^2 is sub-exponential.

Proof. If X is sub-Gaussian, then

$$\mathbb{E}e^{X^2/K^2} \leq 2$$

for some $K > 0$. This is exactly an exponential-integrability condition for X^2. □

The one-sided Bernstein condition results in a corresponding concentration inequality.

Theorem 6.1.6 (Bernstein upper-tail bound)**.** Let X be a random variable satisfying the one-sided Bernstein condition with parameters (σ^2, c):

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \le \exp\left(\frac{\sigma^2\lambda^2}{2(1-c\lambda)}\right), \qquad 0 \le \lambda < \frac{1}{c}.$$

Then for every $\varepsilon \ge 0$,

$$\mathbb{P}(X - \mathbb{E}X \ge \varepsilon) \le \exp\left(-\frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)}\right).$$

Equivalently, for every $\delta \in (0,1)$,

$$\mathbb{P}\left(X - \mathbb{E}X \ge \sqrt{2\sigma^2 \log(1/\delta)} + c\log(1/\delta)\right) \le \delta.$$

Proof. By Chernoff's bound, for any $0 \le \lambda < 1/c$,

$$\mathbb{P}(X - \mathbb{E}X \ge \varepsilon) \le e^{-\lambda\varepsilon}\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \le \exp\left(-\lambda\varepsilon + \frac{\sigma^2\lambda^2}{2(1-c\lambda)}\right).$$

Choose

$$\lambda = \frac{\varepsilon}{\sigma^2 + c\varepsilon}.$$

Then $0 \le \lambda < 1/c$, and

$$1 - c\lambda = 1 - \frac{c\varepsilon}{\sigma^2 + c\varepsilon} = \frac{\sigma^2}{\sigma^2 + c\varepsilon}.$$

Substituting gives

$$-\lambda\varepsilon + \frac{\sigma^2\lambda^2}{2(1-c\lambda)} = -\frac{\varepsilon^2}{\sigma^2 + c\varepsilon} + \frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)} = -\frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)}.$$

Hence

$$\mathbb{P}(X - \mathbb{E}X \ge \varepsilon) \le \exp\left(-\frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)}\right).$$

For the high-probability form, let $u = \log(1/\delta)$ and set

$$\varepsilon = \sqrt{2\sigma^2 u} + cu.$$

Then

$$\frac{\varepsilon^2}{2(\sigma^2 + c\varepsilon)} \ge u,$$

so the first inequality implies

$$\mathbb{P}(X - \mathbb{E}X \ge \varepsilon) \le e^{-u} = \delta.$$

□

Remark. Bernstein's inequality interpolates between two regimes:

$$\varepsilon \ll \sigma^2/c \quad \Rightarrow \quad \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right) \quad \text{(Gaussian regime)},$$

and

$$\varepsilon \gg \sigma^2/c \quad \Rightarrow \quad \exp\left(-\frac{\varepsilon}{2c}\right) \quad \text{(exponential regime)}.$$

Similar to the sub-exponential setting, the Bernstein condition is invariant under scaling and convex combinations.

Proposition 6.1.7 (Bernstein condition is preserved under nonnegative weighted sums)**.** Let $X_1, \dots, X_n$ be independent centered random variables, and suppose that each X_i satisfies

$$\mathbb{E}e^{\lambda X_i} \le \exp\left(\frac{\sigma_i^2 \lambda^2}{2(1 - c_i\lambda)}\right), \qquad 0 \le \lambda < 1/c_i.$$

Let $a_i \ge 0$ and define

$$S = \sum_{i=1}^n a_i X_i.$$

Then S satisfies a one-sided Bernstein condition with parameters

$$\sigma_S^2 = \sum_{i=1}^n a_i^2 \sigma_i^2, \qquad c_S = \max_{1 \le i \le n} a_i c_i.$$

Proof. By independence,

$$\mathbb{E}e^{\lambda S} = \prod_{i=1}^n \mathbb{E}e^{\lambda a_i X_i} \le \prod_{i=1}^n \exp\left(\frac{a_i^2 \sigma_i^2 \lambda^2}{2(1 - a_i c_i \lambda)}\right).$$

If $0 \le \lambda < 1/c_S$, then $1 - a_i c_i \lambda \ge 1 - c_S \lambda$ for all i, so

$$\mathbb{E}e^{\lambda S} \le \exp\left(\sum_{i=1}^n \frac{a_i^2 \sigma_i^2 \lambda^2}{2(1 - c_S\lambda)}\right) = \exp\left(\frac{\sigma_S^2 \lambda^2}{2(1 - c_S\lambda)}\right).$$

□

Corollary 6.1.8 (Bernstein inequality for sample means)**.** Let $X_1, \dots, X_n$ be i.i.d. with mean μ, and assume that $X_1 - \mu$ satisfies the one-sided Bernstein condition with parameters (σ^2, c). Then

$$\bar{X}_n - \mu = \frac{1}{n}\sum_{i=1}^n (X_i - \mu)$$

satisfies the one-sided Bernstein condition with parameters

$$\frac{\sigma^2}{n}, \qquad \frac{c}{n}.$$

Hence, for all $\varepsilon \ge 0$,

$$\mathbb{P}(\bar{X}_n - \mu \ge \varepsilon) \le \exp\left(-\frac{n\varepsilon^2}{2(\sigma^2 + c\varepsilon)}\right).$$

Equivalently, for every $\delta \in (0, 1)$,

$$\mathbb{P}\left(\bar{X}_n - \mu \ge \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}} + \frac{c \log(1/\delta)}{n}\right) \le \delta.$$

Proof. Apply the previous proposition with $a_i = 1/n$. Then

$$\sigma_S^2 = n \cdot \frac{\sigma^2}{n^2} = \frac{\sigma^2}{n}, \qquad c_S = \frac{c}{n}.$$

Now apply Bernstein's upper-tail bound. □

Corollary 6.1.9 (Bounded observations: Bernstein and Hoeffding forms)**.** Let $X_1, \dots, X_n$ be i.i.d. with mean μ.

(a) If $X_i - \mu \le c$ almost surely, then

$$\mathbb{P}(\bar{X}_n - \mu \ge \varepsilon) \le \exp\left(-\frac{n\varepsilon^2}{2(\mathrm{Var}(X_1) + c\varepsilon)}\right).$$

(b) If $X_i \in [a, b]$ almost surely, then Hoeffding's inequality gives

$$\mathbb{P}(\bar{X}_n - \mu \ge \varepsilon) \le \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Proof. Part (a) follows from the standard fact that bounded upper support implies a Bernstein-type mgf bound. Part (b) is Hoeffding's inequality. □

Remark. Hoeffding's inequality uses only boundedness. Bernstein's inequality uses both boundedness and variance, so it is often sharper when the variance is much smaller than the squared range.

6.2 Lecture 21: Concentration of other types

6.2.1 The bounded difference inequality

We demonstrate that a function that is bounded in each of its coordinates also give rise to the concentration inequalities.

Definition 6.2.1 (Bounded differences property)**.** A function $f : \mathcal{X}^n \to \mathbb{R}$ satisfies the bounded differences property with constants $L_1, \dots, L_n$ if for every pair $x = (x_1, \dots, x_n)$ and $x' = (x'_1, \dots, x'_n)$ that differ only in the ith coordinate,

$$|f(x) - f(x')| \le L_i.$$

Theorem 6.2.2 (McDiarmid's inequality)**.** Let $X_1, \dots, X_n$ be independent random variables and define

$$Z = f(X_1, \dots, X_n),$$

where f satisfies bounded differences with constants $L_1, \dots, L_n$. Then for all $t \ge 0$,

$$\mathbb{P}(Z - \mathbb{E}Z \ge t) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n L_i^2}\right),$$

and therefore

$$\mathbb{P}(|Z - \mathbb{E}Z| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n L_i^2}\right).$$

Equivalently, $Z - \mathbb{E}Z$ is sub-Gaussian with variance proxy

$$\frac{1}{4}\sum_{i=1}^n L_i^2.$$

Proof sketch. Define the Doob martingale

$$M_i = \mathbb{E}[Z \mid X_1, \ldots, X_i], \qquad i = 0, \ldots, n,$$

with $M_0 = \mathbb{E}Z$ and $M_n = Z$. Then

$$Z - \mathbb{E}Z = \sum_{i=1}^{n} (M_i - M_{i-1}).$$

Changing only X_i can change f by at most L_i, so each martingale increment is almost surely bounded by L_i in absolute value. Applying the Azuma–Hoeffding inequality to the martingale gives

$$\mathbb{P}(Z - \mathbb{E}Z \geq t) \leq \exp\left(-\frac{2t^2}{\sum_i L_i^2}\right).$$

Applying the same argument to $-Z$ yields the two-sided version. □

Remark. If no single observation can change the statistic very much, then the statistic is sharply concentrated around its mean.

Example (Kernel density estimation in L^1)**.** Let $X_1, \ldots, X_n$ be i.i.d., let $K \geq 0$ be a kernel with $\int K(u)\, du = 1$, and define

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right).$$

Set

$$Z = \int_{\mathbb{R}} \left| \hat{f}_n(x) - \mathbb{E}\hat{f}_n(x) \right| dx.$$

Then

$$\mathbb{P}\big(|Z - \mathbb{E}Z| \geq t\big) \leq 2e^{-nt^2/2}.$$

Proof. If only X_i is replaced by X_i', then the estimator changes by

$$\Delta_i(x) = \frac{1}{nh}\left[K\left(\frac{x - X_i}{h}\right) - K\left(\frac{x - X_i'}{h}\right)\right].$$

Hence

$$\int |\Delta_i(x)|\, dx \leq \frac{1}{nh} \int K\left(\frac{x - X_i}{h}\right) dx + \frac{1}{nh} \int K\left(\frac{x - X_i'}{h}\right) dx = \frac{2}{n}.$$

So Z satisfies bounded differences with $L_i = 2/n$. Therefore

$$\sum_{i=1}^{n} L_i^2 = n \cdot \frac{4}{n^2} = \frac{4}{n}.$$

Applying McDiarmid's inequality gives

$$\mathbb{P}(|Z - \mathbb{E}Z| \geq t) \leq 2\exp\left(-\frac{2t^2}{4/n}\right) = 2e^{-nt^2/2}.$$

□

Remark. Although the KDE is a function of all data points, each observation contributes only order $1/n$ in L^1, so the global estimator concentrates.

Example (Uniform deviation of the empirical measure)**.** Let $\mathcal{A}$ be a collection of measurable sets and define

$$\mathbb{P}_n(A) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \in A\}}, \qquad P(A) = \mathbb{P}(X \in A).$$

Consider

$$Z = \sup_{A\in\mathcal{A}} |\mathbb{P}_n(A) - P(A)|.$$

Then

$$\mathbb{P}(|Z - \mathbb{E}Z| \geq t) \leq 2e^{-2nt^2},$$

and consequently, with probability at least $1-\delta$,

$$Z \leq \mathbb{E}Z + \sqrt{\frac{\log(2/\delta)}{2n}}.$$

Proof. If only the ith sample point is changed, then for every fixed A,

$$\left| \frac{1}{n}\sum_{j=1}^{n} \mathbb{1}_{\{X_j \in A\}} - \frac{1}{n}\sum_{j=1}^{n} \mathbb{1}_{\{X'_j \in A\}} \right| \leq \frac{1}{n}.$$

Taking the supremum over $A \in \mathcal{A}$ preserves this bound, so $L_i = 1/n$ for all i. Hence

$$\sum_{i=1}^{n} L_i^2 = n \cdot \frac{1}{n^2} = \frac{1}{n}.$$

McDiarmid gives

$$\mathbb{P}(|Z - \mathbb{E}Z| \geq t) \leq 2\exp\left(-\frac{2t^2}{1/n}\right) = 2e^{-2nt^2}.$$

Set the right-hand side equal to δ and solve for t to obtain the stated high-probability bound. □

Remark. The oscillation of the empirical measure over a whole class $\mathcal{A}$ is still controlled because changing one observation can alter every empirical probability by at most $1/n$.

These tools are the standard bridge from probability inequalities to non-asymptotic statistics. They naturally generalize to the following result:

Proposition 6.2.3 (Concentration of bounded U-statistics)**.** Let $g : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be symmetric and bounded:

$$\|g\|_\infty \leq b.$$

Given i.i.d. $X_1, \ldots, X_n$, define the order-2 U-statistic

$$U_n = \frac{1}{\binom{n}{2}} \sum_{1\leq i<j\leq n} g(X_i, X_j).$$

Then

$$\mathbb{E}U_n = \mathbb{E}g(X_1, X_2),$$

and for every $t \geq 0$,

$$\mathbb{P}\big(|U_n - \mathbb{E}U_n| \geq t\big) \leq 2\exp\left(-\frac{nt^2}{8b^2}\right).$$

Proof. Unbiasedness is immediate by symmetry:

$$\mathbb{E}U_n = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \mathbb{E}g(X_i, X_j) = \mathbb{E}g(X_1, X_2).$$

To prove concentration, write $U_n = f(X_1, \ldots, X_n)$. Replace only X_i by X_i'. Then only the $(n-1)$ pairs involving index i can change, so

$$|f(x) - f(x')| \le \frac{n-1}{\binom{n}{2}} \cdot 2b = \frac{4b}{n}.$$

Thus McDiarmid's inequality applies with $L_i = 4b/n$ for all i. Since

$$\sum_{i=1}^{n} L_i^2 = n \cdot \frac{16b^2}{n^2} = \frac{16b^2}{n},$$

we obtain

$$\mathbb{P}(|U_n - \mathbb{E}U_n| \ge t) \le 2\exp\left(-\frac{2t^2}{16b^2/n}\right) = 2\exp\left(-\frac{nt^2}{8b^2}\right).$$

□

Proposition 6.2.4 (Bounded empirical-process suprema)**.** Let $\mathcal{F}$ be a class of measurable functions such that

$$\sup_{f\in\mathcal{F}} \|f\|_\infty \le b.$$

Given i.i.d. $X_1, \ldots, X_n$, define

$$Z = \|\mathbb{P}_n - P\|_{\mathcal{F}} := \sup_{f\in\mathcal{F}} \left| \frac{1}{n}\sum_{i=1}^{n} f(X_i) - \mathbb{E}f(X_1) \right|.$$

Then for every $t \ge 0$,

$$\mathbb{P}(Z \ge \mathbb{E}Z + t) \le \exp\left(-\frac{nt^2}{2b^2}\right).$$

Equivalently, with probability at least $1-\delta$,

$$Z \le \mathbb{E}Z + b\sqrt{\frac{2\log(1/\delta)}{n}}.$$

Proof. Write $Z = f(X_1, \ldots, X_n)$. If only X_i is replaced by X_i', then for each fixed $f \in \mathcal{F}$,

$$\left| \frac{1}{n}\sum_{j=1}^{n} f(X_j) - \frac{1}{n}\sum_{j=1}^{n} f(X_j') \right| \le \frac{1}{n}|f(X_i) - f(X_i')| \le \frac{2b}{n}.$$

Taking the supremum over $f \in \mathcal{F}$ preserves the same bound:

$$|Z(x) - Z(x')| \le \frac{2b}{n}.$$

Thus McDiarmid applies with $L_i = 2b/n$. Therefore

$$\mathbb{P}(Z \ge \mathbb{E}Z + t) \le \exp\left(-\frac{2t^2}{n(2b/n)^2}\right) = \exp\left(-\frac{nt^2}{2b^2}\right).$$

Solving $\exp(-nt^2/(2b^2)) = \delta$ yields the stated high-probability form. □

Remark. Even though Z is a supremum over a possibly huge class $\mathcal{F}$, one observation still changes every empirical average by at most $2b/n$, which is enough for concentration.

6.2.2 Concentration of Lipschitz functions

We demonstrate that Gaussian variables demonstrate sub-Gaussianity under Lipschitz transformations, which is quite generaly and admits many interesting applications.

Theorem 6.2.5 (Gaussian log-Sobolev inequality)**.** Let γ_d denote the standard Gaussian measure on $\mathbb{R}^d$,

$$d\gamma_d(x) = (2\pi)^{-d/2} e^{-\|x\|_2^2/2}\, dx.$$

Then for every smooth function $g : \mathbb{R}^d \to \mathbb{R}$ such that e^g and $|\nabla g|^2 e^g$ are integrable with respect to γ_d,

$$\mathrm{Ent}_{\gamma_d}(e^g) := \int e^g g\, d\gamma_d - \Big(\int e^g\, d\gamma_d\Big) \log\Big(\int e^g\, d\gamma_d\Big) \le \frac{1}{2} \int |\nabla g|^2 e^g\, d\gamma_d.$$

Equivalently, for every smooth positive function u,

$$\mathrm{Ent}_{\gamma_d}(u) \le \frac{1}{2} \int \frac{|\nabla u|^2}{u}\, d\gamma_d.$$

Proof. Let $(P_t)_{t\ge 0}$ be the Ornstein–Uhlenbeck semigroup, defined by

$$P_t f(x) = \mathbb{E}\Big[f\big(e^{-t}x + \sqrt{1-e^{-2t}}\, Z\big)\Big], \qquad Z \sim N(0, I_d).$$

Its generator is

$$Lf = \Delta f - x \cdot \nabla f,$$

and γ_d is invariant under P_t, i.e.

$$\int P_t f\, d\gamma_d = \int f\, d\gamma_d.$$

Moreover, $P_t f \to \int f\, d\gamma_d$ as $t \to \infty$ for sufficiently integrable f.
We first recall the Gaussian integration by parts identity: for smooth f, h with sufficient decay,

$$\int f\, Lh\, d\gamma_d = -\int \nabla f \cdot \nabla h\, d\gamma_d.$$

Indeed, writing $d\gamma_d(x) = \rho(x)\, dx$ with

$$\rho(x) = (2\pi)^{-d/2} e^{-\|x\|_2^2/2}, \qquad \nabla\rho(x) = -x\rho(x),$$

we compute

$$\int f\, Lh\, d\gamma_d = \int f(\Delta h - x \cdot \nabla h)\rho\, dx = -\int \nabla(f\rho) \cdot \nabla h\, dx - \int f\, x \cdot \nabla h\, \rho\, dx,$$

and the terms involving $x \cdot \nabla h$ cancel, leaving

$$\int f\, Lh\, d\gamma_d = -\int \rho\, \nabla f \cdot \nabla h\, dx = -\int \nabla f \cdot \nabla h\, d\gamma_d.$$

Now let $u > 0$ be smooth, and define

$$F(t) := \mathrm{Ent}_{\gamma_d}(P_t u).$$

Since P_t preserves the Gaussian integral,

$$\int P_t u \, d\gamma_d = \int u \, d\gamma_d =: m,$$

and therefore

$$F(t) = \int P_t u \log(P_t u) \, d\gamma_d - m \log m.$$

Differentiating in t,

$$F'(t) = \int \partial_t(P_t u) \log(P_t u) \, d\gamma_d + \int \partial_t(P_t u) \, d\gamma_d.$$

The second term vanishes because $\int P_t u \, d\gamma_d$ is constant in t, and $\partial_t P_t u = L P_t u$, so

$$F'(t) = \int L P_t u \, \log(P_t u) \, d\gamma_d.$$

Applying the Gaussian integration by parts identity with $f = \log(P_t u)$ and $h = P_t u$, we obtain

$$F'(t) = -\int \nabla \log(P_t u) \cdot \nabla(P_t u) \, d\gamma_d = -\int \frac{|\nabla P_t u|^2}{P_t u} \, d\gamma_d.$$

Hence $F'(t) \leq 0$, and integrating from 0 to ∞ gives

$$\mathrm{Ent}_{\gamma_d}(u) = \int_0^\infty \int \frac{|\nabla P_t u|^2}{P_t u} \, d\gamma_d \, dt,$$

since $P_t u \to m$ as $t \to \infty$ and constants have zero entropy.

It remains to bound the integrand. By differentiating the Mehler formula under the expectation,

$$\nabla P_t u = e^{-t} P_t(\nabla u).$$

Therefore,

$$\frac{|\nabla P_t u|^2}{P_t u} = e^{-2t} \frac{|P_t(\nabla u)|^2}{P_t u}.$$

By Cauchy–Schwarz under the probability kernel defining P_t,

$$|P_t(\nabla u)|^2 \leq P_t\left(\frac{|\nabla u|^2}{u}\right) P_t u,$$

and thus

$$\frac{|\nabla P_t u|^2}{P_t u} \leq e^{-2t} P_t\left(\frac{|\nabla u|^2}{u}\right).$$

Integrating with respect to γ_d and using invariance of γ_d,

$$\int \frac{|\nabla P_t u|^2}{P_t u} \, d\gamma_d \leq e^{-2t} \int P_t\left(\frac{|\nabla u|^2}{u}\right) \, d\gamma_d = e^{-2t} \int \frac{|\nabla u|^2}{u} \, d\gamma_d.$$

Substituting into the entropy representation,

$$\mathrm{Ent}_{\gamma_d}(u) \leq \int_0^\infty e^{-2t} \, dt \int \frac{|\nabla u|^2}{u} \, d\gamma_d = \frac{1}{2} \int \frac{|\nabla u|^2}{u} \, d\gamma_d.$$

Finally, taking $u = e^g$, we have

$$\nabla u = e^g \nabla g, \qquad \frac{|\nabla u|^2}{u} = |\nabla g|^2 e^g,$$

so

$$\mathrm{Ent}_{\gamma_d}(e^g) \leq \frac{1}{2} \int |\nabla g|^2 e^g \, d\gamma_d.$$

This proves the claim. $\square$

Corollary 6.2.6 (Sub-Gaussian mgf bound for Lipschitz functions). Let $X \sim N(0, I_d)$, and let $f : \mathbb{R}^d \to \mathbb{R}$ be L-Lipschitz. Then

$$\mathbb{E}\exp\big(\lambda(f(X) - \mathbb{E}f(X))\big) \le \exp\left(\frac{\lambda^2 L^2}{2}\right), \qquad \lambda \in \mathbb{R}.$$

Proof. Assume first that f is smooth. Define

$$M(t) := \mathbb{E}e^{tf(X)}, \qquad \psi(t) := \log M(t).$$

Applying the Gaussian log-Sobolev inequality with $g = tf$, we get

$$\mathrm{Ent}_{\gamma_d}(e^{tf}) \le \frac{t^2}{2}\int |\nabla f|^2 e^{tf}\, d\gamma_d \le \frac{t^2 L^2}{2}\,\mathbb{E}e^{tf(X)}.$$

On the other hand,

$$\mathrm{Ent}_{\gamma_d}(e^{tf}) = \mathbb{E}[tf(X)e^{tf(X)}] - M(t)\log M(t) = M(t)\big(t\psi'(t) - \psi(t)\big).$$

Hence

$$t\psi'(t) - \psi(t) \le \frac{t^2L^2}{2}.$$

For $t \neq 0$,

$$\left(\frac{\psi(t)}{t}\right)' = \frac{t\psi'(t) - \psi(t)}{t^2} \le \frac{L^2}{2}.$$

Integrating from 0 to $\lambda > 0$,

$$\frac{\psi(\lambda)}{\lambda} - \lim_{t\downarrow 0}\frac{\psi(t)}{t} \le \frac{\lambda L^2}{2}.$$

Since

$$\lim_{t\downarrow 0}\frac{\psi(t)}{t} = \psi'(0) = \mathbb{E}f(X),$$

it follows that

$$\psi(\lambda) \le \lambda\mathbb{E}f(X) + \frac{\lambda^2L^2}{2},$$

or equivalently,

$$\mathbb{E}e^{\lambda(f(X)-\mathbb{E}f(X))} \le e^{\lambda^2L^2/2}, \qquad \lambda \ge 0.$$

Applying the same argument to $-f$ yields the same estimate for $\lambda < 0$.
For a merely Lipschitz f, one may approximate f by smooth mollifications f_ε, apply the estimate to f_ε, and then pass to the limit using dominated convergence. ☐

Theorem 6.2.7 (Gaussian concentration for Lipschitz functions). Let $X \sim N(0, I_m)$ and let $f : \mathbb{R}^m \to \mathbb{R}$ be L-Lipschitz. Then

$$f(X) - \mathbb{E}f(X)$$

is sub-Gaussian with variance proxy L^2. In particular,

$$\mathbb{P}\big(|f(X) - \mathbb{E}f(X)| \ge t\big) \le 2\exp\left(-\frac{t^2}{2L^2}\right), \qquad t \ge 0.$$

Proof sketch. The log-Sobolev inequality gives

$$\mathbb{E}\exp(\lambda(f(X) - \mathbb{E}f(X))) \leq \exp\left(\frac{\lambda^2 L^2}{2}\right), \qquad \lambda \in \mathbb{R}.$$

Then Chernoff's bound gives

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq \inf_{\lambda > 0} \exp\left(-\lambda t + \frac{\lambda^2 L^2}{2}\right).$$

Optimizing at $\lambda = t/L^2$ yields

$$\mathbb{P}(f(X) - \mathbb{E}f(X) \geq t) \leq e^{-t^2/(2L^2)}.$$

Apply the same bound to $-f$ and combine. □

Remark. For Gaussian inputs, Lipschitz observables behave as if they were one-dimensional Gaussians. The only parameter that matters is the Lipschitz constant.

This result is powerful and has the following important corollaries:

Corollary 6.2.8 (Concentration of the Euclidean norm)**.** Let $X \sim N(0, I_n)$ and define

$$f(x) = \frac{\|x\|_2}{\sqrt{n}}.$$

Then f is $1/\sqrt{n}$-Lipschitz, and hence for every $t \geq 0$,

$$\mathbb{P}\left(\left|\frac{\|X\|_2}{\sqrt{n}} - \mathbb{E}\frac{\|X\|_2}{\sqrt{n}}\right| \geq t\right) \leq 2\exp\left(-\frac{nt^2}{2}\right).$$

Proof. By the reverse triangle inequality,

$$\left|\frac{\|x\|_2}{\sqrt{n}} - \frac{\|y\|_2}{\sqrt{n}}\right| \leq \frac{\|x-y\|_2}{\sqrt{n}}.$$

Hence f is $1/\sqrt{n}$-Lipschitz. Apply the Gaussian concentration theorem. □

Corollary 6.2.9 (Upper-tail bound for χ_n^2)**.** Let $Y = \|X\|_2^2 \sim \chi_n^2$. Then for every $t \geq 0$,

$$\mathbb{P}\left(\frac{Y}{n} \geq 1 + 2t + t^2\right) \leq \exp\left(-\frac{nt^2}{2}\right).$$

Proof. Since $\mathbb{E}\,\|X\|_2 \leq \sqrt{\mathbb{E}\,\|X\|_2^2} = \sqrt{n}$, the one-sided version of the previous result yields

$$\mathbb{P}\left(\frac{\|X\|_2}{\sqrt{n}} \geq 1 + t\right) \leq \exp\left(-\frac{nt^2}{2}\right).$$

Squaring the event gives

$$\left\{\frac{\|X\|_2}{\sqrt{n}} \geq 1 + t\right\} = \left\{\frac{Y}{n} \geq (1+t)^2\right\} = \left\{\frac{Y}{n} \geq 1 + 2t + t^2\right\}.$$

□

Remark. Concentration of $\sqrt{Y}$ is often easier than concentration of Y itself, because the square root is Lipschitz when there is a large deviation to the mean.

Corollary 6.2.10 (Concentration of Gaussian order statistics)**.** Let $X_1, \ldots, X_n$ be i.i.d. $N(0,1)$, and let

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$$

be the order statistics. Then for each $k \in \{1, \ldots, n\}$,

$$\mathbb{P}\big(|X_{(k)} - \mathbb{E}X_{(k)}| \ge t\big) \le 2\exp\left(-\frac{t^2}{2}\right), \qquad t \ge 0.$$

Proof. It is a standard fact that the sorting map is non-expansive in ℓ_2: if $x_{(k)}$ and $y_{(k)}$ denote the kth order statistics of x and y, then

$$|x_{(k)} - y_{(k)}| \le \|x - y\|_2 .$$

Thus the function $f_k(x) = x_{(k)}$ is 1-Lipschitz. Apply the Gaussian concentration theorem. □

Remark (Main idea)**.** Although order statistics are highly nonlinear, each individual order statistic is still a Lipschitz functional of the sample.

Corollary 6.2.11 (Concentration of singular values)**.** Let $W \in \mathbb{R}^{n\times d}$ have i.i.d. $N(0,1)$ entries, and let

$$\sigma_1(W) \ge \sigma_2(W) \ge \cdots \ge \sigma_{\min(n,d)}(W) \ge 0$$

be its singular values. Then for every k and every $t \ge 0$,

$$\mathbb{P}\left(\left|\sigma_k\left(\frac{W}{\sqrt{n}}\right) - \mathbb{E}\sigma_k\left(\frac{W}{\sqrt{n}}\right)\right| \ge t\right) \le 2\exp\left(-\frac{nt^2}{2}\right).$$

Proof. By Weyl's inequality,

$$|\sigma_k(A) - \sigma_k(B)| \le \|A - B\|_{\mathrm{op}} \le \|A - B\|_F .$$

Hence the map $A \mapsto \sigma_k(A)$ is 1-Lipschitz with respect to the Frobenius norm. Therefore

$$A \mapsto \sigma_k(A/\sqrt{n})$$

is $1/\sqrt{n}$-Lipschitz. Since W is a standard Gaussian vector in $\mathbb{R}^{nd}$ after vectorization, the Gaussian concentration theorem applies. □

Remark. This is one of the most useful ways to get concentration for random matrices: identify the matrix quantity of interest as a Lipschitz function of the entries.

Lemma 6.2.12 (Whitened covariance error)**.** Let $X_1, \ldots, X_n \in \mathbb{R}^d$ be i.i.d. $N(0,\Sigma)$, and let

$$\widehat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top .$$

Write

$$X_i = \Sigma^{1/2} W_i, \qquad W_i \sim N(0, I_d).$$

If $W \in \mathbb{R}^{n\times d}$ has rows $W_i^\top$, then

$$\Sigma^{-1/2}(\widehat{\Sigma} - \Sigma)\Sigma^{-1/2} = \frac{1}{n}W^\top W - I_d.$$

Consequently,

$$\left\|\Sigma^{-1/2}(\widehat{\Sigma} - \Sigma)\Sigma^{-1/2}\right\|_{\text{op}} = \sup_{v\in S^{d-1}} \left| \frac{1}{n}\sum_{i=1}^n \left((v^\top W_i)^2 - 1\right)\right|.$$

Proof. Using $X_i = \Sigma^{1/2}W_i$,

$$\widehat{\Sigma} = \frac{1}{n}\sum_{i=1}^n \Sigma^{1/2}W_iW_i^\top\Sigma^{1/2} = \Sigma^{1/2}\left(\frac{1}{n}W^\top W\right)\Sigma^{1/2}.$$

Subtract $\Sigma = \Sigma^{1/2}I_d\Sigma^{1/2}$ and whiten:

$$\Sigma^{-1/2}(\widehat{\Sigma} - \Sigma)\Sigma^{-1/2} = \frac{1}{n}W^\top W - I_d.$$

For a symmetric matrix A,

$$\|A\|_{\text{op}} = \sup_{v\in S^{d-1}} |v^\top Av|.$$

Applying this with $A = \frac{1}{n}W^\top W - I_d$ gives

$$v^\top Av = \frac{1}{n}\sum_{i=1}^n (v^\top W_i)^2 - 1.$$

□

Remark. For each fixed v, the random variable $(v^\top W_i)^2 - 1$ is sub-exponential, so the fixed-direction fluctuations are easy. The hard part is the supremum over all $v \in S^{d-1}$, which is handled by empirical-process methods or by an ε-net argument.

6.3 Lecture 22: Applications in regression

6.3.1 The Gaussian sequence model and Hard thresholding

Definition 6.3.1 (Gaussian sequence model)**.** The Gaussian sequence model is defined by the observations

$$Y_j = \mu_j + \varepsilon_j, \qquad j = 1,\dots,d, \qquad \varepsilon_j \overset{\text{iid}}{\sim} N\left(0, \frac{\sigma^2}{n}\right).$$

Lemma 6.3.2 (Bias-variance decomposition of the risk)**.** For any estimator $\widehat{\mu} = \widehat{\mu}(Y) \in \mathbb{R}^d$,

$$\mathbb{E}\,\|\widehat{\mu} - \mu\|_2^2 = \|\mathbb{E}\widehat{\mu} - \mu\|_2^2 + \operatorname{tr}\big(\operatorname{Cov}(\widehat{\mu})\big).$$

Proof. Write

$$\widehat{\mu} - \mu = (\widehat{\mu} - \mathbb{E}\widehat{\mu}) + (\mathbb{E}\widehat{\mu} - \mu).$$

Squaring and taking expectations,

$$\mathbb{E}\,\|\widehat{\mu} - \mu\|_2^2 = \mathbb{E}\,\|\widehat{\mu} - \mathbb{E}\widehat{\mu}\|_2^2 + \|\mathbb{E}\widehat{\mu} - \mu\|_2^2 + 2\mathbb{E}\,\langle \widehat{\mu} - \mathbb{E}\widehat{\mu}, \mathbb{E}\widehat{\mu} - \mu\rangle\,.$$

The cross term is zero because $\mathbb{E}(\widehat{\mu} - \mathbb{E}\widehat{\mu}) = 0$. Also,

$$\mathbb{E}\,\|\widehat{\mu} - \mathbb{E}\widehat{\mu}\|_2^2 = \operatorname{tr}(\operatorname{Cov}(\widehat{\mu})).$$

□

Corollary 6.3.3 (Risk of the naive estimator)**.** For the estimator $\widehat{\mu} = Y$,

$$\mathbb{E}\,\|Y - \mu\|_2^2 = \sum_{j=1}^{d} \mathbb{E}\varepsilon_j^2 = \frac{\sigma^2 d}{n}.$$

Proof. Since $Y - \mu = \varepsilon$ and the coordinates are independent with variance σ^2/n,

$$\mathbb{E}\,\|Y - \mu\|_2^2 = \sum_{j=1}^{d} \operatorname{Var}(\varepsilon_j) = d\frac{\sigma^2}{n}.$$

□

Assume now that μ is sparse, with support

$$S_\mu := \{j : \mu_j \neq 0\}, \qquad s := |S_\mu|.$$

Proposition 6.3.4 (Oracle estimator when the support is known)**.** If the support S_μ were known, the oracle estimator

$$\widehat{\mu}_j^{\,\mathrm{or}} = Y_j \mathbb{1}_{\{j \in S_\mu\}}$$

would satisfy

$$\mathbb{E}\,\|\widehat{\mu}^{\,\mathrm{or}} - \mu\|_2^2 = \frac{\sigma^2 s}{n}.$$

Proof. If $j \in S_\mu$, then $\widehat{\mu}_j^{\,\mathrm{or}} - \mu_j = Y_j - \mu_j = \varepsilon_j$. If $j \notin S_\mu$, then $\mu_j = 0$ and $\widehat{\mu}_j^{\,\mathrm{or}} = 0$, so the error is zero. Hence

$$\mathbb{E}\,\|\widehat{\mu}^{\,\mathrm{or}} - \mu\|_2^2 = \sum_{j \in S_\mu} \mathbb{E}\varepsilon_j^2 = \frac{\sigma^2 s}{n}.$$

□

Remark. This is the ideal benchmark. If we knew exactly which coordinates were nonzero, we would only estimate those s coordinates and ignore the rest. The risk would then scale like s/n instead of d/n. Since usually $s \ll d$, this can be a dramatic improvement. The real task is therefore to imitate this oracle behavior without knowing the support.

To estimate the support, we applying hard thresholding that uses the observed magnitude $|Y_j|$ as a proxy for whether μ_j is nonzero.

Definition 6.3.5 (Hard-thresholding estimator)**.** For $\tau \geq 0$, define

$$\widehat{\mu}_j^{\tau} = Y_j \mathbb{1}_{\{|Y_j| \geq \tau\}}, \qquad j = 1, \ldots, d.$$

A convenient event is

$$E_1 := \left\{ \max_{1 \leq j \leq d} |\varepsilon_j| \leq \tau_0 \right\}, \qquad \tau_0 = \sigma\sqrt{\frac{2\log(2d/\delta)}{n}}.$$

Remark. The event E_1 says that *all* noise coordinates are simultaneously bounded by τ_0. The logarithmic term appears because we are controlling the maximum over d coordinates, not just one coordinate. This is the usual multiple-comparison price.

The following lemma gives a clean high-probability event on which the rest of the argument becomes deterministic. Once we know that every noise coordinate is at most τ_0, we can compare Y_j and μ_j coordinate by coordinate.

Lemma 6.3.6 (Uniform noise control)**.** For the choice above,

$$\mathbb{P}(E_1) \geq 1 - \delta.$$

Proof. By the Gaussian tail bound,

$$\mathbb{P}(|\varepsilon_j| > \tau_0) \leq 2\exp\left(-\frac{n\tau_0^2}{2\sigma^2}\right) = 2\exp\left(-\log\frac{2d}{\delta}\right) = \frac{\delta}{d}.$$

Apply the union bound:

$$\mathbb{P}(E_1^c) = \mathbb{P}\left(\max_{1\leq j\leq d}|\varepsilon_j| > \tau_0\right) \leq \sum_{j=1}^{d}\mathbb{P}(|\varepsilon_j| > \tau_0) \leq d\cdot\frac{\delta}{d} = \delta.$$

□

Proposition 6.3.7 (Coordinatewise oracle inequality for hard thresholding)**.** Let $\widehat{\mu}^{2\tau_0}$ be the hard-threshold estimator with threshold $2\tau_0$. On the event E_1,

$$\left\|\widehat{\mu}^{2\tau_0} - \mu\right\|_2^2 \leq 9\sum_{j=1}^{d}\min\{\mu_j^2, \tau_0^2\}.$$

Therefore, with probability at least $1-\delta$,

$$\left\|\widehat{\mu}^{2\tau_0} - \mu\right\|_2^2 \leq 9\sum_{j=1}^{d}\min\{\mu_j^2, \tau_0^2\}.$$

Proof. We work on E_1, so that $|\varepsilon_j| \leq \tau_0$ for all j. We consider three cases.

Case 1: $|\mu_j| \leq \tau_0$.
Then

$$|Y_j| \leq |\mu_j| + |\varepsilon_j| \leq \tau_0 + \tau_0 = 2\tau_0.$$

Hence $\widehat{\mu}_j^{2\tau_0} = 0$, so

$$(\widehat{\mu}_j^{2\tau_0} - \mu_j)^2 = \mu_j^2 = \min\{\mu_j^2, \tau_0^2\} \leq 9\min\{\mu_j^2, \tau_0^2\}.$$

Case 2: $|\mu_j| > 3\tau_0$.
Then

$$|Y_j| \geq |\mu_j| - |\varepsilon_j| > 3\tau_0 - \tau_0 = 2\tau_0,$$

so $\widehat{\mu}_j^{2\tau_0} = Y_j$. Therefore

$$(\widehat{\mu}_j^{2\tau_0} - \mu_j)^2 = \varepsilon_j^2 \leq \tau_0^2 \leq 9\min\{\mu_j^2, \tau_0^2\}.$$

Case 3: $\tau_0 < |\mu_j| \leq 3\tau_0$.

In this regime,
$$\widehat{\mu}_j^{2\tau_0} - \mu_j = -\mu_j \mathbb{1}_{\{|Y_j|<2\tau_0\}} + \varepsilon_j \mathbb{1}_{\{|Y_j|\geq 2\tau_0\}}.$$
Hence
$$(\widehat{\mu}_j^{2\tau_0} - \mu_j)^2 \leq \mu_j^2 \vee \varepsilon_j^2 \leq (3\tau_0)^2 = 9\tau_0^2 = 9\min\{\mu_j^2, \tau_0^2\}.$$
Summing the three bounds over $j = 1, \ldots, d$ proves the claim. □

Remark. The reason that we threshold at $2\tau_0$ is that on the event E_1, a pure-noise coordinate satisfies $|Y_j| = |\varepsilon_j| \leq \tau_0$, so it stays below $2\tau_0$ and gets removed; on the other hand, a sufficiently strong signal with $|\mu_j| > 3\tau_0$ must satisfy
$$|Y_j| \geq |\mu_j| - |\varepsilon_j| > 2\tau_0,$$
so it survives thresholding. Thus $2\tau_0$ creates a safety gap between noise-only coordinates and clearly nonzero coordinates.

Remark. The magnitude of the bound
$$\sum_{j=1}^{d} \min\{\mu_j^2, \tau_0^2\}$$
has the following interpretations:

- if μ_j is very small, the contribution is μ_j^2, meaning it is cheap to ignore that coordinate;
- if μ_j is large, the contribution is only τ_0^2, meaning the error is essentially bounded by the noise level.

So the estimator behaves almost like an oracle: it discards weak coordinates and keeps strong ones.

Remark (Main idea)**.** The estimator behaves like an oracle up to a logarithmic threshold. Small coordinates are discarded, large coordinates are kept, and only the middle region requires a rough bound.

Corollary 6.3.8 (Sparse high-probability rate)**.** If $\|\mu\|_0 = s$, then with probability at least $1-\delta$,
$$\left\|\widehat{\mu}^{2\tau_0} - \mu\right\|_2^2 \leq 9s\tau_0^2 = 18\,\frac{\sigma^2 s}{n}\log\left(\frac{2d}{\delta}\right).$$

Proof. If $\|\mu\|_0 = s$, then only s coordinates are nonzero, so
$$\sum_{j=1}^{d} \min\{\mu_j^2, \tau_0^2\} \leq s\tau_0^2.$$
Now apply the previous proposition. □

Remark. This is the standard sparse recovery rate in this model. Compared with the oracle rate $\sigma^2 s/n$, the extra factor $\log d$ is the cost of not knowing the support in advance and having to search among d possible coordinates.

Corollary 6.3.9 (Exact support recovery under a beta-min condition)**.** Assume
$$\min_{j\in S_\mu} |\mu_j| > 3\tau_0.$$

Then on the event E_1,

$$\operatorname{supp}(\widehat{\mu}^{2\tau_0}) = \operatorname{supp}(\mu).$$

In particular, this holds with probability at least $1-\delta$.

Proof. If $j \in S_\mu$, then $|\mu_j| > 3\tau_0$, so on E_1,

$$|Y_j| \geq |\mu_j| - |\varepsilon_j| > 3\tau_0 - \tau_0 = 2\tau_0.$$

Hence $\widehat{\mu}_j^{2\tau_0} = Y_j \neq 0$.
If $j \notin S_\mu$, then $\mu_j = 0$, so on E_1,

$$|Y_j| = |\varepsilon_j| \leq \tau_0 < 2\tau_0.$$

Hence $\widehat{\mu}_j^{2\tau_0} = 0$.
Thus the estimated and true supports coincide. $\square$

Remark. The condition

$$\min_{j \in S_\mu} |\mu_j| > 3\tau_0$$

is a *beta-min condition*: every nonzero signal must be large enough to stand out from the noise. Without such a separation condition, exact support recovery is impossible in general, because very small nonzero coefficients are statistically indistinguishable from zero.

6.3.2 Soft thresholding and hyperparameter tuning

Instead of using hard-thresholds, we may also use soft-thresholds.

Definition 6.3.10 (Soft-threshold map)**.** For $\lambda \geq 0$, define

$$\eta_\lambda(y) = \begin{cases} y - \lambda, & y > \lambda, \\ 0, & |y| \leq \lambda, \\ y + \lambda, & y < -\lambda. \end{cases}$$

Coordinatewise, for $y = (y_1, \ldots, y_d)$,

$$\eta_\lambda(y) = \big(\eta_\lambda(y_1), \ldots, \eta_\lambda(y_d)\big).$$

Proposition 6.3.11 (Soft thresholding is the lasso solution)**.** For any $Y \in \mathbb{R}^d$ and $\lambda \geq 0$,

$$\widehat{\beta}^\lambda := \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \left\{ \frac{1}{2} \|Y - \beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

is given coordinatewise by

$$\widehat{\beta}_j^\lambda = \eta_\lambda(Y_j), \qquad j = 1, \ldots, d.$$

Proof. The objective is separable:

$$\frac{1}{2} \|Y - \beta\|_2^2 + \lambda \|\beta\|_1 = \sum_{j=1}^d \left(\frac{1}{2}(Y_j - \beta_j)^2 + \lambda |\beta_j| \right).$$

Thus it suffices to solve the one-dimensional problem

$$\min_{b\in\mathbb{R}}\left\{\frac{1}{2}(y-b)^2+\lambda|b|\right\}.$$

If $b>0$, differentiate:

$$-(y-b)+\lambda=0 \quad\Longrightarrow\quad b=y-\lambda,$$

which is admissible only when $y>\lambda$.
If $b<0$, differentiate:

$$-(y-b)-\lambda=0 \quad\Longrightarrow\quad b=y+\lambda,$$

which is admissible only when $y<-\lambda$.
If $|y|\le\lambda$, then $b=0$ satisfies the subgradient optimality condition

$$0\in-(y-0)+\lambda[-1,1].$$

Hence the minimizer is exactly

$$b=\begin{cases} y-\lambda, & y>\lambda,\\ 0, & |y|\le\lambda,\\ y+\lambda, & y<-\lambda.\end{cases}$$

Applying this coordinatewise completes the proof. □

Remark. This proposition shows that soft thresholding is not just a heuristic denoising rule: it is exactly the solution of an ℓ_1-penalized least-squares problem. So the lasso can be viewed as a principled optimization formulation of thresholding.

Now we assume the mean of the noisy observations is identical:

$$Y\sim N(\mu,I_n).$$

For a weakly differentiable estimator $\hat{\mu}(Y)$, Stein's unbiased risk estimate is

$$\widehat{R}=-n+\|Y-\hat{\mu}(Y)\|_2^2+2\,\mathrm{div}\hat{\mu}(Y),$$

where

$$\mathrm{div}\hat{\mu}(Y)=\sum_{i=1}^{n}\frac{\partial\hat{\mu}_i(Y)}{\partial Y_i}.$$

Then

$$\mathbb{E}_\mu[\widehat{R}]=\mathbb{E}_\mu\|\hat{\mu}(Y)-\mu\|_2^2.$$

So $\widehat{R}$ is an unbiased estimator of the true risk. For the soft-threshold estimator $\hat{\mu}_i^\lambda=\eta_\lambda(Y_i)$, we have

$$Y_i-\eta_\lambda(Y_i)=\begin{cases}\lambda, & Y_i>\lambda,\\ Y_i, & |Y_i|\le\lambda,\\ -\lambda, & Y_i<-\lambda,\end{cases}$$

hence

$$\big(Y_i-\eta_\lambda(Y_i)\big)^2=\min\{Y_i^2,\lambda^2\}.$$

Also,

$$\frac{\partial}{\partial Y_i}\eta_\lambda(Y_i)=\mathbb{1}\{|Y_i|>\lambda\}$$

away from the threshold points. Therefore,

$$\widehat{R}(\lambda) = -n + \sum_{i=1}^{n} \min\{Y_i^2, \lambda^2\} + 2\sum_{i=1}^{n} \mathbb{1}\{|Y_i| > \lambda\}.$$

Equivalently,

$$\widehat{R}(\lambda) = \sum_{i=1}^{n} \left[1 + \min\{Y_i^2, \lambda^2\} - 2\mathbb{1}\{|Y_i| \le \lambda\}\right].$$

Thus, one can choose

$$\hat{\lambda} = \operatorname*{argmin}_{\lambda \ge 0} \widehat{R}(\lambda),$$

and then use $\hat{\mu}^{\hat{\lambda}}$ as the data-driven estimator. If $\widehat{R}(\lambda)$ is uniformly close to the true risk $R(\lambda)$ over a range of λ, then the selected $\hat{\lambda}$ behaves approximately like the oracle choice

$$\lambda^\star = \operatorname*{argmin}_{\lambda \ge 0} R(\lambda).$$

So the SURE-selected estimator is asymptotically as good as the best estimator in the class.

6.3.3 Lasso in the regression model

Now we move to the regression model

$$Y = X\beta^\star + \varepsilon,$$

where $X \in \mathbb{R}^{n\times d}$ and $\beta^\star \in \mathbb{R}^d$. The least squares estimator is

$$\hat{\beta}^{\mathrm{OLS}} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \|Y - X\beta\|_2^2.$$

If $\operatorname{rank}(X) = d$ and $d \le n$, then

$$\hat{\beta}^{\mathrm{OLS}} = (X^\top X)^{-1} X^\top Y.$$

When $d > n$ or $\operatorname{rank}(X) < d$, the coefficient vector need not be unique, but the fitted values $X\hat{\beta}$ are unique. Among all least-squares solutions, the minimum-norm solution is

$$\hat{\beta}^{\mathrm{MN}} = X^+ Y,$$

where X^+ is the Moore–Penrose pseudoinverse.

> **Remark.** Even when β is not unique, the prediction $X\beta$ may still be uniquely determined by the convex objective.

A general penalised estimator has the form

$$\hat{\beta} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \left\{\|Y - X\beta\|_2^2 + \lambda \operatorname{pen}(\beta)\right\}.$$

Typical penalties are:

$$\ell_0 : \ \|\beta\|_0, \qquad \ell_1 : \ \|\beta\|_1, \qquad \ell_2 : \ \|\beta\|_2^2.$$

> **Remark.** There is also a constrained form:
>
> $$\hat{\beta}_t = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \|Y - X\beta\|_2^2 \quad \text{subject to} \quad \operatorname{pen}(\beta) \le t.$$
>
> For convex penalties, the penalized and constrained formulations are closely related.

The lasso estimator is

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^d}{\operatorname{argmin}} \left\{ \frac{1}{2} \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}.$$

Geometrically, the ℓ_1 ball has corners, and these corners encourage sparse solutions. Because $\|\beta\|_1$ is not differentiable at 0, optimality is expressed by subgradients:

$$X^\top (Y - X\hat{\beta}) = \lambda s, \qquad s \in \partial \|\hat{\beta}\|_1.$$

Coordinatewise,

$$s_j = \begin{cases} \operatorname{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ [-1, 1], & \hat{\beta}_j = 0. \end{cases}$$

Hence:

- if $\hat{\beta}_j \neq 0$, then $X_j^\top (Y - X\hat{\beta}) = \lambda \operatorname{sign}(\hat{\beta}_j)$;
- if $\hat{\beta}_j = 0$, then $|X_j^\top (Y - X\hat{\beta})| \leq \lambda$.

Proposition 6.3.12. Consider the lasso problem

$$\hat{\beta} \in \arg\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}, \qquad \lambda > 0.$$

Then the following hold:

(a). the fitted value $X\hat{\beta}$ is unique;

(b). the coefficient vector $\hat{\beta}$ need not be unique in general;

(c). the subgradient vector $s \in \partial \|\hat{\beta}\|_1$ associated with an optimum is unique.

Proof. Write

$$f(\beta) = \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1.$$

Uniqueness of the fitted value. Suppose $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ are two minimizers. Since f is convex, for any $t \in [0, 1]$,

$$f\left(t\hat{\beta}^{(1)} + (1-t)\hat{\beta}^{(2)}\right) \leq t f(\hat{\beta}^{(1)}) + (1-t) f(\hat{\beta}^{(2)}),$$

and the right-hand side equals the minimum value of f. Hence equality must hold. Subtracting the convex function $\lambda \|\beta\|_1$ from the inequality yields that

$$\left\| y - X\left(t\hat{\beta}^{(1)} + (1-t)\hat{\beta}^{(2)}\right) \right\|_2^2 \geq t \left\| y - X\hat{\beta}^{(1)} \right\|_2^2 + (1-t) \left\| y - X\hat{\beta}^{(2)} \right\|_2^2$$

The strict convexity of the map

$$u \mapsto \frac{1}{2} \|y - u\|_2^2$$

then implies that

$$X\hat{\beta}^{(1)} = X\hat{\beta}^{(2)}.$$

Thus all minimizers have the same fitted value, so $X\hat{\beta}$ is unique.

Uniqueness of the subgradient vector. By the optimality condition for the lasso, $\hat{\beta}$ is optimal if and only if

$$0 \in -X^\top (y - X\hat{\beta}) + \lambda \, \partial \|\hat{\beta}\|_1.$$

Equivalently, there exists $s \in \partial \|\hat{\beta}\|_1$ such that

$$X^\top (y - X\hat{\beta}) = \lambda s.$$

Therefore,

$$s = \frac{1}{\lambda} X^\top (y - X\hat{\beta})$$

is also unique. Thus the subgradient vector associated with any optimum is unique. This completes the proof. $\square$

Remark. We can also show that for all selection sets E where X has full rank on E, it admits a unique $\hat{\beta}$, i.e., for nice selection sets, Lasso only selects but does not estimate. Let $E = \mathrm{supp}(\hat{\beta})$ and $\mathrm{rank}(X_E) = |E|$. Then any lasso solution satisfies

$$\hat{\beta}_{E^c} = 0, \qquad X_E^\top (Y - X_E \hat{\beta}_E) = \lambda s_E.$$

Therefore,

$$\hat{\beta}_E = (X_E^\top X_E)^+ (X_E^\top Y - \lambda s_E) + v, \qquad v \in \mathrm{Null}(X_E).$$

Since $\mathrm{rank}(X_E) = |E|$ implies $v = 0$, we know that $\hat{\beta}$ is unique.

Remark. A sufficient condition for uniqueness is that the columns of X are in *general position*. Informally, this means that no signed column lies in the affine span of too few other signed columns. When the entries of X come from a continuous distribution, the general position condition holds almost surely.

Proposition 6.3.13 (Sparse support of a lasso solution)**.** For any $\lambda > 0$, there exists a lasso solution with at most n nonzero coefficients.

This reflects the fact that the lasso chooses a sparse representation even when $d \gg n$.

6.3.4 Hyperparameter tuning in Lasso

To decide the hyperparameter λ, we need to evaluate the performance of the estimator $\hat{\beta}$. There are several different notions of error for $\hat{\beta}$:

(a). **In-sample prediction risk**

$$\mathbb{E}\left[\frac{1}{n}\|X\hat{\beta} - X\beta^\star\|_2^2\right].$$

(b). **Estimation risk**

$$\mathbb{E}\|\hat{\beta} - \beta^\star\|_2^2.$$

(c). **Out-of-sample prediction risk** for a new covariate vector X_0,

$$\mathbb{E}\left[(X_0^\top \hat{\beta} - X_0^\top \beta^\star)^2\right].$$

Usually, the in-sample prediction error is the easiest to analyse, since it depends only on the observed design matrix.

Proposition 6.3.14. Consider the normalised lasso

$$\hat{\beta} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \left\{ \frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1 \right\},$$

under the model

$$Y = X\beta^\star + \varepsilon.$$

Let

$$\Delta = \hat{\beta} - \beta^\star,$$

and define the event

$$\mathcal{E}_\lambda = \left\{ \left\| \frac{1}{n} X^\top \varepsilon \right\|_\infty \le \frac{\lambda}{2} \right\}.$$

Then on the event $\mathcal{E}_\lambda$,

$$\frac{1}{n}\|X\hat{\beta} - X\beta^\star\|_2^2 = \frac{1}{n}\|X\Delta\|_2^2 \le 3\lambda\|\beta^\star\|_1.$$

In particular, this is a slow-rate oracle inequality for the in-sample prediction risk. Moreover, if the columns of X satisfy

$$\frac{1}{n}\|X_j\|_2^2 \le 1, \qquad j = 1, \dots, d,$$

and the noise is sub-Gaussian with scale parameter σ, then with high probability

$$\left\| \frac{1}{n} X^\top \varepsilon \right\|_\infty \lesssim \sigma \sqrt{\frac{\log(d/\delta)}{n}}.$$

Hence a natural choice is

$$\lambda \asymp \sigma \sqrt{\frac{\log(d/\delta)}{n}},$$

for which

$$\frac{1}{n}\|X\hat{\beta} - X\beta^\star\|_2^2 \lesssim \sigma \|\beta^\star\|_1 \sqrt{\frac{\log d}{n}},$$

up to universal constants.

Proof. By optimality of $\hat{\beta}$,

$$\frac{1}{2n}\|Y - X\hat{\beta}\|_2^2 + \lambda\|\hat{\beta}\|_1 \le \frac{1}{2n}\|Y - X\beta^\star\|_2^2 + \lambda\|\beta^\star\|_1.$$

Substituting $Y = X\beta^\star + \varepsilon$ and writing $\Delta = \hat{\beta} - \beta^\star$, we obtain

$$\frac{1}{2n}\|\varepsilon - X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 \le \frac{1}{2n}\|\varepsilon\|_2^2 + \lambda\|\beta^\star\|_1.$$

Expanding the square and cancelling $\|\varepsilon\|_2^2/(2n)$ gives

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{1}{n}\varepsilon^\top X\Delta + \lambda\big(\|\beta^\star\|_1 - \|\hat{\beta}\|_1\big).$$

Using Hölder's inequality,

$$\frac{1}{n}\varepsilon^\top X\Delta \le \left\| \frac{1}{n} X^\top \varepsilon \right\|_\infty \|\Delta\|_1.$$

Therefore,

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \left\| \frac{1}{n} X^\top \varepsilon \right\|_\infty \|\Delta\|_1 + \lambda\big(\|\beta^\star\|_1 - \|\hat{\beta}\|_1\big).$$

On the event $\mathcal{E}_\lambda$,

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{\lambda}{2}\|\Delta\|_1 + \lambda\big(\|\beta^\star\|_1 - \|\hat{\beta}\|_1\big).$$

Now apply the triangle inequality,

$$\|\Delta\|_1 = \|\hat{\beta} - \beta^\star\|_1 \le \|\hat{\beta}\|_1 + \|\beta^\star\|_1.$$

Substituting this into the previous bound yields

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{\lambda}{2}\big(\|\hat{\beta}\|_1 + \|\beta^\star\|_1\big) + \lambda\big(\|\beta^\star\|_1 - \|\hat{\beta}\|_1\big).$$

After collecting terms,

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{3\lambda}{2}\|\beta^\star\|_1 - \frac{\lambda}{2}\|\hat\beta\|_1 \le \frac{3\lambda}{2}\|\beta^\star\|_1.$$

Multiplying by 2, we conclude that

$$\frac{1}{n}\|X\Delta\|_2^2 \le 3\lambda\|\beta^\star\|_1.$$

Since $X\Delta = X\hat\beta - X\beta^\star$, this proves the desired slow-rate prediction bound.
For the choice of λ, note that if $\|X_j\|_2^2/n \le 1$ for all j and the noise is sub-Gaussian with scale σ, then standard maximal inequalities imply that, with high probability,

$$\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty \lesssim \sigma\sqrt{\frac{\log(d/\delta)}{n}}.$$

Thus taking

$$\lambda \asymp \sigma\sqrt{\frac{\log(d/\delta)}{n}}$$

ensures that the event $\mathcal{E}_\lambda$ holds with high probability, and hence

$$\frac{1}{n}\|X\hat\beta - X\beta^\star\|_2^2 \lesssim \sigma\|\beta^\star\|_1\sqrt{\frac{\log d}{n}}.$$

□

Remark. This slow-rate bound does not require sparsity of $\beta^\star$ or a strong structural condition on the design matrix X. Its conclusion is only for prediction:

$$\frac{1}{n}\|X\hat\beta - X\beta^\star\|_2^2 \lesssim \lambda\|\beta^\star\|_1.$$

It does not by itself imply a corresponding bound on $\|\hat\beta - \beta^\star\|_1$ or $\|\hat\beta - \beta^\star\|_2$.

To obtain a sharper bound, we need stronger assumptions.

Proposition 6.3.15. Suppose the spare model is given by

$$Y = X\beta^\star + \varepsilon, \qquad S^\star = \mathrm{supp}(\beta^\star), \qquad |S^\star| = s.$$

and let the lasso estimator be defined by

$$\hat\beta \in \arg\min_{\beta\in\mathbb{R}^d}\left\{\frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1\right\}.$$

Define the estimation error

$$\Delta = \hat\beta - \beta^\star,$$

and the event

$$\mathcal{E}_\lambda = \left\{\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty \le \frac{\lambda}{2}\right\}.$$

Assume moreover that the restricted eigenvalue condition holds: there exists $\kappa > 0$ such that

$$\frac{1}{n}\|Xu\|_2^2 \ge \kappa\|u\|_2^2 \qquad \text{for all } u \in C_3(S^\star),$$

where

$$C_3(S^\star) = \left\{ u \in \mathbb{R}^d : \|u_{(S^\star)^c}\|_1 \le 3\|u_{S^\star}\|_1 \right\}.$$

Then on the event $\mathcal{E}_\lambda$, one has

$$\Delta \in C_3(S^\star),$$

and

$$\|\Delta\|_2 \le \frac{3\sqrt{s}\lambda}{\kappa}, \qquad \|\Delta\|_1 \le \frac{12s\lambda}{\kappa}, \qquad \frac{1}{n}\|X\Delta\|_2^2 \le \frac{9s\lambda^2}{\kappa}.$$

In particular, if

$$\lambda \asymp \sigma\sqrt{\frac{\log d}{n}},$$

then

$$\frac{1}{n}\|X\hat{\beta} - X\beta^\star\|_2^2 \lesssim \frac{\sigma^2 s \log d}{n},$$

up to universal constants and the restricted eigenvalue factor.

Proof. Since $\hat{\beta}$ minimizes the lasso objective and $\beta^\star$ is feasible for comparison,

$$\frac{1}{2n}\|Y - X\hat{\beta}\|_2^2 + \lambda\|\hat{\beta}\|_1 \le \frac{1}{2n}\|Y - X\beta^\star\|_2^2 + \lambda\|\beta^\star\|_1.$$

Using $Y = X\beta^\star + \varepsilon$ and $\Delta = \hat{\beta} - \beta^\star$, this becomes

$$\frac{1}{2n}\|\varepsilon - X\Delta\|_2^2 + \lambda\|\hat{\beta}\|_1 \le \frac{1}{2n}\|\varepsilon\|_2^2 + \lambda\|\beta^\star\|_1,$$

hence

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{1}{n}\varepsilon^\top X\Delta + \lambda\|\beta^\star\|_1 - \lambda\|\hat{\beta}\|_1.$$

On the event $\mathcal{E}_\lambda$,

$$\frac{1}{n}\varepsilon^\top X\Delta \le \left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty \|\Delta\|_1 \le \frac{\lambda}{2}\|\Delta\|_1.$$

Therefore

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{\lambda}{2}\|\Delta\|_1 + \lambda\|\beta^\star\|_1 - \lambda\|\hat{\beta}\|_1.$$

Because $\beta^\star_{(S^\star)^c} = 0$, we have

$$\|\beta^\star\|_1 - \|\hat{\beta}\|_1 = \|\beta^\star_{S^\star}\|_1 - \|\hat{\beta}_{S^\star}\|_1 - \|\hat{\beta}_{(S^\star)^c}\|_1 \le \|\Delta_{S^\star}\|_1 - \|\Delta_{(S^\star)^c}\|_1.$$

Substituting this into the previous inequality gives

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{\lambda}{2}\left(\|\Delta_{S^\star}\|_1 + \|\Delta_{(S^\star)^c}\|_1\right) + \lambda\|\Delta_{S^\star}\|_1 - \lambda\|\Delta_{(S^\star)^c}\|_1.$$

After collecting terms,

$$\frac{1}{2n}\|X\Delta\|_2^2 \le \frac{3\lambda}{2}\|\Delta_{S^\star}\|_1 - \frac{\lambda}{2}\|\Delta_{(S^\star)^c}\|_1.$$

Rearranging yields

$$\frac{1}{2n}\|X\Delta\|_2^2 + \frac{\lambda}{2}\|\Delta_{(S^\star)^c}\|_1 \le \frac{3\lambda}{2}\|\Delta_{S^\star}\|_1.$$

In particular,

$$\|\Delta_{(S^\star)^c}\|_1 \le 3\|\Delta_{S^\star}\|_1,$$

so

$$\Delta \in C_3(S^\star).$$

Now apply the restricted eigenvalue condition:

$$\frac{1}{n}\|X\Delta\|_2^2 \geq \kappa\|\Delta\|_2^2.$$

On the other hand, from the basic inequality above,

$$\frac{1}{n}\|X\Delta\|_2^2 \leq 3\lambda\|\Delta_{S^\star}\|_1 \leq 3\lambda\sqrt{s}\,\|\Delta\|_2.$$

Combining these gives

$$\kappa\|\Delta\|_2^2 \leq 3\lambda\sqrt{s}\,\|\Delta\|_2,$$

and therefore

$$\|\Delta\|_2 \leq \frac{3\sqrt{s}\lambda}{\kappa}.$$

Since $\Delta \in C_3(S^\star)$,

$$\|\Delta\|_1 = \|\Delta_{S^\star}\|_1 + \|\Delta_{(S^\star)^c}\|_1 \leq 4\|\Delta_{S^\star}\|_1 \leq 4\sqrt{s}\,\|\Delta\|_2 \leq \frac{12s\lambda}{\kappa}.$$

Finally,

$$\frac{1}{n}\|X\Delta\|_2^2 \leq 3\lambda\sqrt{s}\,\|\Delta\|_2 \leq \frac{9s\lambda^2}{\kappa}.$$

If now $\lambda \asymp \sigma\sqrt{\log d/n}$, then

$$\frac{1}{n}\|X\hat{\beta} - X\beta^\star\|_2^2 = \frac{1}{n}\|X\Delta\|_2^2 \lesssim \frac{\sigma^2 s\log d}{n},$$

up to universal constants and the restricted eigenvalue factor. □

Remark. This is the standard fast-rate prediction bound for the lasso. Under sparsity and a restricted eigenvalue condition, the prediction error scales as

$$\frac{\sigma^2 s\log d}{n},$$

which is substantially sharper than the slower approximation-type bounds that hold without such structural assumptions.

Remark. Prediction consistency is weaker than exact variable-selection consistency. To recover the support $S^\star$ exactly, one usually needs:

- a design condition stronger than RE, often an irrepresentable-type condition;
- a *beta-min* condition ensuring that the nonzero coefficients are large enough:

$$\min_{j\in S^\star}|\beta_j^\star| \quad \text{must dominate the estimation error level.}$$

Thus, exact support recovery is much harder than obtaining a small prediction error.

Remark (Choosing the tuning parameter in practice)**.** Two standard strategies were emphasized:

(a). **SURE**, when a Gaussian sequence-model approximation is appropriate.

(b). **Cross-validation**, especially in regression problems where prediction performance is the main criterion.

The tuning parameter controls the amount of regularization:

- too small λ gives instability and overfitting;
- too large λ gives excessive shrinkage and underfitting.

More generally, if the true regression function is not exactly linear, say

$$Y = f^\star + \varepsilon,$$

and we fit the constrained lasso

$$\hat{\beta}_t = \operatorname*{argmin}_{\|\beta\|_1 \le t} \frac{1}{n}\|Y - X\beta\|_2^2,$$

then one can show an approximation-type bound of the form

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \lesssim \inf_{\|\beta\|_1 \le t} \frac{1}{n}\|X\beta - f^\star\|_2^2 + \sigma t\sqrt{\frac{\log d}{n}}.$$

So the lasso trades off approximation error and stochastic error.

Remark (Connection to nonparametric regression)**.** A useful way to connect nonparametric regression with high-dimensional linear methods is through a basis expansion. Let $Z_i \in \mathbb{R}$ be a scalar input, and choose basis functions

$$\phi_1, \phi_2, \ldots, \phi_m.$$

Define the feature vector

$$X_i = \big(\phi_1(Z_i), \phi_2(Z_i), \ldots, \phi_m(Z_i)\big) \in \mathbb{R}^m.$$

Then any function represented in this basis has the form

$$f_\beta(z) = \sum_{j=1}^{m} \beta_j \phi_j(z),$$

and the regression model becomes

$$Y_i = f_0(Z_i) + \varepsilon_i \approx X_i^\top \beta + \varepsilon_i.$$

This converts a function-estimation problem into a linear regression problem in a possibly high-dimensional dictionary. If the function is well approximated by only a few basis coefficients, then sparse methods such as the Lasso are natural.

However, Lasso can still give an oracle inequality in this case.

Proposition 6.3.16. Assume the data obey

$$Y = f^\star + \varepsilon \in \mathbb{R}^n,$$

where $f^\star \in \mathbb{R}^n$ is an arbitrary regression signal and $\varepsilon \in \mathbb{R}^n$ is a noise vector. Let

$$\hat{\beta}_t \in \arg\min_{\|\beta\|_1 \le t} \frac{1}{n}\|Y - X\beta\|_2^2,$$

where $X \in \mathbb{R}^{n \times d}$ is the design matrix.

Then for every $t \geq 0$, with probability at least $1 - c_1 d^{-c_2}$,

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \leq \inf_{\|\beta\|_1 \leq t} \frac{1}{n}\|X\beta - f^\star\|_2^2 + C\sigma t\sqrt{\frac{\log d}{n}},$$

provided that the columns of X are normalized so that

$$\frac{1}{n}\|X_j\|_2^2 \leq 1, \qquad j = 1, \dots, d,$$

and the noise coordinates are mean-zero sub-Gaussian with scale parameter σ.

Proof. Since $\hat{\beta}_t$ minimizes the constrained empirical risk over the set $\{\beta : \|\beta\|_1 \leq t\}$, for every β with $\|\beta\|_1 \leq t$,

$$\frac{1}{n}\|Y - X\hat{\beta}_t\|_2^2 \leq \frac{1}{n}\|Y - X\beta\|_2^2.$$

Substituting $Y = f^\star + \varepsilon$, we get

$$\frac{1}{n}\|f^\star + \varepsilon - X\hat{\beta}_t\|_2^2 \leq \frac{1}{n}\|f^\star + \varepsilon - X\beta\|_2^2.$$

Expanding both sides and cancelling the common term $\|\varepsilon\|_2^2/n$ yields

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \leq \frac{1}{n}\|X\beta - f^\star\|_2^2 + \frac{2}{n}\varepsilon^\top X(\hat{\beta}_t - \beta).$$

Using Hölder's inequality,

$$\frac{2}{n}\varepsilon^\top X(\hat{\beta}_t - \beta) \leq 2\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty \|\hat{\beta}_t - \beta\|_1.$$

Since both $\hat{\beta}_t$ and β belong to the ℓ_1-ball of radius t,

$$\|\hat{\beta}_t - \beta\|_1 \leq \|\hat{\beta}_t\|_1 + \|\beta\|_1 \leq 2t.$$

Therefore,

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \leq \frac{1}{n}\|X\beta - f^\star\|_2^2 + 4t\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty.$$

Taking the infimum over all β with $\|\beta\|_1 \leq t$, we obtain the deterministic bound

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \leq \inf_{\|\beta\|_1 \leq t} \frac{1}{n}\|X\beta - f^\star\|_2^2 + 4t\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty.$$

It remains to control the stochastic term. Under the column normalization, $\|X_j\|_2^2/n \leq 1$ and the sub-Gaussian assumption on ε, a standard maximal inequality gives

$$\left\|\frac{1}{n}X^\top\varepsilon\right\|_\infty \leq C'\sigma\sqrt{\frac{\log d}{n}}$$

with probability at least $1 - c_1 d^{-c_2}$. Substituting this estimate into the previous display gives

$$\frac{1}{n}\|X\hat{\beta}_t - f^\star\|_2^2 \leq \inf_{\|\beta\|_1 \leq t} \frac{1}{n}\|X\beta - f^\star\|_2^2 + C\sigma t\sqrt{\frac{\log d}{n}},$$

for a universal constant $C > 0$. This proves the claim. $\square$

Remark. The bound shows that the constrained lasso balances two effects:

- the *approximation error*
$$\inf_{\|\beta\|_1 \le t} \frac{1}{n}\|X\beta - f^\star\|_2^2,$$
which measures how well the true regression function can be approximated by a linear predictor with ℓ_1-norm at most t;
- the *stochastic error*
$$\sigma t \sqrt{\frac{\log d}{n}},$$
which grows with the size of the feasible set.

Hence larger values of t reduce approximation error but increase stochastic error, while smaller values of t do the reverse.